Does Option Volume Convey Incremental Information? Evidence from Synthetic Stock Benchmarks*

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Abstract

If option volume conveys incremental information, it should forecast the spread between actual and synthetic (option-implied) stock returns—not just actual returns. We test this implication around earnings announcements, material event filings, and week-by-week throughout the calendar, and find that both signed and unsigned volumes consistently fail to predict the spread. When predictability appears, it is statistically weak, economically negligible, or reverses sign. A noisy rational expectations model with informed trading across the stock and options markets yields this testable implication. By not benchmarking against synthetic returns, prior studies likely overstated the informational content of option volume.

Keywords: Option trading volume, stock return predictability, information.

JEL Classification: G12, C13, G14

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Abstract

If option volume conveys incremental information, it should forecast the spread between actual and synthetic (option-implied) stock returns—not just actual returns. We test this implication around earnings announcements, material event filings, and week-by-week throughout the calendar, and find that both signed and unsigned volumes consistently fail to predict the spread. When predictability appears, it is statistically weak, economically negligible, or reverses sign. A noisy rational expectations model with informed trading across the stock and options markets yields this testable implication. By not benchmarking against synthetic returns, prior studies likely overstated the informational content of option volume.

1 Introduction

A central question in asset pricing is whether option markets convey information that is not yet reflected in underlying stock prices. The conventional view is that they do. Because options provide embedded leverage, downside protection, and low-cost access to short positions, they are often regarded as a preferred vehicle for informed trading. Empirical studies documenting that option prices and trading volumes predict subsequent stock returns are typically interpreted as evidence that options contain incremental information. We argue that this interpretation is incomplete. The fact that option-based variables forecast *actual* stock returns is a necessary, but not sufficient, condition for options to be informationally superior. Because stock and option prices are connected by no-arbitrage, option-based predictors may appear to forecast stock returns mechanically, even if options themselves contain no independent information.

To separate genuine informational content from mechanical linkages, we focus on the *spread* between actual and synthetic stock returns implied by put–call parity. This spread removes the component of stock return variation that is mechanically tied to option prices through no-arbitrage. Hence, it isolates the residual variation that is more plausibly attributable to option-specific information.

Building on this insight, we revisit the evidence on stock return predictability derived from option trading volumes. Prior work has largely emphasized price-based predictors such as implied volatility spreads (Cremers and Weinbaum, 2010), implied skewness (Xing et al., 2010), and changes in implied volatility (An et al., 2014). However, mounting evidence suggests that much of the predictability in these measures reflects mechanical effects rather than information-based trading. For example, Goncalves-Pinto et al. (2020) show that apparent predictability in implied volatility and skewness can arise from temporary price pressure in the underlying stocks; skipping a day between portfolio formation and holding eliminates most of these predictability patterns. Similarly, Muravyev et al. (2025) argue that the predictive content of implied volatilities stems from their failure to adjust for stock borrowing costs: because OptionMetrics volatility estimates do not account for borrow fees, implied volatilities effectively proxy for those fees—known predictors of stock returns—and lose explanatory power once fee adjustments are applied. Both temporary price pressure and borrowing fees distort put–call parity, mechanically affecting implied volatilities even in the absence of option trading activity.[1]

[1] Goncalves-Pinto et al. (2020) show that the predictive power of implied volatility remains intact even within

In contrast, volume-based predictors are less susceptible to distortions arising from no-arbitrage conditions or trading frictions. Unsigned measures such as the option-to-stock (O/S) and put-to-call (P/C) volume ratios have been shown to predict subsequent stock returns (e.g., Roll et al. 2010; Johnson and So 2012; Ge et al. 2016; Pan and Poteshman 2006). More recent studies use signed volume data, which distinguish buyer- from seller-initiated trades, and find that option trading activity helps forecast the direction of stock price movements (e.g., Ge et al. 2016; Cremers et al. 2022). Despite this evidence, it remains unclear whether the observed predictability reflects information originating in the options market or simply the mechanical transmission of stock-market signals through no-arbitrage relationships. To distinguish between these channels, we develop a methodology that focuses on the component of stock returns unexplained by synthetic replication, thereby isolating the information that is unique to option markets.

This methodology is motivated by cross-market predictability implications derived from a stylized noisy rational expectations model that builds on and extends the framework in An et al. (2014). It features an informed investor, noise traders, and a market maker. The informed investor observes a private signal and trades in both the stock and its options to exploit this information. Noise traders generate random demands for stock and options. The market maker clears both markets and sets equilibrium prices, eliminating any arbitrage opportunities. This model yields three key insights. First, the put-call parity condition always holds in equilibrium, regardless of whether—or in which market—the informed investor trades. As a result, actual and synthetic stock prices remain aligned at all times, and an external observer cannot determine the venue through which private information is incorporated into prices.

Second, we show that even in this setting where actual and synthetic prices fully reflect the same information—rendering options informationally *irrelevant*—option prices and volumes can still correlate with future stock returns. We derive conditions under which this "predictability" emerges, not from violations of no-arbitrage constraints, but from prices only partially converging to fundamental values due to the presence of noise traders. This result highlights a key identification challenge: return predictability alone does not imply incremental information in options, as the model assumes that information is impounded simultaneously and symmetrically across both markets. In addition, the model shows that elevated option volume is not necessarily a signal of informed trading, as it may instead arise from noise trading or speculative demand, complicating

a subsample of stocks with no option trading. This evidence challenges the view that such predictability reflects informed trading in options, since by construction no option trading occurs in that subsample.

the interpretation of volume-based predictors.

Third, the model identifies a *sufficient* condition for incremental information: option signals must forecast the spread between actual and synthetic stock returns. For a positive (negative) private information signal, the spread would be positive (negative) when options lead in price discovery, negative (positive) when stocks lead, and zero when both markets incorporate information simultaneously. In the model, this latter condition always holds—actual and synthetic prices are equal at all times—reflecting the fact that information is impounded symmetrically across markets. As a result, the spread correctly indicates that option signals contain no incremental information in this framework. This highlights that the actual stock return is a necessary but not sufficient object for testing incremental information in the options market. The extra step of considering the *actual-minus-synthetic* return spread is essential, since it isolates the predictive component that goes beyond what is mechanically embedded in stock prices. This is the main testable implication from this theoretical framework that we take to the data.

We construct a comprehensive dataset of U.S. stocks and their listed options over the 1996–2021 period and examine the predictive ability of both signed and unsigned option volume measures across the full calendar year, and around scheduled and unscheduled corporate events. To assess the origin of price discovery, we compare the percentage change in synthetic stock prices to the actual return on the underlying stock. Synthetic stock returns are derived from the no-arbitrage bounds implied by option prices—specifically, the midpoint between the upper and lower bounds of put-call parity for American-style options, adjusted for dividends and bid-ask spreads. Actual stock returns are computed using mid-quote prices to ensure comparability.

Using this dataset, we replicate and extend key findings from the literature on the predictive content of option volume. In the replication stage, we follow the standard approach of regressing future actual stock returns on option volume-based signals. We then extend the analysis in two directions. First, we examine whether these signals also predict synthetic stock returns. Second, we test whether option volumes forecast the spread between actual and synthetic stock returns, which, as our model suggests, is the appropriate measure for identifying incremental information in the options market.

To provide a comprehensive assessment across time horizons and volume measures, we begin our empirical analysis by adopting the cross-sectional regression framework of Ge et al. (2016). We use both signed and unsigned O/S volume ratios to forecast one-week-ahead stock returns. Consistent with their findings, we document significant stock return predictability over their original sample period. However, when we replace actual stock returns with synthetic stock returns as the dependent variable, the direction and magnitude of predictability remain largely unchanged. As a result, the O/S volume measures exhibit little to no predictive power for the actual-minus-synthetic return spread. In the few instances where some statistical significance persists, the effects are economically negligible—typically an order of magnitude smaller than those for actual stock returns—and/or directionally inconsistent with the presence of incremental information in options.

Next, we examine the informativeness of option volume around major corporate news events, focusing separately on scheduled earnings announcements and unscheduled 8-K filings. These periods are particularly conducive to informed trading, as the release of material information heightens information asymmetry and strengthens incentives to acquire and trade on private signals. Not surprisingly, they are commonly used in the literature to validate proxies for informed trading (e.g., Amin and Lee (1997), Roll et al. (2010), Johnson and So (2012)). Consistent with prior findings, we observe that both the unsigned O/S and P/C volume ratios strongly predict actual stock returns around these events. However, these same measures also predict synthetic stock returns to a nearly identical degree, and the return spreads between actual and synthetic prices remain statistically indistinguishable from zero. This result holds across both scheduled and unscheduled announcements.

To further probe the timing of information incorporation, we repeat the analysis using two alternative windows for portfolio formation: the day before the event and the event day itself. As expected, return predictability is stronger when predictors are measured on the event day. Yet, in both configurations, the actual-minus-synthetic stock return spread remains unpredictable, suggesting that option traders are neither systematically better informed nor faster at processing news than stock market participants.

Lastly, we examine the predictive content of *signed* option volume measures around scheduled and unscheduled corporate events. Following the methodology of Cremers et al. (2022), we find that seller-initiated option volume predicts negative stock returns ahead of scheduled earnings announcements, while buyer-initiated activity is more predictive around unscheduled 8-K filings. In addition, return predictability is stronger when volume signals are measured on the announcement day itself, rather than the day prior. However, when we repeat the analysis using synthetic stock returns as the dependent variable, we find that the predictive patterns remain nearly identical in both magnitude and statistical significance. As a result, the spread between actual and synthetic stock returns is again either statistically indistinguishable from zero or economically negligible, further reinforcing the view that option volume does not convey any incremental information beyond what is already embedded in stock prices.

We show that our results are not mechanically driven by the tight link between stock and option prices. Because options on individual stocks are American-style, the no-arbitrage conditions imply an interval of feasible stock prices bounded by the upper and lower limits of put-call parity. Actual stock prices must lie within this range to preclude arbitrage. One concern is that these bounds might be so narrow that actual and synthetic prices are forced to remain close by construction. However, our portfolio sorts indicate that this is not the case. For example, in the top decile of the O/S distribution, the average width of the no-arbitrage band is approximately 0.8% of the midpoint price. In contrast, for stocks in the bottom decile, this range can reach 1.8%. When sorting on the P/C ratio, the top and bottom deciles exhibit average price bands of about 1.3% and 2.0%, respectively. These are wide enough to allow meaningful divergence between actual and synthetic prices. Thus, the systematic lack of predictability in the actual-minus-synthetic return spread cannot be attributed to tight arbitrage bounds.

We also verify that our results are not driven by stock borrowing costs. Using option-implied borrowing fee data from Muravyev et al. (2022), we replicate all analyses while controlling for stock borrow fees and excluding hard-to-borrow stocks. Any predictability of option volume–based measures arising from their relation to borrowing fees should disappear once fees are accounted for. Across unsigned and signed O/S volume specifications, option volumes continue to significantly predict both actual and synthetic stock returns, even after controlling for borrow fees or excluding hard-to-borrow stocks. The predictability of the spread remains negligible across all specifications and predictors. Together, these results indicate that the information in option volumes is distinct from that embedded in borrowing fees and remains robust to their inclusion. Consistent with Muravyev et al. (2022, 2025), we verify that implied borrow fees do not predict synthetic stock returns, which are already fee-adjusted, but negatively predict actual stock returns, which we have not adjusted for borrowing fees.

Taken together, our findings suggest that prior research may have overstated the informational advantage of options by overlooking a critical step—benchmarking actual stock returns against their synthetic counterparts. By focusing solely on the predictability of actual stock returns, earlier studies implicitly attribute observed patterns to information revealed through option trading, without considering that the same information may already be embedded in stock prices. This interpretation neglects the possibility that stock and option markets incorporate information simultaneously, leaving no incremental role for options in the price discovery process. This challenges the prevailing view that the options market systematically leads the stock market.

Our contribution is twofold. First, we develop a sharper theoretical link between stock return predictability and incremental information by showing that the spread between actual and synthetic stock returns is the appropriate object for identification. While prior studies infer informational content from the predictability of actual stock returns alone, we demonstrate that such evidence is insufficient. Second, we propose a simple and flexible empirical strategy that leverages synthetic return benchmarks to cleanly test for the presence of incremental information in option trading activity. Our approach can be readily applied across different contexts, including various option-based signals and event settings, and offers a more rigorous framework for evaluating the informational role of options.

Our paper contributes to the literature that finds little evidence of price discovery in the options market. For instance, Stephan and Whaley (1990) and Chan et al. (1993) find no evidence that price changes in the options market lead to price changes in the stock market. Vijh (1990) finds that large option trades have relatively small price effects, which is inconsistent with such trades being related to information. Chan et al. (2002) study the cross-market lead-lag effects and show that stock volume has predictive ability for both stock and option prices, but option volume has no incremental predictive ability. Muravyev et al. (2013) show that, when a deviation from the put-call parity occurs, it is the option quote that adjusts to correct the disagreement, while the stock quote does not. More recently, Collin-Dufresne et al. (2021) focus their analysis on stock and option trades by Schedule 13D filers and find that volatility information is more likely to be reflected in options, while price information is more likely to be reflected in stocks.

More generally, our paper is also related to the literature that argues against the superior informational efficiency of the options market. Cao et al. (2022) show that option returns are predictable using a variety of underlying stock characteristics and firm fundamentals, such as idiosyncratic volatility, past stock returns, profitability, cash holding, new share issuance, and dispersion of analyst forecasts. They also conclude that such predictability patterns have important implications for the efficiency of the options market.

The studies mentioned above are all empirical in nature. The theoretical research on this topic is more incipient. The theoretical models in An et al. (2014) and Easley et al. (1998) are often interpreted as suggesting that informed traders prefer to initiate trades in the options market, with information subsequently spilling over to the underlying stock. However, this interpretation is not supported by the structure of the models themselves. In fact, these frameworks imply that information revealed through the history of trades is impounded into both markets simultaneously. As a result, while stock return predictability may still arise, it cannot be attributed to incremental information in option markets over and above what is already reflected in stock prices.

To our knowledge, no existing study benchmarks actual stock returns against synthetic stock returns implied by put-call parity—a necessary step for isolating truly incremental information in options. By introducing this necessary comparison, our framework distinguishes between predictability driven by shared information across markets and predictability arising from option-specific signals.

The remainder of our paper is organized as follows. Section 2 introduces a stylized example that illustrates the key theoretical mechanisms and yields testable empirical predictions. A formal noisy rational expectations model—in which option signals forecast future stock returns even when options are informationally irrelevant—is presented in the Appendix. Section 3 describes the data and empirical methodology, including cross-sectional regressions and portfolio sorts conducted both in calendar time and around firm-specific events such as earnings announcements and 8-K filings. We employ both signed and unsigned measures of option volume throughout the analysis. Section 4 concludes.

2 Theoretical Framework

In this section, we develop the theoretical framework that underpins our empirical analysis. We proceed in two steps. First, we introduce a stylized example to build a basic intuition for the key mechanisms of the paper and to motivate the more general model. Second, in Appendix A, we present a noisy rational expectations model that extends the framework of An et al. (2014), featuring an informed investor, noise traders, and a market maker. The informed investor observes a private signal and trades in both the stock and options markets based on that information. Noise traders generate random demands in both markets, while the market maker enforces no-arbitrage by clearing both markets and setting equilibrium prices.

As a preview of our main result, the model generates stock return predictability from option prices and volumes even when options are informationally *irrelevant*. In other words, option-based signals can forecast stock returns simply because of the mechanical transmission of stock-market signals to options through no-arbitrage linkages.

2.1 Stylized Example and Model Intuition

If informed investors prefer to trade in options, and stock and options markets are not tightly interconnected because of market frictions, then synthetic stock prices should adjust to the fundamental stock values to a larger extent than actual stock prices. Therefore, one should be able to use option signals to predict future *actual* stock returns, but there should be weaker or no predictability regarding future *synthetic* stock returns, depending on the extent of noise trading in options. To clarify this mechanism, Figure 1 presents a stylized example.

[Insert Figure 1 about here]

Panel A of Figure 1 shows the case in which an informed investor prefers to trade in options, and noise trading is absent from the options market. To take advantage of a positive signal received at time $t = 1$, the informed investor purchases call options and sells put options. This pushes up the synthetic stock price immediately to the fundamental value of the stock. This is because of the assumption that noise trading is absent from the options market. As the new information gets revealed to the stock traders, the actual stock price converges to the fundamental value at time $t = 2$. The increase in the synthetic stock price relative to the actual stock price at time $t = 1$ signals that there is positive information in the options market not yet reflected in stock prices. In this case, and one should buy the stock because it is expected to increase in the near future. Moreover, given that the synthetic stock price is already at its efficient level at time $t = 1$, and the fundamental stock value is a random variable, there should be no predictability for synthetic stock returns.

However, as emphasized by Grossman and Stiglitz (1980), the scenario depicted in Panel A of Figure 1 cannot represent an equilibrium. If the options market were fully efficient and prices adjusted instantaneously to reflect all available information, informed traders would be unable to earn positive expected profits. In such a setting, the incentive to acquire costly information would vanish, undermining the very foundation of informed trading. Therefore, there needs to be a certain degree of inefficiency in the options market to encourage informed investors to participate.

Panel B illustrates the same case, but with noise traders. Like in Panel A, at time $t = 1$ the informed investor trades on a positive signal by purchasing call and selling put options, thus pushing the synthetic stock price towards the fundamental stock value. However, due to the presence of noise traders, prices only adjust partially towards their efficient levels.[2] This means that the informed

[2]In the example shown in Figure 1, we assume that noise trading leads to underreaction, such that the price

investor can profit from the trades and is willing to participate. The increase in the synthetic price of the stock relative to its actual price signals that there is positive information in options not yet reflected in the stock, and one should buy the stock as its price is expected to increase in the near future—i.e., there is predictability in actual stock returns. There is also predictability in synthetic stock returns, because option-implied prices did not adjust fully to their efficient levels at time $t = 1$, and are expected to continue adjusting towards time $t = 2$. However, the predictability in actual stock returns should be *stronger* than the predictability in synthetic stock returns, because synthetic prices *already* moved toward fundamentals at time $t = 1$, while the actual stock price has yet to adjust, as in Panel A.

To distinguish the different degrees of market efficiency across Panels A and B, one can examine how much the option signals at time $t = 1$ predict actual compared to synthetic stock returns. In other words, if one *subtracts the synthetic stock return from the actual stock return*, Panel A shows that option signals at time $t = 1$ should predict actual stock returns and the difference between these returns *equally*. However, in Panel B, option signals should be strong predictors of actual stock returns, but *weaker* predictors of the difference in returns. Therefore, the weaker the predictability of the difference in returns, the lower the efficiency of the options market. At the extreme, if option signals *fail* to predict the difference in returns, this can be interpreted as stock and options markets being equally *inefficient*, and options failing to convey any incremental information relative to stocks.

Panels A and B assume that the stock and options markets are not fully interconnected. This occurs when there exist market frictions, such as transaction costs, and when options are American-type, which is typically the case for options on individual equities. The two markets are not tightly linked because the put-call parity is not an equality but a set of inequalities (Cox and Rubinstein (1985)). Therefore, the synthetic and actual stock prices can diverge considerably from each other, without leading to arbitrage opportunities.

Panel C illustrates the case in which the two markets are fully interconnected (no frictions), informed investors can trade in both markets, and there exist noise traders in both markets. This situation can occur when options are European-type, because the put-call parity is an equality, and any deviation from this relation leads to an arbitrage opportunity. Therefore, even if the informed

adjustment is too small relative to the change in fundamental value. The same logic applies to overreaction, where noise trading causes prices to overshoot. In such cases, prices would decline from time 1 to 2, rather than continue to rise.

investor prefers to trade in options, the options market maker will arbitrage between the two markets such that the put-call parity will always hold. If informed investors instead prefer to trade in the stock market, the results would be identical. Therefore, synthetic and actual stock prices are always equal to each other in Panel C, and there is no incremental information in options that is not simultaneously reflected in the stock. This creates a peculiar situation in which the activity observed in the options market at time $t = 1$ can be used to predict future movements in the actual stock price, despite the fact that options do *not* convey any additional information compared to the stock, i.e., options are informationally irrelevant (Chabakauri et al. (2021)). Moreover, by construction, in Panel C the predictability of actual stock returns is exactly the same as that of synthetic stock returns, which results in a zero difference between returns.

By the same token, the case in which the informed investor receives a negative signal at time $t = 1$ and exploits it by purchasing puts and writing calls is simply the mirror image of the scenarios illustrated in Panels A, B, and C of Figure 1.

Overall, the example illustrated in Figure 1 has two important implications. First, if the options market is informationally more efficient than the stock market, then one should be able to use option signals to predict future *actual* stock returns, but the predictability of future synthetic stock returns should be *weaker* (Panel B) or *null* (Panel A). Second, if the two markets are equally inefficient, and options do not convey any incremental information compared to stocks, then option signals should predict both actual and synthetic stock returns to the *same* extent, and they should fail to predict the difference (Panel C).

In Appendix A, we examine this question more formally using a noisy rational expectations model with informed investors who can trade simultaneously in both markets. The framework is borrowed from An et al. (2014), which is an analytic representation of the situation illustrated in Panel C of Figure 1. Specifically, in the model presented in Appendix A, there exists an informed investor who can trade simultaneously in a stock, a call option, and a put option; there are noise traders in both markets, and there is a market maker who arbitrages between the two markets. The options are European-type, which means the put-call parity is a strict equality, and the market maker ensures that this relation is never violated. The main takeaway from the model is that it is possible to use option signals to predict future actual stock returns even when options are informationally irrelevant.

Our framework reveals that stock return predictability based on option signals does not, in itself, imply the presence of incremental information in options. In fact, such predictability may emerge even when options convey no additional information beyond what is already embedded in stock prices. This arises from the behavior of noise traders, whose actions create pricing distortions, and from market makers, who—by enforcing no-arbitrage constraints such as put–call parity—mediate these distortions without transmitting truly novel information. Consequently, for option signals to reflect genuinely incremental information, they must predict not just future stock returns, but specifically the divergence between observed stock prices and their synthetic counterparts implied by option markets. The absence of predictive power over this return differential would again suggest that option signals lack independent informational value. This insight provides a sharp empirical test: if options are indeed informative, their signals should anticipate movements in the actual-minus-synthetic stock return spread. We examine this empirically in Section 3, but first discuss how the presence of American-style option features for individual stocks affects the construction of our empirical measures.

2.2 Effects of American-Style Features

Options on individual stocks are typically American-style—as illustrated in Panels A and B of Figure 1—in which case the put-call parity condition holds as a pair of inequalities. As a result of this relaxed arbitrage constraint, the relationship between stock and option prices continues to be shaped by put-call parity and market competition, but it no longer implies a strict equality. Instead, as shown in Cox and Rubinstein (1985), the put-call parity condition defines a no-arbitrage interval. Within this framework, the actual stock price must remain within an upper and a lower bound to preclude arbitrage opportunities.

The lower bound (L) on the ask price of the synthetic stock is given by:

$$S_1^{*,\text{ask}} \geq S^{*,L} \equiv C_1^{\text{bid}} + Ke^{-r\tau} - P_1^{\text{ask}}, \tag{1}$$

where τ denotes the time to maturity. This inequality ensures that arbitrageurs cannot profit by purchasing the actual stock while simultaneously shorting a synthetic stock created via options and risk-free borrowing. If the inequality is violated, i.e., if $S_1^{*,\text{ask}} < C_1^{\text{bid}} + Ke^{-r\tau} - P_1^{\text{ask}}$, then an arbitrageur can construct a riskless profit by purchasing the put and the stock, writing the call, and borrowing the present value of the strike price. If the call is exercised early—perhaps to capture a large dividend D_1—the arbitrageur delivers the stock and realizes a payoff of $P_1 + K - Ke^{-r\tau} > 0$. If early exercise does not occur, the arbitrageur retains the stock and receives the dividend, with the net payoff at maturity given by $P_2 + S_2 + D_1 e^{r\tau} - C_2 - K = D_1 e^{r\tau} > 0$.

The upper bound (U) on the bid price of the synthetic stock is similarly defined as:

$$S_1^{*,\text{bid}} \leq S^{*,U} \equiv C_1^{\text{ask}} + K + D_1 - P_1^{\text{bid}}, \tag{2}$$

which precludes arbitrage from shorting the actual stock while going long the synthetic one. If this inequality is violated, i.e., if $S_1^{*,\text{bid}} > C_1^{\text{ask}} + K + D_1 - P_1^{\text{bid}}$, an arbitrageur can buy the call, purchase a risk-free bond with present value $K + D_1$, write the put, and short the stock. This strategy again yields a riskless profit. Because the arbitrageur is short the stock, they must deliver the dividend. If the put is exercised early, the arbitrageur's portfolio at maturity is worth $C_2 + Ke^{r\tau} - S_2 > 0$. If the put is not exercised, the final payoff is $C_2 + Ke^{r\tau} - P_2 - S_2 = Ke^{r\tau} - K > 0$.

Overall, as long as the bid and ask prices of the stock remain within the bounds defined by Equations (1) and (2), no arbitrage opportunities exist. Importantly, this means that—unlike in the benchmark model of An et al. (2014) (illustrated in Panel C of Figure 1)—the stock and options markets are no longer tightly synchronized. The existence of no-arbitrage bounds—rather than a strict put-call parity condition—creates room for informational asymmetries to generate temporary divergences between actual and synthetic stock prices. In particular, if informed traders act primarily in the options market, their activity will alter call and put prices, thus moving the synthetic stock price, but without immediately affecting the underlying. For example, a trader with positive (negative) private information may bid up (down) calls and bid down (up) puts, thereby increasing (decreasing) the synthetic stock price relative to the observed spot price. The actual stock price then adjusts with a delay, as the information eventually diffuses into the equity market. This mechanism is exactly what is illustrated in Panels A and B of Figure 1.

Extending our frameworks—both the stylized example and the formal model described in Appendix A—to incorporate more realistic features such as American-style options, market frictions, or heterogeneous information speeds presents substantial challenges. Accounting for early-exercise features in American contracts, for instance, complicates the agent's optimization problem in Appendix A and may undermine the existence and uniqueness of equilibrium. Introducing frictions such as margin requirements or trading costs would further reduce tractability. Consequently, incorporating these elements lies beyond the scope of this paper. By contrast, differences in information processing speeds are less relevant in our setting, as we rely on closing prices. Although our theoretical model does not accommodate American-style options, we can incorporate American-style characteristics empirically when constructing synthetic stock prices.

We now turn to the construction of the actual-synthetic spread. This construction proceeds in three steps. First, the actual stock price S_t is defined as the mid-quote derived from bid and ask prices:

$$S_t = \frac{S_t^{ask} + S_t^{bid}}{2}. \tag{3}$$

The actual stock return is then computed as the percent change in this mid-quote price across consecutive trading periods.

Second, synthetic stock prices S_t^* are estimated using the midpoint between the upper and lower no-arbitrage bounds implied by the American-style put-call parity conditions. Specifically:

$$S_t^* = \frac{S_t^{*,ask} + S_t^{*,bid}}{2}, \tag{4}$$

where $S_t^{*,ask} = C_t^{bid} + Ke^{-r\tau} - P_t^{ask}$ and $S_t^{*,bid} = C_t^{ask} + K + D_t - P_t^{bid}$. Analogous to actual stock returns, synthetic stock returns are computed as the percent change in this synthetic price S^* between consecutive trading periods.

To minimize the noise in the estimation, synthetic stock prices are computed by aggregating across all put-call pairs using the inverse of option bid-ask spreads in each pair as weights. This approach gives less weight to option pairs with wider bid-ask spreads, consistent with the methodology of Goncalves-Pinto et al. (2020).[3]

Lastly, the spread between actual and synthetic stock returns over a generic period that goes from t to $t+1$ is defined as $(S_{t+1} - S_t)/S_t - (S_{t+1}^* - S_t^*)/S_t^*$.

3 Empirical Analysis

The theoretical discussion described above leads to several implications that can be tested empirically. In particular, if an option-based measure reflects incremental information, then it should predict future *actual* stock returns, but it should not predict (or predict weakly) future *synthetic* stock returns. This can be implemented by subtracting the synthetic stock return from the actual stock return, and then testing if the option-based measures predict this difference. If it does, then the options market is (plausibly) informationally more efficient than the stock market, and there is

[3] Our results remain qualitatively unchanged if we instead weight option pairs by open interest, as in Cremers and Weinbaum (2010).

incremental information in options. If it does not, then this would be evidence inconsistent with incremental information reflected in option measures.

3.1 Sample Construction

The option data, which consists of option prices, implied volatilities, open interest, and option volume, is extracted from the OptionMetrics database. Signed option volumes are extracted from the International Securities Exchange (ISE) Open/Close Trade Profile. The sample used in this paper covers the period from 1996 to 2021 when using only OptionMetrics, and from May 2005 to December 2021 when using ISE data.

The stock data, including prices, stock returns, volume, and the number of shares outstanding, for common stock (share codes 10 and 11) traded on the NYSE, NASDAQ, and AMEX (exchange codes 1, 2, and 3), are extracted from CRSP.

Earnings announcement dates are from I/B/E/S, and 8-K filing dates are from the SEC Analytics Suite. After applying some filters to the options data, merging I/B/E/S with OptionMetrics results in a final sample of about 90,000 earnings announcements. These are assumed to be pre-scheduled corporate events.[4] The merging of the SEC Analytics Suite with OptionMetrics results in about 77,000 8-K filings that are unrelated to earnings. Only the first filing of every month is included in the sample, and only if it is not within a week before or after an earnings announcement date. These are all considered to be unscheduled corporate events.

When we merge I/B/E/S and SEC Analytics Suite with the signed option volume data from ISE Open/Close Trade Profile, the number of earnings events and 8-K filings falls to about 50,000. Following Gao and Ritter (2010), reported volume data for stocks trading on NASDAQ is adjusted in the period before 2004. The stock data is merged with daily options data from OptionMetrics. Pairs of call and put options with the same maturities and strike prices are formed whenever both options satisfy the following restrictions: (i) their expiration date is at least 10 days but not more than 30 days away, (ii) their annualized implied volatility does not exceed 250%, (iii) their bid prices are non-missing and are strictly positive, (iv) their ask prices are non-missing and are strictly greater than their bid prices, (v) their open interest is greater than zero, and (vi) their deltas are non-missing. We remove option pairs for which the difference between the call delta and the put

[4]It is typical to classify earnings announcement dates as pre-scheduled and non-discretionary since the SEC requires firms to report earnings within 35 (60) days after quarter (year) end. Prior research finds little flexibility: Bagnoli et al. (2002) find that only 1.8% (1.6%) of firms are more than 7 days late (early) compared with their pre-announced release dates, and that being late leads to a negative market reaction.

delta falls outside the interval $[0.9, 1.1]$. While not explicitly designed to do so, these option filters remove penny stocks (i.e., stocks with closing prices below \$5 (\$1) before (after) April 2001) from the final sample.

To compute the option-implied bounds in Equations (1) and (2), continuously compounded risk-free rates are extracted from the OptionMetrics Zero Coupon Yield Curve dataset and are used to calculate the present value of the strike price and the present value of dividends with ex-dates prior to the maturity of the options in each put-call pair. The present value of each dividend is calculated by discounting it back from its payment date.[5] A firm is defined to be a dividend payer if its distribution in the OptionMetrics Dividend dataset is of type 1 (i.e., cash dividends) as this is most likely to affect the option-implied stock price bounds.[6]

3.2 Descriptive Statistics

Table 1 reports descriptive statistics for the various samples employed in our empirical analysis and is organized into three panels. Panel A summarizes the weekly variables used in our baseline cross-sectional regressions. This dataset contains over 500,000 firm-week observations spanning the period from May 2005 to December 2021. The sample is restricted to this window due to the availability of signed option volume data from the ISE Open/Close Trade Profile database. Panel B focuses on data surrounding corporate news events. Panel B1 examines earnings announcements sourced from I/B/E/S, while Panel B2 focuses on 8-K filings obtained from the SEC Analytics Suite. Both subpanels use the unsigned option volume data from OptionMetrics, which allows coverage from 1996 through 2021. Panel C requires signed option volume data from the ISE Open/Close Trade Profile and thus is limited to the post-May 2005 period. All variables reported in Table 1 are defined in detail in the Appendix.

[Insert Table 1 about here]

Panel A begins with summary statistics for the weekly O/S volume ratios, both signed and unsigned. The distributions are broadly consistent with those reported in Ge et al. (2016), although some differences arise due to our extended sample period, which runs from May 2005 through

[5] The standardized set of maturities for risk-free rates in OptionMetrics is linearly interpolated for other maturities. The realized dividends are then discounted at the risk-free rate. The dividends in this sample are almost certain since they have almost always been announced and are payable within 22 calendar days.

[6] The final sample excludes all stocks with a liquidating dividend (LIQUID_FLAG=1), a dividend cancellation (CANCEL_FLAG=1), a stock dividend, a stock split or a special dividend (DISTR_TYPE = 2, 3, or 5, with AMOUNT > 0) before the option expiration date, as these events may give rise to adjustments in the terms of the option contracts.

December 2021, whereas their data end in August 2012. The mean of the signed ISE-based ratio (ISEO/S) is approximately one-fourth the mean of the unsigned OptionMetrics-based ratio (OMO/S), which aligns with the ISE accounting for roughly 30% of total option volume in the OptionMetrics database. A comparison of the signed call and put ratios (ISEC/S and ISEP/S, respectively) indicates that the call trading volume is, on average, not only higher but also more volatile than the put volume. When disaggregating the option volume into open and close trades, we observe that open volumes are two to three times larger than close volumes—an expected result given that option positions can be held through expiration. For instance, the mean of the open buy call ratio (OBC/S) is 0.011, nearly three times that of the close buy call ratio (CBC/S), which averages 0.004. In addition to these trading activity measures, Panel A also reports summary statistics for the control variables used in the cross-sectional regressions, including Δeqvol, Δopvol, illiquidity measure (Amihud), Impl_vol, B/M, Size, Momentum, and Skewness.

Regarding the returns variables, and compared with Ge et al. (2016), we use a wider range of variables as we are interested in testing, not only the predictability of the O/S volume ratios for actual stock returns, but also for synthetic stock returns, as well as the spread between the two. Thus, we use as dependent variable one of these three measures: (i) CAAR1(%), the one-week cumulative abnormal actual stock return in the week following the week in which the O/S measures are calculated, (ii) CASR1(%), the one-week cumulative abnormal synthetic stock return in the week following O/S measurement, and (iii) AMS_Spread1(%), which denotes the actual minus synthetic spread on the week after the O/S measurement, i.e., this is simply the difference between CAAR1(%) and CASR1(%). For all these measures, we use the CRSP value-weighted market return as the benchmark in calculating the abnormal return every day, and then sum all those abnormal returns across all days in a week. Finally, we also control for CAAR0(%), which captures the cumulative abnormal actual return during the same week in which the O/S measures are constructed.

Panel A shows that CAAR1 (%) has a mean of -0.018% and a standard deviation of 4.890%, and that CASR1 (%) exhibits similar distributional properties. As a consequence, AMS_Spread1 (%), which captures the difference between actual and synthetic stock returns, has a mean close to zero (0.002%) and a standard deviation of 0.511. Consistent with the hypothesis that little incremental information originates in the options market beyond what is already impounded in stock prices, the mean of AMS_Spread1 is economically negligible. AMS_Spread1 represents only 11% of CAAR1(%), i.e., it is about one order of magnitude smaller, suggesting that option-based signals are largely subsumed by the information already embedded in equity prices.

Panel B reports descriptive statistics for the tests around news events when using unsigned option volumes from OptionMetrics. Panel B1 reports that our final sample includes about 90,000 earnings announcements from I/B/E/S. Of these, approximately 83,000 observations have option volume data from OptionMetrics (OMO) available both on the day prior to the earnings announcement ($t = -1$) and on the announcement day itself ($t = 0$). Relative to the full sample in Panel A, OMO/S ratios are significantly higher around earnings announcements, as shown in Panel B1. For instance, compared to the mean value of 0.208 in the weekly sample on Panel A, the mean OMO/S is 0.242 (16% larger) for day $t = -1$ and is 0.323 (55% larger) for day $t = 0$ relative to the earnings announcement. This is consistent with the results in Roll et al. (2010), which show that O/S volume ratios increase significantly before earnings.

As shown in Panel B2, OMO/S ratios around 8-K filings are even higher than those observed around earnings announcements, with mean values of 0.329 on the day prior to the filing ($t = -1$) and 0.346 on the filing day itself ($t = 0$).

Regarding P/C ratios calculated with unsigned OptionMetrics data (i.e., OMP/OMC), their mean values and standard deviations are quite large, both around earnings (Panel B1) and around 8-K filings (Panel B2), with a value close to 3. This implies that the put volume is, on average, three times the call volume around news events. Standard deviations are more than 10 times that, especially on the day of the news releases. In contrast, the mean OMP/S is lower than the mean OMC/S in the weekly sample of Panel A, suggesting that, in normal times, the P/C ratio is lower than 1. Thus, the volume of the put increases significantly around news in comparison to other periods.

Panel B also reports actual and synthetic stock returns for a range of periods, including the day the O/S and P/C ratios are measured, and three windows that either include the news day or skip it. A consistent pattern observed across both Panels B1 and B2 is that the mean and median spreads between actual and synthetic stock returns are essentially zero. Once more, this finding aligns with our working hypothesis that the options market does not convey incremental information beyond what is already reflected in stock prices. Still in Panel B, we also report statistics for the market capitalization of the stocks in our sample of news events. The average stock has a market cap of $15 billion in Panel B1 and $19 billion in Panel B2. We use these market caps to compute value-weighted returns for the portfolios formed by sorting on O/S and P/C volume ratios.

Panel C of Table 1 reports descriptive statistics for the sample of news events after merging with the signed option volume data from ISE. The number of events decreases to approximately 50,000 in each Panel C1 (earnings) and Panel C2 (8-K filings). The open-buy (OB) and open-sell (OS) ratios are very similar across the two panels and for different measurement days. For this analysis of signed option volumes around events, we follow Cremers et al. (2022). Like them, we use the OB and OS ratios to predict the following day's return. In contrast, while they just focused on the predictability of actual stock returns on the following day, we also report results for the predictability of synthetic stock returns, and the spread between actual and synthetic stock returns. Interestingly, both the mean and median values of the spreads are zero in Panels C1 and C2, suggesting ex-ante that neither OB nor OS is likely to exhibit predictive power for next-day spread movements.

Lastly, Panels A, B and C report the no-arbitrage price range (NAPR) for the three different datasets measured at different points in time. If option and stock markets were tightly linked, synthetic stock prices would closely track actual stock prices. This is, by construction, the case under European-style options, where the put-call parity imposes a strict pricing relation, and any deviation implies arbitrage. However, the equity options analyzed in this paper are American-style, and wide bid-ask spreads imply that put-call parity defines an inequality-based no-arbitrage range (Cox and Rubinstein (1985)).

Defined as the difference between the upper ($S^{*,U}$) and the lower ($S^{*,L}$) no-arbitrage bounds from Equations 1 and 2, expressed as a percentage of their mid-point $(S^{*,U} + S^{*,L})/2$, the NAPR is well above zero for all datasets we used, with mean values ranging from 0.016 to 0.020 across the three panels. These values imply that actual stock prices can vary by 1.6% to 2% around the synthetic (option-implied) prices without triggering arbitrage. Such sizable ranges suggest that the connection between the option and stock markets is not tight, thus aligning with an economy as the one represented in Panel B of Figure 1.

3.3 Predictability Results

In this section, we revisit some of the main tests available in the literature that use option volume-based measures to predict future actual stock returns. We first replicate them, and then extend the analysis to synthetic stock returns.

We start with the multivariate study by Ge et al. (2016), which runs cross-sectional regressions on a weekly basis and uses both unsigned and signed option volume measures as predictors, controlling for a long list of variables.

We then turn to studies that focus on the periods around corporate events (e.g. Cremers et al. (2022). The period around the release of material corporate news provides investors with stronger incentives to gather private information and trade on such information. These periods are then typically used to validate informed trading proxies, like in Amin and Lee (1997), Roll et al. (2010), and Johnson and So (2012), among others. We thus run our empirical analysis focusing on the period around scheduled earnings announcements as well as unscheduled 8-K filings, using both signed and unsigned option volume indicators.

3.3.1 Weekly Cross-Sectional Regressions and Unsigned Option Volumes

Table 2 reports the results from testing the extent to which O/S volume ratios predict future actual and synthetic stock returns. Our sample includes only stocks with signed option volume in ISE, meaning that it covers the period starting from May 2005. Following Ge et al. (2016), we estimate weekly cross-sectional regressions of the form:

$$Ret(1) = \alpha + \beta_1 \text{DecO/S} + \gamma' X + \epsilon \tag{5}$$

where the dependent variable $Ret(1)$ is replaced in Panel A with the cumulative abnormal actual (mid-quote) return $CAAR1(\%)$ in the week subsequent to the week from which trading volumes are used to construct O/S, while in Panels B and C the dependent variable is replaced with $CASR1(\%)$ and $AMS\_Spread1(\%)$, respectively. The main regressor of interest, DecO/S, is the decile variable constructed weekly from the various O/S volume ratios considered. When the O/S volume ratio is constructed using weekly option volumes aggregated across all exchanges (sourced from OptionMetrics), the corresponding variables are prefixed with OM. If volumes are restricted to the International Securities Exchange (ISE), variable names are prefixed with ISE. When the analysis isolates call or put volumes, the numerator of the O/S volume ratio is replaced with C or P, respectively. Finally, X denotes a vector of stock characteristics used as controls, which are described in detail in Appendix 1. We exclude expiration weeks (third week of each month) and the fourth quarter of 2008, to avoid the potential impact of the short-sale ban during the financial crisis on option market trading volumes.

[Insert Table 2 about here]

We report results using both the original sample period analyzed in Ge et al. (2016)—spanning May 2005 to August 2012 (Panels A1, B1, and C1)—and an extended sample that adds nine additional years through December 2021 (Panels A2, B2, and C2). While the predictive power of O/S volume ratios over the out-of-sample (OOS) period from August 2012 to December 2021 is considerably weaker than in the original study, our primary objective is not to provide a formal OOS validation. Rather, we aim to reassess the interpretation of their findings. In particular, we revisit their conclusion that option volumes convey incremental information, a claim based solely on the predictability of actual stock returns, without considering whether such predictability extends beyond what is already embedded in synthetic stock returns.

Panel A1 of Table 2 confirms that all unsigned option volume ratios—whether computed using OptionMetrics (OM) or ISE data—significantly predict actual stock returns in the subsequent week. The estimated coefficients are uniformly negative and statistically significant at the 1% level, consistent with the findings in Johnson and So (2012), who argue that this could be because informed investors prefer to trade options when holding negative signals. However, when the sample is extended by nine years, from August 2012 through December 2021 (Panel A2), the predictive power weakens considerably. The unsigned ISE-based ratios lose statistical significance entirely, while among the OM-based ratios, only OMO/S and OMP/S retain their negative signs, with OMP/S remaining significant at conventional levels and OMO/S falling to marginal significance at the 10% level.

In Panel B, we replace the dependent variable with next-week cumulative abnormal synthetic stock returns, $CASR1(\%)$. The results closely track those in Panel A, both in terms of coefficient magnitudes and statistical significance. Given this similarity, the findings in Panel C are predictably muted: when the dependent variable is defined as the return spread between actual and synthetic stock returns— AMS_Spread1(%) = CAAR1(%) - CASR1(%)—there is little remaining variation to be explained, and the coefficients are largely economically negligible and statistically insignificant. Overall, the evidence in Table 2 supports the interpretation that unsigned option volume ratios, whether from OM or ISE, do not convey incremental information beyond what is already reflected in the stock price, consistent with the implications of our theoretical framework.

In the following section we repeat this analysis using signed option volumes.

3.3.2 Weekly Cross-Sectional Regressions and Signed Option Volumes

We now repeat the previous analysis, but this time using signed O/S volume ratios as our key explanatory variables:

$$\begin{aligned} Ret(1) = \alpha + \beta_1 \text{DecOBC/S} + \beta_2 \text{DecOSC/S} + \beta_3 \text{DecCBC/S} + \beta_4 \text{DecCSC/S} + \\ + \beta_5 \text{DecOBP/S} + \beta_6 \text{DecOSP/S} + \beta_7 \text{DecCBP/S} + \beta_8 \text{DecCSP/S} + \gamma' X + \epsilon \end{aligned} \tag{6}$$

This specification includes all eight signed O/S volume components simultaneously, allowing us to isolate the marginal predictive content of each variable while controlling for the others.

[Insert Table 3 about here]

Panel A1 replicates the results reported in Panel A of Table 5 in Ge et al. (2016), with the primary difference being that we compute $CAAR1(\%)$ using mid-quote returns on the left-hand side. This methodological adjustment is expected to attenuate the magnitude of coefficient estimates, which is broadly consistent with what we observe. The signs and relative magnitudes of the coefficients remain largely in line with those reported by Ge et al. (2016). For example, the coefficient on DecOBC/S in column (1) of our Panel A1 is 0.013 and statistically significant at the 5% level, compared to their estimate of 0.021, significant at the 1% level. Similarly, our estimate for DecOSC/S is -0.023 (significant at the 1% level), versus -0.033 in their study, also significant at the 1% level. These findings are consistent with theoretical predictions: higher open-buy call volume (OBC/S) is associated with positive future returns, while higher open-sell call volume (OSC/S) predicts lower subsequent returns.

The full model in column (7) shows the results for a regression with all eight components. Consistent with the results in Ge et al. (2016), our results show that the opening trades that establish synthetic long positions in the underlying stock positively predict subsequent stock returns. DecOBC/S has the largest estimated coefficient of 0.028 significant at the 1% level. The second largest coefficient is 0.025 for DecOSP/S, implying that open-sell put volume is also a strong predictor of positive returns the following week.

Panel A2 shows slightly different results when considering the extended sample that ends in December 2021. Focusing on the full model of column (14), most of the coefficients lose their statistical significance, except for the signed volumes CBC, CSC, OSP, and CBP. The largest coefficient in column (14) is that for the open-sell put volume (OSP), which remains positive at 0.016 and significant at the 1% level, dropping from 0.025 in column (7).

Panel B replicates the analysis from Panel A, replacing the dependent variable with $CASR1(\%)$, the cumulative abnormal synthetic stock return over the week following the construction of the signed O/S volume ratios. Notably, both the magnitude and statistical significance of the coefficients in Panels B1 and B2 closely track those observed in Panel A. This similarity suggests that signed option volumes predict synthetic stock returns to a comparable extent as they do actual stock returns.

Lastly, Panel C regresses the spread $AMS\_Spread1(\%) = CAAR1(\%) - CASR1(\%)$ on the signed option volumes. Interestingly, some of the coefficients remain statistically significant, but their magnitudes are about an order of magnitude smaller compared to using actual stock returns as the dependent variable (Panel A). For example, column (7) of Panel C1 reports a coefficient of 0.002 for DecOBC/S, which represents approximately 7% of its counterpart in Panel A1. This suggests that, although OBC/S may predict the spread between actual and synthetic stock returns, the incremental information it conveys—if the spread is interpreted as such—is less than one-tenth of what might be inferred from its relation to actual stock returns alone. The effect on the coefficient for DecOSC/S is even stronger: in Panel C1 its value is −0.001, which is only 5.5% of its value in Panel A1. None of the other coefficients in the full model of column (7) in Panel C1 remain statistically significant. Turning to the full model in column (14) of Panel C2 the scenario is even gloomier, as only the coefficient on DecOBC/S remains significant at 5% level, but it is hard to interpret because these coefficients are insignificant in column (14) of both Panels A2 and B2.

Overall, the results suggest that signed option volumes generally fail to predict the spread between actual and synthetic stock returns, indicating that they do not convey incremental information beyond what is already incorporated into market prices. In the few cases where spread predictability does appear—such as the coefficients on DecOBC/S and DecOSC/S in Panel C1—the estimated effects are an order of magnitude smaller than those observed when using actual stock returns. These findings imply that analyses relying solely on the predictability of actual stock returns (as in Panel A) may overstate the informational content of signed option volume measures.

3.3.3 Effects of Stock Borrow Fees

In this section, we examine whether stock borrow fees affect our previous results. We replicate the analyses in Tables 2 and 3, this time explicitly controlling for stock borrow fees.

We expect stock borrow fees to have little effect on our baseline results. This expectation follows from Muravyev et al. (2022) and Muravyev et al. (2025), who show that stock borrow fees do not predict fee-adjusted stock returns. Because synthetic stock returns are already fee-adjusted, borrow fees should not predict them. If option volumes merely reflected borrowing costs, their predictive power for synthetic stock returns would disappear once fees were accounted for. Yet, as shown in previous sections, option volumes strongly predict both actual and synthetic stock returns, indicating that they are not highly correlated with stock borrow fees. We formally test this hypothesis below, and find that stock borrow fees have a negligible impact on the predictive power of volume–based option measures.

To implement these tests, we use the option-implied stock borrow fee data from Muravyev et al. (2022), which covers the period from July 2006 to August 2015.[7] We then merge the option-implied stock borrow fee data with our sample. Because the option-implied fee series ends in 2015, we restrict the main analysis to the period examined by Ge et al. (2016), which spans May 2005 through August 2012. After merging the two datasets, our sample covers July 2006 to August 2012.[8]

Table 4 reports the results for the unsigned O/S volume predictors. For ease of comparison, subpanels A1, B1, and C1 reproduce the actual, implied, and spread results from Table 2, respectively, but restricted to the subsample of stocks with available implied-fee data. The number of observations in this subsample is about 80% of that in Table 2, reflecting both the later start date in July 2006 (rather than May 2005) and the fact that implied-fee data are unavailable for some stocks because Muravyev et al. (2022) apply additional filters in constructing their sample. Subpanels A2, B2, and C2 present the corresponding results when option-implied stock borrow fees are included as a control variable in the cross-sectional regressions, while subpanels A3, B3, and C3 exclude hard-to-borrow stocks (i.e., stocks with implied fees greater than 1%), further reducing the number of observations by roughly one third.

[Insert Table 4 about here]

Despite the smaller samples used in these analyses, the main findings are robust across specifications. Option volumes continue to significantly predict both actual and synthetic stock returns

[7] We thank Dmitriy Muravyev for making the option-implied stock borrow fee data publicly available on his website. Using option-implied rather than realized stock borrow fees should not materially affect our results, as Muravyev et al. (2022) document that the two series are highly correlated and differ only modestly.

[8] Results using the full sample through 2015 are qualitatively similar and available from the authors upon request. We omit them for brevity.

to a similar extent, but not the spread. When statistical significance for the spread does arise, the associated economic magnitudes are negligible relative to those for actual stock returns. These findings confirm that our main results are neither driven by, nor attributable to, stock borrowing fees.

Consistent with Muravyev et al. (2022) and Muravyev et al. (2025), option-implied fees do not predict synthetic stock returns (subpanel B2), as synthetic stock returns are already fee-adjusted. In contrast, they strongly and negatively predict actual stock returns (subpanel A2), which are not adjusted for stock borrowing fees.

The same conclusion holds when we consider the entire battery of signed O/S volume predictors. Table 5 shows that, across all subsamples and for all definitions of signed volume, the predictability of the spread between actual and synthetic stock returns is either statistically indistinguishable from zero or at least an order of magnitude smaller in economic terms—whether without accounting for implied fees (subpanel C1) or when accounting for fees in subpanels C2 and C3. Hence, even for the more granular signed predictors, the results cannot be attributed to stock borrow fees. This evidence highlights a key distinction between volume- and price-based predictors: the latter are closely tied to borrowing costs and effectively capture them (Muravyev et al., 2025), whereas option volume-based measures retain predictive power that is largely unaffected by such fees.

[Insert Table 5 about here]

3.3.4 Scheduled News and Unsigned Option Volumes

While Sections 3.3.1 and 3.3.2 presented a complete weekly analysis over the entire sample period, this section adopts a more granular approach, by concentrating on scheduled corporate events—periods in which investors have particularly strong incentives to gather private information and trade on it. For instance, Johnson and So (2012) show that the O/S volume ratio predicts future firm-specific earnings news, and argue that this is consistent with the idea that this measure reflects private information. Similarly, Pan and Poteshman (2006) argue that option trading volume contains information about future stock prices and show that stocks with low P/C ratios outperform stocks with high P/C ratios.

Table 6 reports, in Panels A1 and A2, the predictability derived from the O/S volume ratio and, in Panels B1 and B2, the predictability derived from the P/C ratio. Both of these ratios are computed using unsigned volumes from OptionMetrics from 1996 to 2021.

For both indicators, we construct long-short portfolios as follows. Each quarter, we sort stocks into deciles based on their O/S (or P/C) values. The portfolio takes a long position in the top decile (stocks with high O/S or P/C) and a short position in the bottom decile (stocks with low O/S or P/C). All portfolio returns are obtained by weighting individual stock returns using their market capitalization on the day prior to the news release. Panels A1 and B1 report returns from portfolios formed one trading day prior to the announcement date ($t = -1$), while Panels A2 and B2 report returns from portfolios formed on the announcement day itself ($t = 0$). Economically, the $t = -1$ results (Panels A1 and B1) shed light on whether option traders possess superior ex ante forecasting ability. In contrast, the $t = 0$ results (Panels A2 and B2) capture how efficiently option traders process newly released public information, as in Engelberg et al. (2012) and Cremers et al. (2022).

Actual stock returns are calculated as percentage changes in mid-quote prices as defined in Equation (3) and are reported in rows (1) through (3). Synthetic stock returns are computed analogously, but as the midpoint between the upper and lower no-arbitrage bounds as defined in Equation (4). These synthetic stock returns are reported in rows (4) through (6). Lastly, the spread between actual and synthetic stock returns is calculated for each event window and presented separately, allowing for a direct assessment of whether observed price movements deviate from model-implied bounds.

[Insert Table 6 about here]

Panel A1 shows that, when we sort stocks based on O/S measured on the day before an earnings announcement ($t = -1$), this strongly predicts actual stock returns during the subsequent 3-day announcement window $[0, 2]$, and even after skipping the announcement day itself $[1, 3]$.[9] The actual value-weighted return of the long-short portfolio during the announcement window $[0, 2]$ in row (2) is -92 basis points (bps). Similarly, the actual return of the long-short portfolio for the period $[1, 3]$ in row (3) is -54 bps. The O/S volume measure consistently predicts returns negatively, which aligns with the results in Table 2 and is consistent with the findings of Johnson and So (2012).

Next, we present the predictability results for synthetic stock returns. These results closely resemble those for actual stock returns, both during the announcement window (row (5), -95 bps) and in the post-announcement period $[1, 3]$ (row (6), -60 bps). More importantly, when testing for

[9]The timeline used in this section is different from the one used in Figure 1. An earnings announcement day is denoted as $t = 0$, and the day before is denoted as $t = -1$. In Figure 1, the announcement date, which is the date on which the information is revealed and becomes public, corresponds to $t = 2$.

differences between actual and synthetic stock returns, we find no statistically significant deviations. According to our theoretical argument, any incremental information conveyed by options beyond what is already reflected in stocks should be captured by the spread between actual and synthetic stock returns. Our results in Panel A1 support the notion that the O/S measure, observed the day before earnings announcements, does not provide additional information beyond what is already incorporated in stock prices.

The evidence is similar in Panel B1, where we use the P/C ratio from the day prior to earnings announcements to form portfolios. According to Pan and Poteshman (2006), the P/C ratio should predict returns negatively, as higher P/C ratios imply that the put option volume is relatively higher than the call option volume. We find that the long–short portfolio of actual stock returns formed using P/C ratios at $t = -1$ yields consistently negative returns—-72.4 bps over period -1 (row (1)) and -44.2 bps over the announcement window $[0, 2]$ (row (2)). However, the P/C volume ratio does not seem to predict subsequent 3-day returns when we skip the announcement day itself, i.e., for the window $[1, 3]$. Similarly to the O/S results, the P/C ratio measured on the day prior to earnings announcements is *unable* to significantly predict the spread between actual and synthetic stock returns. Returns on the days before earnings announcements ($t = -1$) show that stocks with high P/C ratios typically experience significant negative price drops on those days. That is, high put option volumes relative to call option volumes are contemporaneously accompanied by negative stock returns. The opposite is true for stocks with low P/C ratios, which are accompanied by positive and significant returns. However, returns are not significant on sorting days when sorting stocks based on O/S in Panel A1.

We now turn to Panels A2 and B2, in which O/S and P/C volume ratios are measured on the day of the earnings announcement ($t = 0$) instead of the day prior. These panels show some important differences relative to Panels A1 and B1 discussed above. First, note that both O/S and P/C volume ratios are contemporaneously accompanied by significant returns on the days they are measured ($t = 0$). In particular, stocks with high O/S volume ratios experience positive and significant returns (both actual and synthetic) on the earnings day, which subsequently reverse in $[1, 3]$. While both legs of the long-short strategy using O/S ratios explain the strong predictability in $[1, 3]$, for the period $[2, 5]$, only the long leg shows a significant contribution.

We highlight an interesting result in Panel A2: when the O/S volume ratio is measured on the earnings announcement day, it predicts the return spread between actual and synthetic prices over the $[2, 5]$ window. The long-short spread return is 5.8 basis points, statistically significant at the

5% level, and primarily driven by a significant −4.6 bps from the long leg. Importantly, however, this finding not only fails to undermine our central argument—it actually lends support to it, for several reasons.

First, the economic magnitude is modest: the corresponding long-short return based on actual stock returns alone (row 3 of Panel A2) is −34.4 bps—nearly six times larger in absolute value. Second, and more critically, the sign of the spread return contradicts theoretical expectations under the hypothesis that the options market incorporates information more efficiently than the stock market. If option prices lead, actual stock returns should adjust more than synthetic stock returns, resulting in larger movements in actual prices. However, Panel A2 shows the opposite: both the long and short legs of the synthetic stock return strategy (row 6) exhibit greater magnitude than those of the actual stock return strategy (row 3), which is inconsistent with the premise of informational leadership by options.

A similar pattern arises in Panel B2: when the P/C ratio is measured on the earnings announcement day, low P/C ratios predict a return spread of −4.3 basis points over the $[2, 5]$ window, significant at the 5% level. As with the O/S results, however, this implies that the synthetic price adjusts more than the actual price—directly contradicting the theoretical argument that synthetic prices, being closer to fundamental value, should respond less in equilibrium.

To validate these results, it is important to assess whether the stock and option markets are loosely or tightly interconnected. Specifically, Panel C of Figure 1 suggests that the failure of option-based measures to predict the difference between actual and synthetic stock returns could be driven by tight interconnectedness between the two markets.

If the two markets are tightly linked—as is the case with European options—synthetic prices cannot deviate from actual stock prices without generating an arbitrage opportunity. In contrast, for American options, this strict equality is replaced by a set of no-arbitrage bounds.

Table 7 reports the difference between the upper and lower no-arbitrage bounds, expressed as a percentage of the midpoint, for the portfolios of stocks constructed in Table 6.

[Insert Table 7 about here]

Panel A reports the no-arbitrage bounds at time $t = -1$ and $t = 0$ for portfolios sorted by the O/S volume ratio, measured one day before and on the day of the earnings announcement, respectively. Panel B presents analogous results for portfolios sorted by the P/C ratio.

These bounds represent the interval within which actual stock prices must lie to avoid arbitrage violations. For stocks in the top decile of the O/S distribution (Panel A), the average width of the no-arbitrage band is approximately 0.8% of the midpoint price. In contrast, stocks in the bottom decile exhibit a wider band of about 1.8%. A similar pattern emerges when sorting on the P/C ratio (Panel B), with average price bands of roughly 1.3% and 2.0% in the top and bottom deciles, respectively. These ranges are sufficiently wide to allow economically meaningful deviations between actual and synthetic prices. Therefore, the lack of return spread predictability observed in Table 6 cannot be explained by mechanical price alignment within narrow no-arbitrage bounds.

3.3.5 Unscheduled Corporate Events and Unsigned Option Volumes

The previous section focused on the period surrounding (scheduled) earnings announcements. We now extend the analysis to the days around the release of (unscheduled) non-earnings 8-K filings, as identified through the SEC Analytics Suite.[10] By occurring without prior notice, these events may lead to different dynamics in how information is incorporated across the stock and options markets.

Table 8 replicates the analysis in Table 6 for the period around the release of non-earnings 8-K filings.[11]

[Insert Table 8 about here]

Panel A1 shows that the actual stock return of the long-short portfolio during the announcement window—defined as the filing day plus the two subsequent trading days ($[0, 2]$, row (2))—is -52 basis points (bps), significant at the 1% level. Similarly, for the $[1, 3]$ window in row (3), the return is -44 bps, also significant at the 1% level. As in the case of scheduled earnings announcements discussed in the previous section, the O/S volume ratio continues to negatively predict returns around unscheduled 8-K filings. Consistent with this pattern, we find no significant contemporaneous return on the day the O/S ratio is measured, i.e., the trading day immediately preceding the 8-K filing ($t = -1$).

[10] Alternative sources of unscheduled corporate events have been used in prior work. For example, Jin et al. (2012) rely on the Capital IQ Key Developments database, Augustin et al. (2019) employ Dow Jones news from RavenPack, while Cremers et al. (2022) use Thomson Reuters News Analytics. While these datasets offer valuable insights, they typically cover only the more recent period. In contrast, 8-K filings span a much longer time horizon, encompassing the full sample period available in OptionMetrics. In addition, 8-Ks are filed directly by the firm and disclose material events that are more likely to reflect genuine shifts in firm fundamentals. By comparison, media-based datasets contain third-party coverage that may not be tied to fundamental news and often include noise unrelated to firm-driven disclosures.

[11] The final sample of non-earnings 8-K filings includes only the first filing of every month, and only if it is not within a week before or after an earnings announcement date.

Regarding our main variables of interest, i.e., the differences between actual and synthetic long-short returns, results clearly show that, once more, they are not significantly different from zero in either of the two periods following the release of 8-K filings. Once more, these results are inconsistent with the idea that the options market is informationally more efficient than the stock market, and that the relative trading activity in options versus stocks conveys any incremental information that is not already reflected in stocks.

In Panel A2, the O/S ratio is measured on the day of the 8-K filing ($t = 0$). Overall, the results are weaker in this case. If we skip two days between the measurement and the portfolio holding (i.e., window $[2, 5]$ in row (3)), the actual stock return of the long-short portfolio is -49 bps, and only significant at the 10% level. Rows (4) to (6) show that the synthetic stock returns of the O/S long-short portfolio formed on the day of the 8-K filing are very similar to those using actual stock returns, resulting in an insignificant spread as predicted by our theoretical argument.

Together, the results in Panel A show that, when portfolios are formed ahead of earnings days (Panel A1), the O/S ratio is a strong predictor of both actual and synthetic long-short returns, implying that the O/S ratio does not convey any incremental information that is not already captured by the stock market. They also show that option traders are not any better at processing information once it is made public, and that both markets embed the new information into prices at equal speed (Panel A2).

Next, we repeat this analysis for the P/C ratios in Panel B. We find that the difference between actual and synthetic stock returns is, once more, never statistically significant across the board. Thus, the P/C ratio also does not convey any incremental information not already present in the stock market around the release of unscheduled 8-K filings. Most of the action occurs on the day of the measurement of the P/C ratios, i.e., $t = -1$ in Panel B1 and $t = 0$ in Panel B2. Stocks with low (high) P/C values also experience positive (negative) and significant returns on the same day P/C is measured. However, in Panel B1, all subsequent windows show insignificant returns. In Panel B2, the long-short portfolio shows significant predictive power for the window $[1, 3]$, at -21.6 bps for actual stock returns, and -24.6 bps for synthetic stock returns, both significant at the 5% level. However, these values are not significantly different from each other in rows (2) - (5), i.e., the P/C ratio cannot predict the actual-synthetic spread around 8-K filings.

Overall, the null hypothesis that actual and synthetic long-short returns are equal cannot be rejected in the empirical tests of Table 8, like in the case of earnings announcements in Table 6. This underscores the idea that, even around corporate events—periods when gathering and trading on private information might be more profitable—the options market does not appear to be more informationally efficient than the stock market. This finding confirms that our results are not driven by a broad, generic analysis over the entire year but also hold true around specific, information-sensitive events.

3.3.6 Signed Option Volumes and Information Events

As a final test, we keep the time windows fixed but increase granularity by leveraging signed-volume indicators, allowing us to analyze the volume dimension more closely around both scheduled and unscheduled events.

Cremers et al. (2022) argue that, since scheduled events tend to reduce volatility more than unscheduled ones, option purchases should be more informative around unscheduled events, while option sales should be more informative around scheduled events. However, like much of the existing literature, their analysis relies solely on actual stock returns. We revisit their tests and extend the analysis to include synthetic stock returns and the return spread. To do so, we follow their methodology to distinguish between option purchases and sales using data from the International Securities Exchange (ISE). Due to data availability, this portion of the analysis covers the period from May 2005 to December 2021—shorter than the broader sample used in previous sections, which begins in 1996 and is based on OptionMetrics data.

We define the open buy indicator as:

$$OB_{i,t} = \frac{PB_{i,t}}{PB_{i,t} + CB_{i,t}} \tag{7}$$

where, for each stock i and for each time t, the quantities $PB_{i,t}$ and $CB_{i,t}$ represent the number of put and call contracts bought by all players in the market but market makers, respectively. Similarly, we define the open sell indicator as:

$$OS_{i,t} = \frac{PS_{i,t}}{PS_{i,t} + CS_{i,t}} \tag{8}$$

where, for each stock i and for each time t, the quantities $PS_{i,t}$ and $CS_{i,t}$ represent the number of put and call contracts sold by all players in the market but market makers, respectively.

Following the methodology of Cremers et al. (2022), we construct equally weighted stock portfolios on a quarterly basis, sorting stocks into quintiles based on the O/S and O/B measures. We then compute the returns of these portfolios for the trading day immediately following the day the predictor is measured. The results are reported in Table 9.

[Insert Table 9 about here]

Panel A examines the predictive power of the indicators around (scheduled) earnings announcements, while Panel B focuses on (unscheduled) 8-K filings. Each panel is further divided into two subpanels: Subpanels A1 and B1 use the open-sell (OS) indicator defined in Equation (8), while Subpanels A2 and B2 use the open-buy (OB) indicator from Equation (7). Within each, Subpanels A1 and B1 report results for portfolios formed on the day prior to the announcement ($t = -1$), whereas Subpanels A2 and B2 analyze portfolios constructed on the announcement day itself ($t = 0$).

The results closely mirror those of Cremers et al. (2022), with short option volume predicting returns prior to scheduled announcements, and long option volume predicting returns around unscheduled announcements and on the event day. Specifically, for scheduled earnings, OS-based portfolios formed on $t = -1$ yield positive and marginally significant actual stock returns (13 bps), while OB-based portfolios produce statistically insignificant results. For unscheduled 8-K filings, OB-based portfolios formed on $t = 0$ generate statistically significant and economically meaningful negative actual stock returns (–12 bps), whereas OS-based portfolios remain insignificant.

We also find that portfolios formed on the announcement day exhibit stronger return predictability when volume-based signals are measured contemporaneously—that is, on the news day itself—rather than the day prior. In particular, both OS and OB portfolios yield significantly negative actual stock returns of at least 1% for both scheduled and unscheduled events, consistent with heightened informational relevance of trading activity on the event day.

We next turn to synthetic stock returns. Across all specifications, synthetic stock returns closely track actual stock returns, and the predictive content of volume-based signals largely disappears when applied to the actual-minus-synthetic return spread.

In fact, none of the volume indicators significantly predicts the return spread, with a single marginal exception in Subpanel B1 of Panel A. There, the forecasted effect is economically trivial—roughly 90% smaller than the corresponding actual return (1.0 bps vs. 10.9 bps)—and statistically weak, attaining significance only at the 10% level for the spread while remaining insignificant for actual stock returns.

4 Conclusion

We examine whether option trading volume contains incremental information beyond what is already reflected in stock prices. To address this question, we propose a novel empirical strategy that focuses on the spread between actual and synthetic stock returns—an approach that offers a more stringent test of informational content than standard predictability regressions that use only actual stock returns.

Our findings challenge the prevailing view that option volume-based indicators convey incremental information. While these signals frequently predict actual stock returns, they fail to predict the spread between actual and synthetic stock returns. In the few cases where predictability of the spread emerges, the results are statistically weak, economically negligible, or directionally inconsistent with the presence of incremental information in options.

Our findings are robust to stock-borrow fees: controlling for fees or excluding hard-to-borrow stocks leaves the predictability of option volume for actual and synthetic stock returns essentially unchanged, indicating that these effects are driven by forces distinct from those captured by borrow fees.

We reinforce our empirical results with a noisy rational expectations model in which informed traders operate in both stock and options markets. In this setting, return predictability can arise even when options are informationally irrelevant—that is, when price movements reflect noise or common shocks rather than option-specific information.

Overall, our results underscore the importance of benchmarking actual stock returns against their synthetic counterparts to isolate truly option-specific information. To avoid overstating the informational content of option signals, future research should explicitly compare forecasts of actual stock returns with those based on synthetic stock returns, ensuring that any apparent predictability reflects genuine information from the options market rather than mechanical cross-market linkages.

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Appendix

A Theoretical Model

This section builds on and extends An et al. (2014). They propose a theoretical model in which informed investors are allowed to trade simultaneously in stocks and options. The extent to which they trade depends on the amount of noise trading present in these markets. The prices of stocks and options are determined by a market maker, who can arbitrage between the two markets. Like in a typical noisy rational expectations model, the prices of stocks and options change via the trading of the informed investors, but they are not fully revealing, or the informed investors would not profit from gathering information. This creates predictability from option signals (prices and volumes) to future stock returns, even when information flows simultaneously to both option and stock markets. Below, we describe the main ingredients and outputs of this parsimonious model. Importantly, and differently from An et al. (2014), in the description of the main ingredients of the model we introduce the put option, and derive all quantities related to it. This will be of crucial importance, as through the put options we are able to complete the put-call parity and infer the synthetic stock prices, thus linking the option to the stock market.

A.1 Economy

The firm is born at date $t = 0$, investors trade the stock and options at date $t = 1$, and the firm's cash flows F are realized at time $t = 2$. The distribution of F is binomial, where $F = F_H$ with probability 0.5, and $F = F_L$ with probability 0.5. The prices of the stock at times $t = 0$ and $t = 1$ are denoted as S_0 and S_1, respectively.

There exist call and put options written on the stock. Their strike price is K, where $F_L < K < F_H$, and the options mature at time $t = 2$. The prices of the call (put) option at times $t = 0$ and $t = 1$ are denoted as C_0 and C_1 (P_0 and P_1), respectively. The payoffs of the call and the put at time $t = 2$ are $C_2 = \max(F - K, 0)$ and $P_2 = \max(K - F, 0)$, respectively.

There exist informed traders, noise traders, and a market maker, all with CARA utility, and with risk aversion γ. Informed traders receive a signal θ just before date $t = 1$. This signal takes the value $\theta = 1$ with probability of 0.5, and the value $\theta = 0$ with probability of 0.5. If $\theta = 1$, the cash flow of the firm is equal to F_H with probability ω, and it is equal to F_L with the probability $1 - \omega$. If $\theta = 0$, the cash flow of the firm is equal to F_H with probability $1 - \omega$, and it is equal to F_L with probability ω. The parameter ω captures the quality of the signal θ, where $\frac{1}{2} < \omega < 1$. Therefore, the probability that the firm cash flow is F_H, conditional on the signal θ, is given by $p(\theta) = \omega\theta + (1 - \omega)(1 - \theta)$. The conditional probability of a firm cash flow of F_L is, therefore, $1 - p(\theta)$.

The informed investor can trade both stock and options. Her demand for stock is denoted as q_I, and her demands for call and put options are denoted as d_I and u_I, respectively. The corresponding demands for stock and options for the market maker are denoted as q_D, d_D, and u_D, respectively. There exist noise traders in both the stock market and the options market. Their demand for stock is denoted as $z \sim N(0, \sigma_z^2)$, and their demands for call and put options are denoted as $\nu_c \sim N(0, \sigma_c^2)$ and $\nu_p \sim N(0, \sigma_p^2)$, respectively. Noise traders cannot trade across stock and options markets, which means that z is independent of ν_c and ν_p. However, their demands for calls and puts can be correlated. The market-clearing conditions for the stock and options markets are as follows:

$$q_I + q_D + z = 1 \tag{A.1}$$

$$d_I + d_D + \nu_c = 0 \tag{A.2}$$

$$u_I + u_D + \nu_p = 0 \tag{A.3}$$

which means that the options are in zero net supply, and there is one share of stock.

A.2 Equilibrium

If the informed investor receives no signal, and there are no demand shocks in either market at time $t = 0$, the informed investor and the market maker are identical; they buy half of the stock each at price S_0, and there is no trading in options.

If the informed investor receives a signal just prior to time $t = 1$, her objective is to maximize CARA utility from terminal wealth:

$$\max_{q_I, d_I, u_I} E\left[-\frac{1}{\gamma}\exp(-\gamma W_I) \mid \theta\right] \tag{A.4}$$

subject to:

$$q_I S_1 + d_I C_1 + u_I P_1 = \frac{1}{2} S_1 \tag{A.5}$$

where $W_I = q_I(F - S_1) + d_I(C_2 - C_1) + u_I(P_2 - P_1)$, and γ is the absolute risk aversion of the informed investor. Taking the first order conditions with respect to q_I, d_I, and u_I, gives the following relation:

$$q_I(F_H - F_L) + d_I(F_H - K) - u_I(K - F_L) = -\frac{1}{\gamma} \log\left(\frac{1 - p(\theta)}{p(\theta)} \ \frac{S_1 - F_L}{F_H - S_1}\right) \tag{A.6}$$

and the prices of the call and the put at time $t = 1$ are given by:

$$C_1 = \frac{F_H - K}{F_H - F_L}(S_1 - F_L) \tag{A.7}$$

$$P_1 = \frac{K - F_L}{F_H - F_L}(F_H - S_1) \tag{A.8}$$

Both call and put prices are linear with respect to the stock price. This is because of the assumption of a binomial distribution for the firm's cash flows.

The marker maker does not observe the signal θ and is assumed to have unlimited wealth, which means that she has no budget constraint. The first order condition for the market maker is as follows:

$$q_D(F_H - F_L) + d_D(F_H - K) - u_D(K - F_L) = -\frac{1}{\gamma} \log\left(\frac{S_1 - F_L}{F_H - S_1}\right) \tag{A.9}$$

The price of the stock is derived from the sum of (A.6) and (A.9), which results in:

$$S_1(\theta, z, \nu_c, \nu_p) = \frac{G(\theta, z, \nu_c, \nu_p)}{1 + G(\theta, z, \nu_c, \nu_p)} F_H + \frac{1}{1 + G(\theta, z, \nu_c, \nu_p)} F_L \tag{A.10}$$

where the function $G(\theta, z, \nu_c, \nu_p)$ is given by:

$$G(\theta, z, \nu_c, \nu_p) = \sqrt{\frac{p(\theta)}{1 - p(\theta)}} \exp\left(-\frac{\gamma}{2}\left((1 - z)(F_H - F_L) - \nu_c(F_H - K) + \nu_p(K - F_L)\right)\right)$$

The equilibrium stock price (A.10) is essentially a weighted average of the high and low firm cash flows to be realized at time $t = 2$, where the ratio $\frac{G(.)}{1+G(.)}$ represents the weight allocated to the high cash flow F_H, the remaining being allocated to the low cash flow F_L. The numerical example below shows that $G(.)$ is typically lower than 1, which makes S_1 skewed towards F_L.

A.3 Numerical Example

Following the calibration used in An et al. (2014), the options are taken to be approximately at the money, i.e., $K = E(F)$, and the baseline parameter values used in this numerical example are as follows: $F_H = 103$, $F_L = 97$, $K = 100$, $\omega = 0.7$, and $\gamma = 1.5$.

Informed investors trade the stock and the two options, and the extent to which they trade depends on the amount of noise trading in these markets. As they trade to take advantage of their private information, the prices of the stock and the options change at time $t = 1$. Figure 2 plots the prices of the stock, call, and put options, as functions of noise trading demands, conditional on a positive signal $\theta = 1$ arriving just before time $t = 1$. Panel A plots the price of the stock as a solid line for stock demand shocks z, while keeping the call and put demand shocks at $\nu_c = 0$ and $\nu_p = 0$. The dash-dotted line represents the stock price as a function of call demand shocks ν_c, while holding the stock and put demand shocks at $z = 0$ and $\nu_p = 0$. The dotted line represents the stock price as a function of put demand shocks ν_p, holding the stock and call demand shocks at $z = 0$ and $\nu_c = 0$. Panels B and C repeat this exercise but plot instead the prices of the call and the put options, respectively.

[Insert Figure 2 about here]

Compared to Figure A1 in An et al. (2014), this figure includes the effect of put demand shocks on stock, call, and put prices. As expected, the higher the noise trading demand for the put option, the higher the put price, but the lower the prices of the stock and the call option.

In this example, the put option is significantly more expensive than the call option. The put prices range between 2.88 and 2.98, while the call prices range between 0.02 and 0.12. This is because the equilibrium stock price S_1 in Panel A is very close to the low cash flow F_L. Therefore, at time $t = 1$, the call option is out-the-money, and the put option is in-the-money, given the strike price $K = 100$.

Comparative Statics

Figure 2 shows that the curves intersect at the point in which all the noise trading shocks are equal to zero (i.e., $z = 0$, $\nu_c = 0$, and $\nu_p = 0$). This section examines the sensitivity of that intersection point to changes in some parameter values.

Figure 3 shows the sensitivity of the stock price (Panel A), the call price (Panel B), and the put price (Panel C), to changes in the probability of the high cash flow (ω), given a positive signal ($\theta = 1$). The parameter ω can also be interpreted as the quality of the signal. Its baseline value is $\omega = 0.7$, and Figure 3 takes ω to vary in the interval $]0, 1[$. As explained in Section 2.A.1, for ω to represent the quality of the signal θ it needs to be bounded (i.e., $\frac{1}{2} < \omega < 1$). Therefore, the results in Figure 3 for $\omega \leq \frac{1}{2}$ can be ignored.

[Insert Figure 3 about here]

Not surprisingly, as the quality of the positive signal ($\theta = 1$) improves (i.e., ω increases), the equilibrium prices of the stock and the call increase, and the put price decreases. The sensitivity of the call price is the strongest. Its price increases by 6 times from 0.05 to 0.3 as the quality of the signal ω changes from its baseline value of 0.70 to 0.99. The price of the put option decreases by 8.5% from 2.95 to 2.70, and the price of the stock increases by only 0.5% from 97.1 to 97.6, for the same change in ω. The reason for this differential sensitivity is that the equilibrium stock price is very close to the low cash flow value of $F_L = 97$. Given the strike price of the options of $K = 100$, the call option is out-the-money, and the put option is in-the-money. The delta of the call is positive, and its gamma increases as the stock price approaches the strike price.

The results in Figure 3 suggest that an informed investor with a positive signal about the stock can profit the most when purchasing the out-of-the-money call option and writing the in-the-money put option. Moreover, these results suggest that there is money to be made from collecting information to improve the quality of the private signal.

Figure 4 examines the relation between equilibrium prices and changes in the risk tolerance of the informed investor (γ), keeping everything else in the baseline model unchanged.

[Insert Figure 4 about here]

The results show that, as the informed investor's risk aversion increases, the prices of the stock and the call decrease, and the price of the put increases. The analytic solution for the equilibrium stock price (A.10) shows that the value of the function $G(\theta, z, \nu_c, \nu_p)$, which determines the weight on the high cash flow F_H, decreases as γ increases. Therefore, as γ increases, the stock price converges to the low cash flow $F_L = 97$, the call price converges to zero, and the put price converges to $K - S_1 = 3$.

Lastly, Figure 5 examines the sensitivity of prices to changes in the volatility of the cash flows at the terminal date $t = 2$. The cash flow volatility can be captured by the distance between the high and the low firm cash flows, i.e., $F_H - F_L$. The variation in cash flows is kept symmetric in relation to the strike price $K = 100$, which means that $F_H = K + a$ and $F_L = K - a$. In the baseline model, $F_H - F_L = 103 - 97 = 6$, which corresponds to the case $a = 3$. Figure 5 plots the results for values of a between 0.05 and 5.05.

[Insert Figure 5 about here]

In the baseline case (i.e., $a = 3$), the equilibrium stock price, when noise trading is absent (i.e., $z = 0$, $\nu_c = 0$, and $\nu_p = 0$), is $S_1 = 97.10$ (see Figure 2, Panel A). This is a value close to the low cash flow of $F_L = 97$. The results in Figure 5 (Panel A) show that, as the dispersion in cash flows increases around the strike price (i.e., a increases), the equilibrium stock price decreases as it is skewed towards the low cash flow. For instance, for $a = 2$, $F_L = 98$ and $S_1 = 98.28$, and for $a = 5$, $F_L = 95$ and $S_1 = 95.01$.

For lower values of the cash-flow dispersion (a), the equilibrium stock price S_1 converges to the strike price $K = 100$. The gamma of the call option is highest the closer the option is at-the-money. This explains the hump-shaped results in Panel B of Figure 5. It shows that the call price increases from 0.029 to a maximum value of 0.255 when a increases from 0.05 to 0.90. For higher values of a, the price of the call decreases to zero. This is because, for high values of a, the stock price tracks the decreasing low cash flow, and the call option goes deep out-of-the-money.

Panel C shows the effect of changes in cash-flow volatility on put prices. It shows a monotonic increase in this price as the dispersion in cash flows increases. This is because, for a given strike price $K = 100$, a higher a leads to a lower S_1, and the put option goes deep in-the-money.

Lastly, Figure 6, depicts, for both the informed investor and the market maker, the demand sensitivity of stock, call and put options, respectively, as a function of uninformed demand shocks, given a positive signal, i.e., $\theta = 1$. In particular, the informed investor's demands for stock, call and put are depicted in Panels A1, B1 and C1, respectively. Panel A2, B2 and C2 repeat the same analysis, but for the market maker. This analysis is of great importance as, throughout the text and in particular in the empirical analysis, we consider the predictability of volume-based indicators. The way in which we produce the various panels in Figure 6 is similar in spirit to the stock price analysis depicted in Figure 2, but where we plot demands instead of prices.

[Insert Figure 6 about here]

Panel A1 of Figure 6 plots the informed demand of the stock as a solid line for uninformed stock demand shocks z, while keeping the uninformed call and put demand shocks at $\nu_c = 0$ and $\nu_p = 0$. The dash-dotted line represents the informed stock demand as a function of the uninformed call demand shocks ν_c, while holding the uninformed stock and put demand shocks at $z = 0$ and $\nu_p = 0$. Lastly, the dotted line represents the informed stock demand as a function of the uninformed put demand shocks ν_p, holding the uninformed stock and call demand shocks at $z = 0$ and $\nu_c = 0$.
Panels B1 and C1 repeat the exercise but plot instead the prices of the call and put options, respectively. The parameter values used to generate these plots are $F_H = 103$, $F_L = 97$, $K = 100$, $\omega = 0.7$, and $\gamma = 1.5$.

Overall, Figure 6 shows that, in the presence of a positive private signal, the informed investor trades more in options than in stock. Panel A1 shows that the informed investor keeps the stock holding nearly flat at the initial value of 1/2 throughout, while the market maker alone offsets most of the noise trading demand shocks. Panel B1 shows that the informed investor buys most calls when the noise traders are selling large amounts of stock (solid line in Panel B1, for $x =$ -0.2). This is when the stock is at its lowest price (see Figure 2), and has the most potential to appreciate in value. This is also the value of x at which the call option is cheapest, so it is intuitive that an informed investor who receives a positive signal is willing to buy more calls. Panel C1 shows how the informed investor sells most put options when the noise trader is selling a large amount of stock (solid line in Panel C1, for $x =$ -0.2). This occurs when the stock price is cheapest, and has the most potential to appreciate in value, which coincides with the value of x at which the put option is most expensive, so it is intuitive that the informed investor would like to collect the premium of the put by going short.

A.4 Predicting Stock Returns with Informationally Irrelevant Options

Next, we show how stock return predictability can arise even in the presence of informationally irrelevant options. Chabakauri et al. (2021) demonstrate that, in a single-stock economy, the introduction of options does not improve the market's ability to infer the distribution of stock payoffs, rendering options informationally irrelevant. This characteristic is shared by several existing models, including An et al. (2014), which is the primary focus of our paper, as well as the widely cited study of Easley et al. (1998). In both frameworks, information from past stock and option trades is fully and simultaneously incorporated into stock and options markets, leaving no incremental role for options in price discovery. As a result, in these settings, any observed predictability of stock prices based on option prices or trading volumes cannot be attributed to an informational advantage in the options market, as none exists. We aim to identify the true mechanisms driving such predictability within the model and derive testable implications, establishing the necessary conditions for the reliable extraction of incremental information from options in empirical settings.

In the model described above, the informed investor is allowed to trade both the underlying stock and European-style call and put options written on it. The synthetic, or option-implied, stock price at time $t = 1$, denoted S_1^*, follows directly from put-call parity:

$$S_1^* = C_1 - P_1 + K, \tag{A.11}$$

where C_1 and P_1 represent the prices of the European call and put, respectively, and K is the common strike price.

Substituting the pricing expressions from Equations (A.7) and (A.8)—namely, $C_1 = \frac{F_H - K}{F_H - F_L}(S_1 - F_L)$ and $P_1 = \frac{K - F_L}{F_H - F_L}(F_H - S_1)$—into Equation (A.11), we find that the synthetic price equals the actual stock price: $S_1^* = S_1$. The same result holds at maturity $t = 2$, when options expire. Specifically:

$$S_2^* = C_2 - P_2 + K = \max(S_2 - K, 0) - \max(K - S_2, 0) + K = S_2. \tag{A.12}$$

Hence, in this framework, the synthetic price S_t^* always equals the observed stock price S_t, for any t. Because the difference between synthetic and observed prices is identically zero, option prices do not contain incremental information beyond what is already reflected in the stock price.

Nevertheless, the model has testable implications for predictability. As shown in An et al. (2014), these arise from the covariance between option prices at time 1 and subsequent changes in the stock price. Specifically, the relevant measure is the covariance between the option price and the future firm value net of the current stock price, $F - S_1$, where F denotes the terminal cash flow realized at $t = 2$ (see Section A.2 for details).

To illustrate, consider the relationship between the call price at time 1, C_1, and the future stock price change $F - S_1$. Since C_1 is a positive linear function of S_1, as derived in Equation (A.7), the following covariance characterizes the predictive relationship:

$$\begin{aligned}\mathrm{Cov}(F - S_1, C_1) &= \mathrm{Cov}(F, S_1) - \mathrm{Var}(S_1) \\ &= E[(F - E(F) - S_1 + E(S_1))(S_1 - E(S_1))]. \end{aligned} \tag{A.13}$$

Similarly, since P_1 is a negative linear function of S_1, as implied by Equation (A.8) and put-call parity, the covariance between the put price and the future stock price change is:

$$\begin{aligned}\mathrm{Cov}(F - S_1, P_1) &= \mathrm{Var}(S_1) - \mathrm{Cov}(F, S_1) \\ &= E[(-F + E(F) + S_1 - E(S_1))(S_1 - E(S_1))]. \end{aligned} \tag{A.14}$$

When compared with Equation (A.13), Equation (A.14) reflects the opposing sensitivity of calls and puts to the underlying stock price.

When the covariance between option prices at time $t = 1$ and future cash flows is nonzero, option prices predict future stock returns. The opposite holds true when this covariance is zero. Equations (A.13) and (A.14) both evaluate to zero when the stock price equals its conditional expectation, i.e., $S_1 = E(S_1)$. This condition defines an indifference point—a state of the world in which option prices at $t = 1$ are uncorrelated with subsequent changes in the stock price, $F - S_1$. At this point, option prices contain no predictive content for future stock returns.

To compute the indifference point, we proceed in two steps. First, we solve for S_1 and $E(S_1)$. The former admits a closed-form expression, given in Equation (A.10), and depends on the informed investor's private signal θ as well as the noise trader demand shocks for the stock (z), call options (ν_c), and put options (ν_p). The conditional expectation $E(S_1)$ is computed numerically using Gaussian Hermite quadrature, which allows for efficient integration by controlling both weights and locations.

Secondly, to identify the roots of Equations (A.13) and (A.14), we search for triplets (z, ν_c, ν_p) that satisfy the equilibrium condition $S_1 = E(S_1)$, holding the signal θ fixed. For example, we can fix the noise trader demand for the stock and for calls, and solve for the implied value of ν_p that ensures the indifference condition holds.

We implement this iterative procedure over a grid of demand shock combinations and report the resulting indifference points as three-dimensional surfaces in Figure 7. Any configuration of demand shocks that lies off these surfaces implies a nonzero covariance between option prices at $t = 1$ and future stock price changes $F - S_1$.

[Insert Figure 7 about here]

The process of computing each point on the indifference surface proceeds as follows. Suppose we begin at a point on the surface where the equilibrium condition $S_1 = E(S_1)$ is satisfied. Now consider a shock to noise trader demand for the stock, specifically $z = -0.2$, which depresses the price of the stock below its expected value. In addition to selling the stock, suppose noise traders also sell call options, $\nu_c = -0.2$, further exerting downward pressure on S_1. To restore the indifference condition, noise traders must offset the combined effects of these short positions by also selling put options, which has a positive impact on the synthetic price S_1.

This particular configuration is illustrated by the lowest point in Panel A of Figure 7, where the level of put option selling required to restore the condition $S_1 = E(S_1)$ is approximately $\nu_p = -0.5$. That is, a sufficiently negative put demand is needed to offset the price-depressing effects of short positions in both the stock and call options.

Given any such combination of uninformed demand shocks, the informed investor will act as counterparty to some trades if doing so is optimal given her private signal θ. Any residual order imbalance is absorbed by the market maker to clear the market. In the case where the informed investor receives a *positive* signal, she has no incentive to buy put options from the noise traders. Consequently, the put options written by noise traders in the above example—corresponding to the lowest point in Panel A—must be absorbed entirely by the market maker. This case is analogous to the dotted line in Panel C2 of Figure 6, except that in that illustration, stock and call option demands are held at zero.

It is important to understand which regions of Figure 7 imply a positive versus a negative covariance. Both panels apply to call and put options, but the sign of the covariance above or below the surface depends on the option under consideration.

For instance, when examining the covariance between call prices and future stock returns (as in Equation (A.13)), coordinates (z, ν_c, ν_p) that lie above the surface in Panel A imply a negative covariance for calls. The same region implies a positive covariance for puts, because Equation (A.14) is simply the negative of Equation (A.13). Conversely, regions below the surface in Panel A imply a positive covariance for calls and a negative covariance for puts.

Panel B provides a variant of the same framework. The only difference is that, in Panel B, we fix the stock and put demands and solve for the level of call demand that satisfies the equilibrium condition $S_1 = E(S_1)$. In this setting, the interpretation of the covariance signs flips: the region *above* the surface implies a *positive* covariance for call prices and a *negative* one for puts, while the region *below* the surface implies a *negative* covariance for calls and a *positive* one for puts.

Given that both call and put options can be traded simultaneously, the covariance between option prices and future stock returns should reflect this joint exposure. Specifically, we examine the covariance between the net position $C_1 - P_1$ and the future stock return $F - S_1$:

$$\begin{aligned}\text{Cov}(F - S_1, C_1 - P_1) &= [\text{Cov}(F, S_1) - \text{Var}(S_1)] - [\text{Var}(S_1) - \text{Cov}(F, S_1)] \\ &= 2\,[\text{Cov}(F, S_1) - \text{Var}(S_1)] \\ &= 2\,E\,[(-F + E(F) + S_1 - E(S_1))(S_1 - E(S_1))]\,. \end{aligned} \tag{A.15}$$

As for the previous cases, as long as $S_1 \neq E(S_1)$, the covariance is nonzero—that is, current option prices predict future stock returns.[12]

We now turn to the question of whether current option prices can predict future *synthetic* (option-implied) stock price changes, rather than actual stock price changes. Specifically, we are interested in the covariance between current option prices and future changes in the synthetic price, $S_2^* - S_1^*$, rather than the realized return $F - S_1$. Equations (A.13) and (A.14) follow through, where S_1 is replaced with S_1^* in the condition for zero covariance, i.e., $(S_1^* - E(S_1^*)) = 0$. However, as demonstrated earlier, in this model, the synthetic price always equals the actual price, i.e., $S_1^* = S_1$. As a result, the zero covariance condition naturally reduces to $(S_1 - E(S_1)) = 0$, just as before.

More generally, suppose we wish to test whether option prices contain information about the spread between future actual and synthetic stock returns. In that case, we can write the following covariance:

$$\text{Cov}((F - S_1) - (S_2^* - S_1^*), \cdot) \tag{A.16}$$

where $\cdot$ can be replaced with the call price, the put price, or a combination of both. As discussed above, in this framework, actual and synthetic stock prices are always identical at any given point in time, i.e., $S_1 = S_1^*$ and $F = S_2 = S_2^*$. This directly implies that $((F - S_1) - (S_2^* - S_1^*)) = 0$, ensuring that Equation (A.16) always holds at zero. This result underscores a key implication of the model: when synthetic prices perfectly track actual prices by construction, option prices cannot contain incremental information about the divergence between actual and option-implied stock returns.

It is worth noting that, although the discussion thus far has focused on covariances between future stock returns and current option *prices*—following the approach in An et al. (2014)—an analogous logic applies when considering *quantities* instead of prices as predictors of returns. The key insight is that option prices are linear functions of the equilibrium stock price S_1, which itself depends on the informed signal θ and the noise trading demands for the stock and options, i.e., $S_1(\theta, z, \nu_c, \nu_p)$.

Using the implicit function theorem, one can map the covariances in Equations (A.13)–(A.16) into covariances between future returns and current quantities—namely, the noise trader demands

[12] This result follows from the put-call parity: $C_1 - P_1 + K = S_1^*$. Since K is constant, it follows that $\text{Corr}(C_1 - P_1, S_1) = \text{Corr}(S_1^*, S_1) = 1$.

z, ν_c, and ν_p. These quantity-based covariances provide an alternative lens for analyzing the informational content of option markets and form the basis for our empirical tests. In particular, the empirical analysis in Section 3 focuses on two observable proxies for option demand: the P/C and O/S volume ratio.

Figure 8 plots unsigned P/C volume ratios for different combinations of noise trading demands. These are unsigned P/C volume ratios at the zero covariance points. Basically, we take Panel A of Figure 7, compute the absolute values of all put and call demands, and divide the absolute put demand by the absolute call demand. For instance, when stock and call demands are fixed at $\nu_c = 0.2$ and $z = 0.2$, respectively, the put demand that sets $(S_1 - E(S_1)) = 0$ is about $\nu_p = 0.7$ (this is the highest point of the surface in Panel A of Figure 7). In this case, the P/C volume ratio results in $0.7/0.2 = 0.35$, which is what is represented by the right-most bar of Figure 8.

[Insert Figure 8 about here]

This graph of model-implied P/C ratios reveals some notable patterns. Most strikingly, it shows that P/C ratios can range widely—from close to zero up to around 10—depending on the specific combination of demand shocks. However, it is important to emphasize that all these P/C ratios result in the same outcome: a zero covariance between current option prices and future stock returns. In other words, despite the variation in P/C ratios, they all imply no return predictability. This suggests that, when we empirically sort stocks in a cross-section based on their P/C ratios, we could effectively be sorting on these zero-covariance P/C ratios none of which implies any predictability.

Second, this observation raises an important caveat for empirical work: cross-sectional portfolios formed solely on the basis of high or low P/C ratios could inadvertently be sorted along these zero-covariance contours, yielding no return predictability. Only deviations from these zero-covariance ratios may convey predictive content. However, even in such cases, any return predictability would apply to future *actual* stock returns, not to the spread between actual and synthetic stock returns, as shown in Equation (A.16). We test this implication empirically in Section 3.

Another relevant question concerns the extent to which the non-zero covariances in Equations (A.13), (A.14), and (A.15) are driven by the informed investor's private signal. To address this, we examine a scenario in which the investor receives no useful information at time $t = 1$, i.e., $p(\theta) = 0.5$, such that the signal is uninformative about the firm's future cash flows. The effect of setting $p(\theta) = 0.5$ is to shift the indifference surface in Panel A of Figure 7 downward, thereby lowering the probability that the covariance between future stock returns and current call prices is positive.

Figure 9 illustrates how a positive private signal affects the put-call volume ratio. These model-implied ratios help isolate the incremental effect of information on trading behavior. For example, when the informed investor holds a positive signal (i.e., $p(\theta) = 0.7$) and there is a positive stock demand shock ($z = 0.20$), she finds it optimal to write put options, thereby exploiting her signal.

These written puts, along with the positive stock shock, exert upward pressure on the equilibrium price S_1, pushing it above $E(S_1)$. To reestablish the zero-covariance condition, noise traders must respond by increasing their demand for puts. The reverse dynamic holds in the presence of a negative stock demand shock ($z = -0.20$).

These results follow intuitively from the equilibrium expression for the stock price S_1 in Equation (A.10). Holding stock and call demands fixed at 0.2, a higher $p(\theta)$ increases S_1, which implies that a higher positive put demand shock is required to absorb the informed investor's put writing and restore the indifference condition.

[Insert Figure 9 about here]

Panel A of Figure 9 builds on Figure 8 by averaging P/C ratios across all call demands for each value of the stock demand, under different values of $p(\theta)$. Conversely, Panel B averages across stock demands for a given call demand. Note that Panel B omits bars for cases where the call demand is zero, as the P/C ratio is undefined in those scenarios. As in Figure 8, the P/C ratios increase mechanically when the absolute value of the call demand is low, since the denominator in the ratio becomes smaller.

Lastly, Figure 10 reports model-implied O/S volume ratios at the zero covariance points. The numerator of the O/S volume ratio in Panel A is the absolute put demand, and in Panel B it is the absolute value of the call demand. In contrast to Figure 8, we now divide those quantities by the absolute stock demand shock.

[Insert Figure 10 about here]

The O/S volume ratio is undefined for values of stock demand shocks equal to zero. Panel A shows that the highest put-to-stock (P/S) ratio is attained for a call demand shock of $\nu_c = 0.2$ and a stock demand shock of $z = 0.05$. This is intuitive because, a high ν_c pushes S_1 above $E(S_1)$, which needs to be counterbalanced with a positive ν_p, which Figure 7 shows to be around 0.4 for that pair of z and ν_c values. This results in a P/S ratio of 8, as reflected by the size of the corresponding bar in Figure 10. Not surprisingly, in Panel B, the highest call-to-stock (C/S) ratio is attained when the put demand shock is $\nu_p = -0.2$ and the stock demand shock is $z = 0.05$. This is where we need to sell a large amount of calls so that we keep $S_1 = E(S_1)$.

To conclude, in the presence of noise trading in both stock and options markets, prices do not fully adjust to their informationally efficient levels, leading to predictability from time $t = 1$ to $t = 2$. Importantly, we demonstrate that this predictability can arise even in an environment where options are *informationally irrelevant.* This observation implies that the existence of predictability from option prices to future actual stock returns is *not* a *sufficient* condition to infer that options contain incremental information not already embedded in stock prices.

An et al. (2014) acknowledge this limitation in Appendix A.5 of their paper, noting that "the predictability of options does not exist after controlling for past stock returns." They argue, however, that in their empirical analysis, they do control for past stock returns, and the predictive power of options remains. However, past stock returns may not constitute a sufficient statistic for the information already incorporated into stock prices, particularly when informed trading is at play.

Panel C of Figure 1 proposes an alternative adjustment to the predictive signal extracted from options. Rather than controlling for past stock returns—which may be an imprecise proxy for informed trading—we suggest subtracting the synthetic stock return from the actual stock return, and testing whether option-based signals can predict this spread. If they cannot, it becomes difficult to argue that the options market contains incremental information beyond what is already priced in by the underlying.

We explore this idea in detail in the empirical analysis of Section 3. According to Equation (A.16), the covariance between option prices and the spread between future actual and synthetic stock returns is identically zero under the model. Hence, we expect no predictability for that spread. However, that condition is derived for European-type options for which the no-arbitrage put-call parity relation is a strict equality. In Section 2.2 we discuss the implications of relaxing some of the assumptions that lead to condition (A.16).

B Empirical Variables:

- Amihud: Amihud illiquidity ratio in the same week as the O/S measures. We multiply the value by one billion for ease of presentation.
- AMS_Spread0(%): The difference between CAAR0(%) and CASR0(%).
- AMS_Spread1(%): The difference between CAAR1(%) and CASR1(%).
- B/M: Natural logarithm of book-to-market ratio measured at the firm's last quarterly financial statement report date.
- CAAR0(%): The underlying stock's cumulative abnormal mid-quote return in the week of the volumes used to calculate the O/S measures used as explanatory variables. The CRSP value-weighted market return is used as the benchmark in calculating the abnormal returns. CAAR0 is in percent.
- CAAR1(%): The underlying stock's cumulative abnormal mid-quote return in the week subsequent to the week of the volumes used to calculate the O/S measures used as explanatory variables. The CRSP value-weighted market return is used as the benchmark in calculating the abnormal returns. CAAR1 is in percent.
- CASR0(%): The underlying stock's cumulative abnormal synthetic return in the week of the volumes used to calculate the O/S measures used as explanatory variables. The CRSP value-weighted market return is used as the benchmark in calculating the abnormal returns. Synthetic return is defined as the percent change in option-implied stock prices between consecutive trading days. CASR0 is in percent.

- CASR1(%): The underlying stock's cumulative abnormal synthetic return in the week subsequent to the week of the volumes used to calculate the O/S measures used as explanatory variables. The CRSP value-weighted market return is used as the benchmark in calculating the abnormal returns. Synthetic return is defined as the percent change in option-implied stock prices between consecutive trading days. CASR1 is in percent.
- CBC/S: Ratio of ISE close buy call volume to stock volume.
- CBP/S: Ratio of ISE close buy put volume to stock volume.
- CSC/S: Ratio of ISE close sell call volume to stock volume.
- CSP/S: Ratio of ISE close sell put volume to stock volume.
- Δeqvol: Change in equity trading volume, defined as the difference between total equity volume traded in the week of the O/S measures and the weekly average equity trading volume over the prior six months, scaled by this average. Deciles of Δeqvol, denoted DecΔeqvol, are used in the regressions.
- Δopvol: Change of option trading volume, defined as the difference between total OptionMetrics trading volume in the week of the O/S measures and the weekly average option trading volume over the prior six months, scaled by this average. Deciles of Δopvol, denoted DecΔopvol, are used in the regressions.
- Impl_fee: The average of the daily option-implied stock borrow fee from Muravyev et al. (2022) in the week of the O/S measures. Implied fee data is obtained from Muravyev's website.
- Impl_vol: The underlying stock's average daily implied volatility during the one month prior to the week of the O/S measures. The daily implied volatility is calculated as open interest-weighted implied volatility for all options traded on that day.
- ISEC/S: Ratio of ISE call options volume to stock volume.
- ISEP/S: Ratio of ISE put options volume to stock volume.
- ISEO/S: Ratio of total options volume in ISE to stock volume.
- Momen: Momentum is calculated as the underlying stock's cumulative abnormal return over the past six months, where we use CRSP value-weighted market return as the benchmark in calculating the abnormal return. Momen is in percent.
- NAPR: No-arbitrage price range is defined as the ratio of the difference between upper and lower no-arbitrage bounds derived from the put-call parity for American options, and their mid-point. That is, $(S^{*,U} - S^{*,L})/MP$, where $MP = (S^{*,U} + S^{*,L})/2$ is also the definition of the option-implied price in Equation (4), and the no-arbitrage bounds $S^{*,U}$ and $S^{*,L}$ are as defined in Section 2.2.
- Open Buy (OB): The open buy put-call volume ratio, like in Cremers et al. (2022), is computed as the ratio of put buy volume to the sum of put and call buy volumes, i.e., put and call contracts purchased by non-market makers to open new positions. This data is obtained from ISE.
- OBC/S: Ratio of ISE open buy call volume to stock volume.

- OBP/S: Ratio of ISE open buy put volume to stock volume.
- OMC/S: Ratio of call options volume in OptionMetrics to stock volume.
- OMO/S: Ratio of total options volume in OptionMetrics to stock volume.
- OMP/S: Ratio of put options volume in OptionMetrics to stock volume.
- Open Sell (OS): The open sell put-call volume ratio is computed as the ratio of put sell volume to the sum of put and call sell volumes, i.e., put and call contracts sold by non-market makers to open new positions. This data is obtained from ISE.
- OSC/S: Ratio of ISE open sell call volume to stock volume.
- OSP/S: Ratio of ISE open sell put volume to stock volume.
- Size: Natural logarithm of market capitalization measured at the firm's last quarterly financial statement report date.
- Skewness: The historical skewness is calculated using daily return during the one month prior to the week of the O/S measures.

Table 1: Descriptive Statistics

Summary statistics for O/S measures and other firm characteristics. The O/S measures OMO/S, OMC/S, and OMP/S are constructed from OptionMetrics volume data, while ISEO/S, ISEC/S, and ISEP/S and the signed O/S measures i.e., OBC/S, OSC/S, etc., are constructed using signed option volume data from the ISE Open/Close Trade Profile. The statistics reported are for the "raw" (not decile) values of the various O/S variables. Panel A and Panel C only consider firms with options traded on the ISE from May 2005 to December 2021. Panel B considers firms with options traded on all exchanges as reported in OptionMetrics. In Panel A, the O/S ratios are computed using weekly volumes, while in Panels B and C, they are computed using daily volumes. Detailed definitions of the variables are in Appendix 1.

Panel A: Weekly Sample for Fama-McBeth Regressions

Variable	Num Obs	Mean	St Dev	Quartile 1	Median	Quartile 3
OMO/S	558,885	0.208	11.105	0.005	0.019	0.054
OMP/S	558,885	0.087	5.311	0.001	0.006	0.021
OMC/S	558,885	0.121	7.078	0.002	0.010	0.031
ISEO/S	558,885	0.050	0.163	0.003	0.014	0.046
ISEC/S	537,872	0.029	0.108	0.002	0.007	0.026
ISEP/S	510,322	0.023	0.090	0.001	0.005	0.020
OBC/S	537,872	0.011	0.044	0.000	0.002	0.009
OSC/S	537,872	0.010	0.040	0.000	0.002	0.008
CBC/S	537,872	0.004	0.021	0.000	0.001	0.003
CSC/S	537,872	0.005	0.023	0.000	0.001	0.003
OBP/S	510,322	0.009	0.038	0.000	0.001	0.007
OSP/S	510,322	0.008	0.036	0.000	0.001	0.006
CBP/S	510,322	0.003	0.018	0.000	0.000	0.002
CSP/S	510,322	0.003	0.017	0.000	0.000	0.002
Δeqvol	558,878	0.033	0.635	-0.296	-0.092	0.192
Δopvol	558,380	0.416	18.333	-0.730	-0.361	0.317
Amihud	558,885	1.150	4.561	0.088	0.273	0.892
Impl_vol	558,821	0.348	0.150	0.248	0.321	0.414
B/M	538,886	-0.954	0.884	-1.432	-0.883	-0.369
Size	557,815	22.311	1.544	21.183	22.193	23.365
Momentum	558,885	0.026	0.260	-0.107	0.018	0.146
Skewness	558,867	0.003	0.043	-0.017	0.002	0.023
CAAR0(%)	558,885	0.047	4.972	-1.997	-0.021	1.999
CAAR1(%)	533,081	-0.018	4.890	-2.030	-0.046	1.949
CASR1(%)	533,081	-0.021	4.878	-2.031	-0.048	1.949
AMS_Spread1(%)	533,081	0.002	0.511	-0.149	0.004	0.151
Average NAPR in Week 0	558,885	0.018	0.019	0.008	0.012	0.021
Average NAPR in Week 1	549,416	0.018	0.020	0.008	0.012	0.021
Impl_fee	198,257	0.010	0.030	0.001	0.007	0.013

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Table 1: Continued

Panel B: Unsigned Option Volumes from OptionMetrics for Earnings and 8-K Filings Tests

	Panel B1: Earnings						Panel B2: 8-K Filings					
Variable	Num Obs	Mean	St Dev	Quartile 1	Median	Quartile 3	Num Obs	Mean	St Dev	Quartile 1	Median	Quartile 3
OMO/S (t=-1)	83,516	0.242	10.588	0.008	0.030	0.089	70,164	0.329	15.461	0.005	0.017	0.055
OMO/S (t=0)	90,066	0.323	19.905	0.007	0.027	0.079	77,189	0.346	17.058	0.004	0.015	0.051
OMP/OMC (t=-1)	79,485	3.334	39.64	0.141	0.531	1.345	66,176	2.803	23.94	0.105	0.485	1.262
OMP/OMC (t=0)	85,917	3.019	30.556	0.176	0.567	1.354	71,747	2.917	36.478	0.077	0.455	1.250
Actual Return (t=-1)	90,066	0.001	0.030	-0.012	0.000	0.014	77,189	0.000	0.032	-0.011	0.000	0.012
Actual Return (t=0)	90,066	0.002	0.053	-0.019	0.001	0.024	77,189	0.001	0.043	-0.013	0.001	0.014
Actual Return (t=0 to 2)	90,066	0.004	0.087	-0.039	0.002	0.046	77,189	0.002	0.063	-0.021	0.001	0.025
Actual Return (t=1 to 3)	90,066	0.002	0.073	-0.030	0.000	0.033	77,189	0.001	0.053	-0.020	0.000	0.022
Actual Return (t=2 to 5)	90,066	0.001	0.055	-0.025	0.000	0.025	77,189	0.001	0.056	-0.022	0.000	0.024
Synthetic Return (t=-1)	90,066	0.001	0.030	-0.012	0.001	0.014	77,189	0.000	0.032	-0.011	0.000	0.012
Synthetic Return (t=0)	90,066	0.002	0.053	-0.020	0.002	0.024	77,189	0.001	0.043	-0.013	0.001	0.014
Synthetic Return (t=0 to 2)	90,066	0.004	0.087	-0.039	0.002	0.046	77,189	0.001	0.063	-0.022	0.002	0.024
Synthetic Return (t=1 to 3)	90,066	0.002	0.073	-0.030	0.000	0.033	77,189	0.001	0.053	-0.020	0.000	0.021
Synthetic Return (t=2 to 5)	90,066	0.001	0.054	-0.024	0.000	0.025	77,189	0.001	0.055	-0.022	0.000	0.024
AMS_Spread (t=-1)	90,066	0.000	0.007	-0.002	0.000	0.001	77,189	0.000	0.006	-0.001	0.000	0.001
AMS_Spread (t=0)	90,066	0.000	0.009	-0.002	0.000	0.002	77,189	0.000	0.007	-0.001	0.000	0.001
AMS_Spread (t=0 to 2)	90,066	0.000	0.015	-0.002	0.000	0.002	77,189	0.000	0.009	-0.001	0.000	0.001
AMS_Spread (t=1 to 3)	90,066	0.000	0.014	-0.002	0.000	0.002	77,189	0.000	0.010	-0.001	0.000	0.001
AMS_Spread (t=2 to 5)	90,066	0.000	0.014	-0.002	0.000	0.002	77,189	0.000	0.013	-0.002	0.000	0.002
Market Cap (Millions)	90,066	14,892	46,269	1,243	3,398	10,987	77,189	18,926	54,949	1,543	4,445	14,798
NAPR at t=0	90,066	0.020	0.019	0.010	0.015	0.024	77,189	0.018	0.020	0.008	0.013	0.021
NAPR at t=-1	90,066	0.020	0.019	0.010	0.015	0.024	77,189	0.018	0.020	0.008	0.013	0.021

Panel C: Signed Option Volumes from ISE for Earnings and 8-K Filings Tests

	Panel C1: Earnings						Panel C2: 8-K Filings					
Variable	Num Obs	Mean	St Dev	Quartile 1	Median	Quartile 3	Num Obs	Mean	St Dev	Quartile 1	Median	Quartile 3
Open Sell (t=0)	41,552	0.456	0.312	0.191	0.455	0.686	36,674	0.431	0.334	0.112	0.412	0.700
Open Buy (t=0)	42,001	0.435	0.315	0.154	0.425	0.667	35,526	0.434	0.353	0.079	0.403	0.743
Open Sell (t=-1)	37,035	0.450	0.315	0.174	0.450	0.680	34,607	0.428	0.334	0.110	0.406	0.696
Open Buy (t=-1)	37,508	0.444	0.318	0.160	0.435	0.683	33,506	0.433	0.353	0.077	0.404	0.743
Actual Return (t=1)	50,643	0.004	0.085	-0.039	0.002	0.046	50,228	0.002	0.063	-0.021	0.002	0.025
Synthetic Return (t=1)	50,643	0.004	0.085	-0.039	0.002	0.045	50,228	0.002	0.063	-0.021	0.002	0.025
AMS_Spread (t=1)	50,643	0.000	0.005	-0.001	0.000	0.002	50,228	0.000	0.005	-0.001	0.000	0.001
Actual Return (t=0)	50,643	0.002	0.054	-0.020	0.001	0.023	50,228	0.001	0.041	-0.012	0.001	0.014
Synthetic Return (t=0)	50,643	0.002	0.053	-0.020	0.002	0.023	50,228	0.001	0.041	-0.012	0.001	0.014
AMS_Spread (t=0)	50,643	0.000	0.005	-0.002	0.000	0.002	50,228	0.000	0.004	-0.001	0.000	0.001
NAPR at t=-1	50,643	0.018	0.017	0.008	0.013	0.021	50,228	0.016	0.019	0.007	0.011	0.019
NAPR at t=0	50,643	0.018	0.017	0.008	0.013	0.021	50,228	0.016	0.019	0.007	0.011	0.019
NAPR at t=+1	48,988	0.016	0.016	0.007	0.012	0.020	49,211	0.016	0.019	0.007	0.011	0.018

Table 2: Fama-MacBeth Regressions of Weekly Returns on Unsigned O/S Volume Measures

Results of weekly Fama-MacBeth regressions of cumulative abnormal returns on the week after the measurement of O/S on deciles of the unsigned O/S measures OMO/S, OMC/S, OMP/S, ISEO/S, ISEC/S, and ISEP/S. Panels A1, B1, and C1 report the results for the subsample used in Ge et al. (2016), which covers the period from May 2005 to August 2012. Panels A2, B2, and C2 cover the full sample from May 2005 to December 2021. These samples consist of firms with options traded on the ISE, they exclude expiration weeks (third week of each month) and observations from the last quarter of 2008 to avoid the potential effect of the short-sale ban on option trading volume. For each firm, we first sum daily option and stock volumes to obtain the weekly volumes. The O/S measures are then computed from the weekly volumes, and the O/S deciles are formed at the end of each week. The dependent variable in Panel A is CAAR1(%), the cumulative abnormal actual return (CAAR) in the week subsequent to the week in which the O/S measures are calculated, and is in percent. In Panel B, the dependent variable is CASR1(%), the cumulative abnormal synthetic return (CASR) in the week subsequent to the week in which the O/S measures are calculated, and is in percent. In Panel C, the dependent variable is the difference between CAAR1(%) and CASR1(%). The other variables are defined in Appendix 1. We estimate a cross-sectional regression for each week, and then report the time-series means of the regression coefficients. Obs is the number of firm-weeks. ***, **, and * represent significance at the 1%, 5%, and 10% levels, respectively. Newey and West (1987) adjusted standard errors with four lags are reported in parentheses.

Panel A: Next Week Actual Stock Returns (CAAR1) as Dependent Variable

	Panel A1: Subsample May 2005 to August 2012						Panel A2: Full Sample May 2005 to December 2021					
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
DecOMO/S	-0.033***						-0.014*					
	(0.008)						(0.007)					
DecOMC/S		-0.027***						-0.007				
		(0.009)						(0.009)				
DecOMP/S			-0.029***						-0.024***			
			(0.007)						(0.007)			
DecISEO/S				-0.014***						0.030		
				(0.005)						(0.040)		
DecISEC/S					-0.013**						-0.004	
					(0.005)						(0.006)	
DecISEP/S						-0.013***						-0.024
						(0.005)						(0.016)
Amihud	0.024	0.025	0.024	0.026*	0.026*	0.026*	-0.032	-0.082	0.089	1.272	0.084	-0.268
	(0.015)	(0.015)	(0.015)	(0.014)	(0.014)	(0.014)	(0.056)	(0.106)	(0.067)	(1.245)	(0.062)	(0.276)
CAAR(0)	-0.980**	-0.940**	-1.090**	-1.019**	-1.000**	-1.053**	-1.263	-1.674	0.234	0.996	-1.061	-1.914
	(0.431)	(0.428)	(0.432)	(0.431)	(0.432)	(0.429)	(0.837)	(1.236)	(0.848)	(1.544)	(0.654)	(1.361)
DecΔeqvol	-0.017***	-0.015**	-0.014**	-0.011*	-0.010*	-0.010*	-0.008*	-0.006	-0.011**	0.062	0.001	-0.025
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.005)	(0.005)	(0.005)	(0.069)	(0.009)	(0.018)
DecΔopvol	0.013*	0.008	0.008	-0.003	-0.004	-0.004	0.003	-0.003	0.007	-0.079	-0.011	0.003
	(0.007)	(0.007)	(0.006)	(0.005)	(0.005)	(0.005)	(0.005)	(0.006)	(0.005)	(0.075)	(0.008)	(0.008)
Impl_vol	-0.294	-0.331	-0.338	-0.479	-0.483	-0.487	0.23	0.253	-0.111	-1.941	-0.11	0.293
	(0.380)	(0.380)	(0.389)	(0.401)	(0.402)	(0.398)	(0.302)	(0.342)	(0.274)	(1.886)	(0.244)	(0.434)
B/M	0.006	0.007	0.008	0.011	0.011	0.011	-0.02	-0.019	-0.029	-0.12	-0.031	-0.012
	(0.050)	(0.049)	(0.050)	(0.049)	(0.049)	(0.049)	(0.028)	(0.028)	(0.029)	(0.103)	(0.030)	(0.029)
Size	-0.017	-0.018	-0.018	-0.025	-0.026	-0.024	0.002	0.000	0.004	-0.002	-0.002	-0.002
	(0.019)	(0.019)	(0.019)	(0.019)	(0.019)	(0.019)	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	(0.014)
Momen	-0.207	-0.206	-0.197	-0.185	-0.185	-0.184	-0.141	-0.139	-0.134	-0.128	-0.127	-0.128
	(0.202)	(0.203)	(0.202)	(0.202)	(0.202)	(0.202)	(0.125)	(0.125)	(0.125)	(0.125)	(0.125)	(0.125)
Skewness	0.261	0.273	0.268	0.258	0.25	0.273	0.141	0.146	0.143	0.134	0.146	0.136
	(0.406)	(0.406)	(0.409)	(0.411)	(0.411)	(0.411)	(0.280)	(0.281)	(0.277)	(0.281)	(0.282)	(0.280)
Intercept	0.62	0.645	0.644	0.818*	0.832*	0.796*	-0.133	-0.099	-0.055	0.109	-0.007	0.125
	(0.447)	(0.448)	(0.450)	(0.446)	(0.446)	(0.445)	(0.382)	(0.377)	(0.371)	(0.364)	(0.371)	(0.368)
Adj R^2	0.073***	0.073***	0.073***	0.072***	0.072***	0.073***	0.081***	0.080***	0.080***	0.079***	0.079***	0.080***
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Obs	219,511	219,511	219,511	219,511	219,511	219,511	454,227	454,227	454,227	454,227	454,227	454,227

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Table 2: Continued

Panel B: Next Week Synthetic Stock Returns (CASR1) as Dependent Variable

	Panel B1: Subsample May 2005 to August 2012						Panel B2: Full Sample May 2005 to December 2021					
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
DecOMO/S	-0.033***						-0.015*					
	(0.009)						(0.007)					
DecOMC/S		-0.027***						-0.006				
		(0.009)						(0.009)				
DecOMP/S			-0.029***						-0.024***			
			(0.007)						(0.007)			
DecISEO/S				-0.015***						0.029		
				(0.005)						(0.038)		
DecISEC/S					-0.013**						-0.004	
					(0.005)						(0.006)	
DecISEP/S						-0.014***						-0.027
						(0.005)						(0.018)
Amihud	0.024	0.025*	0.023	0.026*	0.026*	0.026*	-0.038	-0.085	0.084	1.218	0.073	-0.292
	(0.015)	(0.015)	(0.015)	(0.014)	(0.014)	(0.014)	(0.061)	(0.109)	(0.064)	(1.192)	(0.053)	(0.296)
CAAR(0)	-0.676	-0.636	-0.787*	-0.715*	-0.696	-0.751*	-0.935	-1.328	0.528	1.243	-0.741	-1.706
	(0.431)	(0.427)	(0.432)	(0.430)	(0.431)	(0.429)	(0.843)	(1.226)	(0.811)	(1.458)	(0.666)	(1.477)
DecΔeqvol	-0.016**	-0.014**	-0.014**	-0.010*	-0.01	-0.01	-0.007	-0.005	-0.010**	0.061	0.002	-0.027
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.005)	(0.005)	(0.005)	(0.067)	(0.009)	(0.021)
DecΔopvol	0.012	0.006	0.007	-0.004	-0.006	-0.006	0.002	-0.005	0.006	-0.077	-0.013	0.001
	(0.007)	(0.007)	(0.006)	(0.005)	(0.005)	(0.005)	(0.005)	(0.006)	(0.005)	(0.072)	(0.008)	(0.008)
Impl_vol	-0.319	-0.358	-0.359	-0.5	-0.506	-0.509	0.222	0.239	-0.101	-1.874	-0.11	0.278
	(0.379)	(0.379)	(0.387)	(0.401)	(0.401)	(0.397)	(0.305)	(0.343)	(0.264)	(1.804)	(0.240)	(0.433)
B/M	0.005	0.006	0.007	0.009	0.009	0.009	-0.02	-0.02	-0.029	-0.117	-0.031	-0.013
	(0.049)	(0.049)	(0.049)	(0.049)	(0.049)	(0.049)	(0.028)	(0.027)	(0.028)	(0.100)	(0.029)	(0.028)
Size	-0.019	-0.02	-0.019	-0.027	-0.028	-0.026	0.001	-0.001	0.003	-0.004	-0.003	-0.004
	(0.019)	(0.019)	(0.019)	(0.018)	(0.018)	(0.018)	(0.015)	(0.015)	(0.015)	(0.014)	(0.015)	(0.014)
Momen	-0.2	-0.198	-0.19	-0.178	-0.178	-0.177	-0.131	-0.129	-0.125	-0.118	-0.117	-0.118
	(0.202)	(0.202)	(0.201)	(0.201)	(0.201)	(0.201)	(0.124)	(0.125)	(0.124)	(0.124)	(0.124)	(0.124)
Skewness	0.249	0.262	0.255	0.246	0.239	0.26	0.129	0.133	0.131	0.123	0.133	0.125
	(0.407)	(0.407)	(0.410)	(0.412)	(0.411)	(0.412)	(0.280)	(0.281)	(0.277)	(0.281)	(0.282)	(0.280)
Intercept	0.676	0.703	0.695	0.864*	0.883**	0.841*	-0.1	-0.06	-0.042	0.139	0.029	0.194
	(0.439)	(0.440)	(0.443)	(0.439)	(0.440)	(0.438)	(0.376)	(0.372)	(0.366)	(0.359)	(0.365)	(0.373)
Adj R^2	0.073***	0.073***	0.073***	0.072***	0.072***	0.072***	0.081***	0.080***	0.080***	0.079***	0.079***	0.080***
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Obs	219,511	219,511	219,511	219,511	219,511	219,511	454,227	454,227	454,227	454,227	454,227	454,227

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Table 2: Continued

Panel C: Next Week Spread Between Actual and Synthetic Stock Returns as Dependent Variable

	Panel C1: Subsample May 2005 to August 2012						Panel C2: Full Sample May 2005 to December 2021					
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
DecOMO/S	0.000						0.000					
	(0.001)						(0.001)					
DecOMC/S		0.000						0.000				
		(0.001)						(0.001)				
DecOMP/S			0.000						0.001			
			(0.001)						(0.001)			
DecISEO/S				0.000						0.001		
				(0.001)						(0.002)		
DecISEC/S					0.000						0.000	
					(0.000)						(0.000)	
DecISEP/S						0.000						0.002
						(0.001)						(0.003)
Amihud	0.000	0.000	0.000	0.000	0.000	0.000	0.006	0.003	0.005	0.054	0.010	0.024
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.005)	(0.003)	(0.006)	(0.053)	(0.010)	(0.023)
CAAR(0)	-0.304***	-0.304***	-0.303***	-0.304***	-0.305***	-0.302***	-0.328***	-0.345***	-0.294***	-0.247***	-0.320***	-0.208
	(0.035)	(0.035)	(0.035)	(0.035)	(0.035)	(0.035)	(0.029)	(0.031)	(0.053)	(0.093)	(0.032)	(0.129)
DecΔeqvol	-0.001	-0.001	0.000	0.000	-0.001	-0.001	-0.001**	-0.001***	-0.001***	0.001	-0.001	0.002
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.003)	(0.001)	(0.003)
DecΔopvol	0.001***	0.002***	0.001***	0.001***	0.002***	0.001***	0.001***	0.002***	0.001***	-0.001	0.001***	0.002***
	(0.001)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.003)	(0.000)	(0.000)
Impl_vol	0.025	0.027	0.021	0.022	0.024	0.022	0.008	0.014	-0.01	-0.068	0.000	0.015
	(0.021)	(0.021)	(0.020)	(0.020)	(0.020)	(0.020)	(0.016)	(0.015)	(0.023)	(0.083)	(0.020)	(0.014)
B/M	0.002	0.001	0.002	0.002	0.001	0.001	0.000	0.000	0.000	-0.004	0.000	0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.004)	(0.001)	(0.001)
Size	0.002	0.002	0.002	0.002	0.002	0.001	0.001	0.002	0.001	0.001	0.001	0.001
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Momen	-0.007	-0.008	-0.007	-0.007	-0.008	-0.007	-0.010*	-0.010*	-0.009*	-0.010*	-0.011**	-0.010*
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.006)	(0.005)	(0.006)	(0.005)	(0.005)	(0.005)
Skewness	0.012	0.011	0.013	0.012	0.011	0.013	0.012	0.013	0.012	0.012	0.013	0.011
	(0.018)	(0.018)	(0.018)	(0.018)	(0.018)	(0.018)	(0.017)	(0.017)	(0.016)	(0.016)	(0.016)	(0.016)
Intercept	-0.056	-0.057	-0.05	-0.046	-0.051	-0.045	-0.033	-0.04	-0.014	-0.03	-0.036	-0.069
	(0.043)	(0.044)	(0.043)	(0.044)	(0.044)	(0.043)	(0.029)	(0.029)	(0.029)	(0.029)	(0.029)	(0.047)
Adj R^2	0.040***	0.040***	0.039***	0.039***	0.039***	0.039***	0.047***	0.047***	0.047***	0.046***	0.046***	0.046***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.002)
Obs	219,511	219,511	219,511	219,511	219,511	219,511	454,227	454,227	454,227	454,227	454,227	454,227

Table 3: Fama-MacBeth Regressions of Weekly Returns on Signed O/S Volume Measures

Results of weekly Fama-MacBeth regressions of cumulative abnormal returns on the week after the measurement of O/S on deciles of the signed O/S measures OBC/S, OSC/S, CBC/S, CSC/S, OBP/S, OSP/S, CBP/S, and CSP/S. Panels A1, B1, and C1 report the results for the subsample used in Ge et al. (2016), which covers the period from May 2005 to August 2012. Panels A2, B2, and C2 cover the full sample from May 2005 to December 2021. These samples consist of firms with options traded on the ISE, they exclude expiration weeks (third week of each month) and observations from the last quarter of 2008 to avoid the potential effect of the short-sale ban on option trading volume. For each firm, we first sum daily option volumes from the ISE Open/Close Trade Profile data to obtain the weekly volumes for each category of option trades, and sum the daily stock volumes from CRSP to obtain the weekly stock volumes. The signed O/S measures are then computed from the weekly volumes, and the O/S deciles are formed at the end of each week. Panel A presents the results of regressions in which the dependent variable is the stock cumulative abnormal actual return the week after measurement of O/S (CAAR1(%)). Panel B presents the results of regressions in which the dependent variable is the stock cumulative abnormal synthetic return the week after measurement of O/S (CASR1(%)). Panel C uses the spread between actual and synthetic stock returns as the dependent variable, i.e., the difference between the dependent variables in Panels A and B. The other variables are defined in Appendix 1. We estimate a cross-sectional regression for each week, and then report the time-series means of the regression coefficients. Obs is the number of firm-weeks. ***, **, and * represent significance at the 1%, 5%, and 10% levels, respectively. Newey and West (1987) adjusted standard errors with four lags are reported in parentheses.

Panel A: Next Week Actual Stock Returns (CAAR1) as Dependent Variable

	Panel A1: Subsample May 2005 to August 2012							Panel A2: Full Sample May 2005 to December 2021						
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
DecOBC/S	0.013**				0.024***		0.028***	-0.002				0.014		0.002
	(0.006)				(0.006)		(0.007)	(0.006)				(0.010)		(0.008)
DecOSC/S	-0.023***				-0.015***		-0.018***	-0.009**				-0.005		-0.001
	(0.006)				(0.006)		(0.006)	(0.004)				(0.006)		(0.008)
DecCBC/S		-0.012**			-0.012**		-0.009*		-0.008**			-0.010**		-0.011*
		(0.006)			(0.006)		(0.006)		(0.004)			(0.005)		(0.006)
DecCSC/S		-0.017***			-0.022***		-0.021***		-0.011***			-0.013**		-0.008
		(0.006)			(0.007)		(0.007)		(0.004)			(0.005)		(0.006)
DecOBP/S			-0.026***			-0.021***	-0.023***			-0.017***			-0.010**	-0.011**
			(0.006)			(0.006)	(0.007)			(0.005)			(0.004)	(0.004)
DecOSP/S			0.015***			0.021***	0.025***			0.011**			0.011**	0.016***
			(0.006)			(0.006)	(0.006)			(0.005)			(0.005)	(0.004)
DecCBP/S				-0.009*		-0.012**	-0.007				0.005		-0.011*	-0.011**
				(0.005)		(0.006)	(0.006)				(0.018)		(0.006)	(0.004)
DecCSP/S				-0.013**		-0.010	-0.005				-0.016		-0.004	0.002
				(0.007)		(0.007)	(0.007)				(0.014)		(0.006)	(0.004)
Amihud	0.026*	0.025*	0.026*	0.026*	0.024*	0.025*	0.024*	-0.057	-0.017	-0.007	0.301	0.093	0.027**	0.017
	(0.014)	(0.015)	(0.014)	(0.015)	(0.014)	(0.014)	(0.014)	(0.082)	(0.044)	(0.037)	(0.278)	(0.074)	(0.013)	(0.012)
CAAR(0)	-1.041**	-0.969**	-1.032**	-1.064**	-0.990**	-1.037**	-0.982**	-0.950*	-0.722*	-1.138	0.737	-0.213	-0.537	-0.466
	(0.431)	(0.436)	(0.428)	(0.428)	(0.435)	(0.426)	(0.426)	(0.538)	(0.422)	(0.693)	(1.314)	(0.457)	(0.348)	(0.351)
DecΔeqvol	-0.010	-0.011*	-0.009	-0.010	-0.011*	-0.009	-0.010	-0.007	-0.007*	-0.006	0.006	-0.008*	-0.004	-0.007
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.004)	(0.004)	(0.004)	(0.014)	(0.004)	(0.005)	(0.004)
DecΔopvol	-0.006	0.000	-0.005	-0.003	-0.002	-0.003	-0.001	-0.003	-0.001	-0.003	-0.013	-0.003	-0.004	-0.002
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.004)	(0.004)	(0.004)	(0.010)	(0.003)	(0.003)	(0.004)
Impl_vol	-0.479	-0.336	-0.545	-0.384	-0.331	-0.426	-0.309	0.066	0.108	0.032	-0.522	0.011	-0.043	0.027
	(0.400)	(0.398)	(0.399)	(0.385)	(0.397)	(0.388)	(0.388)	(0.271)	(0.254)	(0.274)	(0.568)	(0.240)	(0.237)	(0.239)
B/M	0.010	0.007	0.011	0.009	0.007	0.010	0.007	-0.020	-0.022	-0.019	-0.020	-0.022	-0.020	-0.022
	(0.049)	(0.049)	(0.049)	(0.049)	(0.049)	(0.049)	(0.049)	(0.028)	(0.027)	(0.028)	(0.028)	(0.027)	(0.028)	(0.028)
Size	-0.026	-0.013	-0.027	-0.015	-0.014	-0.017	-0.01	-0.001	0.007	-0.004	0.005	0.008	0.004	0.010
	(0.019)	(0.019)	(0.019)	(0.019)	(0.019)	(0.019)	(0.019)	(0.016)	(0.015)	(0.015)	(0.015)	(0.016)	(0.016)	(0.017)
Momen	-0.187	-0.178	-0.178	-0.188	-0.174	-0.181	-0.168	-0.128	-0.122	-0.125	-0.139	-0.121	-0.136	-0.124
	(0.202)	(0.201)	(0.202)	(0.201)	(0.201)	(0.201)	(0.199)	(0.125)	(0.124)	(0.125)	(0.125)	(0.124)	(0.125)	(0.124)
Skewness	0.258	0.292	0.285	0.255	0.295	0.275	0.329	0.146	0.178	0.163	0.132	0.178	0.171	0.223
	(0.410)	(0.409)	(0.410)	(0.408)	(0.410)	(0.408)	(0.408)	(0.281)	(0.282)	(0.280)	(0.281)	(0.281)	(0.281)	(0.282)
Intercept	0.819*	0.565	0.856*	0.604	0.576	0.646	0.498	-0.014	-0.166	0.036	-0.036	-0.189	-0.081	-0.196
	(0.449)	(0.441)	(0.441)	(0.447)	(0.445)	(0.444)	(0.442)	(0.388)	(0.370)	(0.369)	(0.375)	(0.393)	(0.379)	(0.400)
Adj R^2	0.072***	0.073***	0.073***	0.073***	0.074***	0.074***	0.075***	0.079***	0.080***	0.080***	0.080***	0.080***	0.081***	0.082***
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.005)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Obs	219,511	219,511	219,511	219,511	219,511	219,511	219,511	454,227	454,227	454,227	454,227	454,227	454,227	454,227

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Table 3: Continued

Panel B: Next Week Synthetic Stock Returns (CASR1) as Dependent Variable

	Panel B1: Subsample May 2005 to August 2012							Panel B2: Full Sample May 2005 to December 2021						
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
DecOBC/S	0.011*				0.022***		0.026***	-0.002				0.015		0.000
	(0.006)				(0.006)		(0.007)	(0.005)				(0.011)		(0.009)
DecOSC/S	-0.022***				-0.014**		-0.017***	-0.008*				-0.005		0.000
	(0.006)				(0.006)		(0.006)	(0.004)				(0.006)		(0.008)
DecCBC/S		-0.013**			-0.012**		-0.010*		-0.008*			-0.009**		-0.010*
		(0.006)			(0.006)		(0.006)		(0.004)			(0.005)		(0.006)
DecCSC/S		-0.017***			-0.022***		-0.020***		-0.011***			-0.013**		-0.008
		(0.006)			(0.007)		(0.007)		(0.004)			(0.005)		(0.006)
DecOBP/S			-0.027***			-0.021***	-0.023***			-0.017***			-0.010**	-0.011**
			(0.006)			(0.007)	(0.007)			(0.005)			(0.004)	(0.004)
DecOSP/S			0.015**			0.021***	0.024***			0.010**			0.010**	0.016***
			(0.006)			(0.006)	(0.007)			(0.005)			(0.005)	(0.004)
DecCBP/S				-0.009		-0.012**	-0.007				0.005		-0.011*	-0.011**
				(0.005)		(0.006)	(0.006)				(0.017)		(0.006)	(0.004)
DecCSP/S				-0.013**		-0.010	-0.005				-0.016		-0.005	0.001
				(0.007)		(0.007)	(0.007)				(0.013)		(0.006)	(0.004)
Amihud	0.025*	0.025*	0.025*	0.025*	0.024*	0.025*	0.023*	-0.061	-0.021	-0.011	0.278	0.101	0.025*	0.015
	(0.014)	(0.014)	(0.014)	(0.015)	(0.014)	(0.014)	(0.014)	(0.085)	(0.047)	(0.040)	(0.256)	(0.082)	(0.013)	(0.012)
CAAR(0)	-0.734*	-0.664	-0.730*	-0.760*	-0.683	-0.734*	-0.676	-0.634	-0.410	-0.807	0.965	0.151	-0.178	-0.129
	(0.430)	(0.435)	(0.427)	(0.427)	(0.434)	(0.426)	(0.425)	(0.554)	(0.435)	(0.696)	(1.212)	(0.477)	(0.348)	(0.352)
DecΔeqvol	-0.010	-0.011*	-0.009	-0.009	-0.011*	-0.009	-0.010	-0.006	-0.006	-0.005	0.007	-0.007	-0.003	-0.005
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.004)	(0.004)	(0.004)	(0.013)	(0.005)	(0.005)	(0.004)
DecΔopvol	-0.007	-0.002	-0.006	-0.005	-0.004	-0.004	-0.002	-0.005	-0.003	-0.005	-0.014	-0.005	-0.005	-0.004
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.004)	(0.004)	(0.004)	(0.010)	(0.003)	(0.003)	(0.004)
Impl_vol	-0.503	-0.361	-0.566	-0.408	-0.356	-0.449	-0.333	0.059	0.100	0.021	-0.497	-0.009	-0.061	0.007
	(0.400)	(0.397)	(0.398)	(0.385)	(0.396)	(0.387)	(0.388)	(0.273)	(0.256)	(0.275)	(0.531)	(0.239)	(0.235)	(0.238)
B/M	0.009	0.005	0.01	0.008	0.005	0.008	0.006	-0.020	-0.022	-0.019	-0.020	-0.022	-0.020	-0.022
	(0.049)	(0.049)	(0.049)	(0.049)	(0.049)	(0.049)	(0.049)	(0.027)	(0.027)	(0.027)	(0.027)	(0.027)	(0.027)	(0.027)
Size	-0.027	-0.014	-0.028	-0.017	-0.015	-0.018	-0.011	-0.002	0.006	-0.005	0.004	0.006	0.003	0.008
	(0.018)	(0.018)	(0.018)	(0.019)	(0.018)	(0.019)	(0.019)	(0.015)	(0.015)	(0.015)	(0.015)	(0.016)	(0.015)	(0.016)
Momen	-0.18	-0.171	-0.172	-0.181	-0.167	-0.175	-0.161	-0.118	-0.112	-0.115	-0.128	-0.110	-0.126	-0.113
	(0.201)	(0.201)	(0.201)	(0.200)	(0.200)	(0.200)	(0.199)	(0.124)	(0.123)	(0.124)	(0.124)	(0.123)	(0.124)	(0.123)
Skewness	0.249	0.279	0.272	0.241	0.284	0.262	0.317	0.135	0.166	0.153	0.118	0.167	0.159	0.213
	(0.411)	(0.410)	(0.411)	(0.409)	(0.411)	(0.409)	(0.409)	(0.280)	(0.282)	(0.280)	(0.280)	(0.281)	(0.281)	(0.282)
Intercept	0.868*	0.616	0.897**	0.654	0.626	0.688	0.542	0.019	-0.129	0.070	-0.006	-0.151	-0.043	-0.153
	(0.442)	(0.434)	(0.435)	(0.440)	(0.438)	(0.437)	(0.435)	(0.381)	(0.365)	(0.362)	(0.368)	(0.386)	(0.373)	(0.394)
Adj R^2	0.072***	0.073***	0.072***	0.073***	0.073***	0.073***	0.075***	0.079***	0.080***	0.080***	0.080***	0.080***	0.081***	0.082***
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.005)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Obs	219,511	219,511	219,511	219,511	219,511	219,511	219,511	454,227	454,227	454,227	454,227	454,227	454,227	454,227

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Table 3: Continued

Panel C: Next Week Spread Between Actual and Synthetic Stock Returns (AMS_Spread1) as Dependent Variable

	Panel C1: Subsample May 2005 to August 2012							Panel C2: Full Sample May 2005 to December 2021						
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
DecOBC/S	0.002***				0.002***		0.002***	0.001				0.000		0.002**
	(0.001)				(0.000)		(0.001)	(0.000)				(0.001)		(0.001)
DecOSC/S	-0.001**				-0.001**		-0.001**	-0.001**				0.000		0.000
	(0.000)				(0.000)		(0.000)	(0.000)				(0.001)		(0.000)
DecCBC/S		0.000			0.000		0.000		0.000			0.000		0.000
		(0.000)			(0.001)		(0.001)		(0.000)			(0.000)		(0.000)
DecCSC/S		0.000			-0.001		-0.001		0.000			0.000		-0.001
		(0.000)			(0.000)		(0.000)		(0.000)			(0.000)		(0.000)
DecOBP/S			0.000			0.000	0.000			0.000			0.000	0.000
			(0.000)			(0.000)	(0.000)			(0.000)			(0.000)	(0.000)
DecOSP/S			0.000			0.000	0.000			0.000			0.000	0.000
			(0.000)			(0.000)	(0.001)			(0.000)			(0.000)	(0.001)
DecCBP/S				0.000		0.000	0.000				0.001		-0.001	0.000
				(0.000)		(0.000)	(0.000)				(0.001)		(0.000)	(0.000)
DecCSP/S				0.000		0.000	0.000				-0.001		0.000	0.000
				(0.000)		(0.000)	(0.000)				(0.001)		(0.000)	(0.000)
Amihud	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.003	0.004	0.004	0.023	-0.008	0.002	0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.003)	(0.004)	(0.004)	(0.022)	(0.008)	(0.002)	(0.001)
CAAR(0)	-0.308***	-0.305***	-0.302***	-0.304***	-0.307***	-0.303***	-0.306***	-0.316***	-0.312***	-0.331***	-0.228**	-0.364***	-0.359***	-0.337***
	(0.035)	(0.035)	(0.035)	(0.035)	(0.035)	(0.035)	(0.035)	(0.035)	(0.036)	(0.029)	(0.109)	(0.041)	(0.038)	(0.029)
DecΔeqvol	-0.001	-0.001	0	-0.001	-0.001	-0.001	-0.001	-0.001***	-0.001***	-0.001***	0.000	-0.001***	-0.001***	-0.001***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.000)	(0.000)	(0.000)
DecΔopvol	0.001***	0.002***	0.001***	0.002***	0.001***	0.001***	0.001***	0.002***	0.002***	0.002***	0.001	0.002***	0.002***	0.002***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.000)	(0.000)	(0.000)
Impl_vol	0.024	0.025	0.021	0.023	0.026	0.022	0.024	0.007	0.008	0.01	-0.025	0.019	0.019	0.021
	(0.020)	(0.021)	(0.020)	(0.020)	(0.021)	(0.020)	(0.021)	(0.016)	(0.017)	(0.015)	(0.043)	(0.015)	(0.015)	(0.015)
B/M	0.001	0.001	0.001	0.001	0.001	0.001	0.002	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Size	0.002	0.002	0.001	0.002	0.002	0.001	0.001	0.001	0.002	0.001	0.001	0.002	0.001	0.002
	(0.002)	(0.002)	(0.002)	(0.001)	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Momen	-0.007	-0.008	-0.006	-0.007	-0.007	-0.006	-0.007	-0.011*	-0.010*	-0.009*	-0.010*	-0.010*	-0.010*	-0.011**
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.006)	(0.005)	(0.005)	(0.005)
Skewness	0.009	0.013	0.012	0.013	0.010	0.013	0.012	0.011	0.012	0.010	0.013	0.011	0.012	0.010
	(0.019)	(0.019)	(0.018)	(0.018)	(0.019)	(0.018)	(0.019)	(0.017)	(0.016)	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)
Intercept	-0.050	-0.051	-0.041	-0.050	-0.050	-0.043	-0.044	-0.033	-0.037	-0.034	-0.030	-0.037	-0.038	-0.043
	(0.044)	(0.043)	(0.042)	(0.040)	(0.042)	(0.040)	(0.039)	(0.029)	(0.028)	(0.029)	(0.029)	(0.029)	(0.029)	(0.029)
Adj R^2	0.039***	0.039***	0.039***	0.039***	0.040***	0.041***	0.042***	0.046***	0.047***	0.046***	0.047***	0.048***	0.048***	0.050***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.002)	(0.003)	(0.003)	(0.003)	(0.003)
Obs	219,511	219,511	219,511	219,511	219,511	219,511	219,511	454,227	454,227	454,227	454,227	454,227	454,227	454,227

Table 4: Unsigned Option Volumes and Stock Borrow Fees

Results of weekly Fama-MacBeth regressions of cumulative abnormal returns on the week after the measurement of O/S on deciles of the unsigned O/S measures OMO/S, OMC/S, and OMP/S. Implied-fee data is obtained from Muravyev et al. (2022) and then merged with the subsample used in Ge et al. (2016), which results in a sample that covers the period from July 2006 to August 2012. Subpanels A1, B1, and C1, run the baselines tests on this subsample, without additional controls. Subpanels A2, B2, and C2 include implied borrow fees as an additional control. Subpanels A3, B3, and C3 exclude hard-to-borrow stock from the sample (i.e., stocks with implied fees greater than 1%). Like in Table 2, the samples used in these tests consist of firms with options traded on the ISE, and they exclude expiration weeks (third week of each month) and observations from the last quarter of 2008 to avoid the potential effect of the short-sale ban on option trading volume. Implied fees are averaged across the days of the week in which the O/S measures are computed. The dependent variables in Panels A, B and C are the same as used in Table 2. All the variables are defined in Appendix 1. We estimate a cross-sectional regression for each week, and then report the time-series means of the regression coefficients. Obs is the number of firm-weeks. ***, **, and * represent significance at the 1%, 5%, and 10% levels, respectively. Newey and West (1987) adjusted standard errors with four lags are reported in parentheses.

Panel A: Next Week Actual Stock Returns (CAAR1) as Dependent Variable

	A1: Subsample with Implied Fee			A2: Controlling for Implied Fee			A3: Subsample without Implied Fee > 1%		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
DecOMO/S	-0.033***			-0.030***			-0.037***		
	(0.01)			(0.01)			(0.01)		
DecOMC/S		-0.027***			-0.025***			-0.027***	
		(0.01)			(0.01)			(0.01)	
DecOMP/S			-0.027***			-0.025***			-0.029***
			(0.01)			(0.01)			(0.01)
Impl_fee				-2.200***	-2.282***	-2.225***			
				(0.78)	(0.79)	(0.78)			
Amihud	0.018*	0.019*	0.018*	0.019*	0.020*	0.019*	0.034**	0.036**	0.032**
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)	(0.02)	(0.02)
CAAR(0)	-0.914*	-0.871*	-1.005**	-0.915*	-0.877*	-0.997**	-0.471	-0.427	-0.552
	(0.49)	(0.49)	(0.49)	(0.49)	(0.49)	(0.49)	(0.62)	(0.62)	(0.62)
DecΔeqvol	-0.015**	-0.013**	-0.012*	-0.015**	-0.013**	-0.012*	-0.006	-0.004	-0.003
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
DecΔopvol	0.020**	0.014*	0.014**	0.018**	0.013	0.012*	0.022**	0.014	0.014**
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Impl_vol	-0.046	-0.090	-0.101	0.027	-0.012	-0.025	0.073	0.003	-0.003
	(0.45)	(0.45)	(0.47)	(0.45)	(0.45)	(0.46)	(0.53)	(0.53)	(0.54)
B/M	-0.015	-0.014	-0.013	-0.016	-0.016	-0.014	-0.024	-0.022	-0.021
	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.04)	(0.04)	(0.04)
Size	-0.012	-0.013	-0.013	-0.015	-0.016	-0.017	0.004	0.000	0.000
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Momen	-0.269	-0.268	-0.258	-0.264	-0.263	-0.253	-0.379	-0.376	-0.368
	(0.24)	(0.24)	(0.23)	(0.24)	(0.23)	(0.23)	(0.26)	(0.25)	(0.25)
Skewness	0.040	0.053	0.061	0.050	0.061	0.075	-0.019	-0.028	0.001
	(0.44)	(0.44)	(0.44)	(0.44)	(0.44)	(0.44)	(0.48)	(0.49)	(0.49)
Intercept	0.359	0.392	0.397	0.408	0.440	0.450	-0.062	0.028	0.038
	(0.50)	(0.50)	(0.52)	(0.50)	(0.50)	(0.51)	(0.54)	(0.54)	(0.54)
Adj R^2	0.076***	0.076***	0.076***	0.079***	0.079***	0.079***	0.083***	0.083***	0.083***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Obs	178,636	178,636	178,636	178,636	178,636	178,636	115,995	115,995	115,995

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Table 4: Continued

Panel B: Next Week Synthetic Stock Returns (CASR1) as Dependent Variable

	B1: Subsample with Implied Fee			B2: Controling for Implied Fee			B3: Subsample without Implied Fee > 1%		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
DecOMO/S	-0.033***			-0.031***			-0.034***		
	(0.01)			(0.01)			(0.01)		
DecOMC/S		-0.026***			-0.025***			-0.024**	
		(0.01)			(0.01)			(0.01)	
DecOMP/S			-0.028***			-0.026***			-0.027***
			(0.01)			(0.01)			(0.01)
Impl_fee				-1.074	-1.160	-1.095			
				(0.76)	(0.76)	(0.75)			
Amihud	0.018*	0.019*	0.018*	0.018*	0.019*	0.018*	0.032**	0.034**	0.030**
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)	(0.02)	(0.02)
CAAR(0)	-0.583	-0.542	-0.674	-0.628	-0.590	-0.714	-0.226	-0.190	-0.300
	(0.49)	(0.49)	(0.49)	(0.48)	(0.48)	(0.48)	(0.62)	(0.62)	(0.62)
DecΔeqvol	-0.014**	-0.013**	-0.012*	-0.014**	-0.012*	-0.011*	-0.006	-0.004	-0.004
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
DecΔopvol	0.019**	0.013	0.012*	0.018**	0.012	0.011*	0.020**	0.012	0.013*
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Impl_vol	-0.084	-0.132	-0.134	-0.044	-0.090	-0.093	0.041	-0.033	-0.022
	(0.45)	(0.45)	(0.47)	(0.45)	(0.45)	(0.47)	(0.53)	(0.53)	(0.54)
B/M	-0.018	-0.017	-0.016	-0.019	-0.018	-0.017	-0.026	-0.025	-0.023
	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.04)	(0.04)	(0.04)
Size	-0.016	-0.017	-0.017	-0.017	-0.018	-0.018	0.006	0.002	0.003
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Momen	-0.266	-0.264	-0.255	-0.262	-0.260	-0.251	-0.386	-0.382	-0.375
	(0.24)	(0.23)	(0.23)	(0.24)	(0.23)	(0.23)	(0.25)	(0.25)	(0.25)
Skewness	0.027	0.041	0.048	0.028	0.039	0.052	-0.009	-0.014	0.011
	(0.45)	(0.45)	(0.45)	(0.44)	(0.44)	(0.45)	(0.49)	(0.49)	(0.49)
Intercept	0.451	0.488	0.484	0.466	0.506	0.502	-0.129	-0.033	-0.046
	(0.50)	(0.50)	(0.51)	(0.49)	(0.49)	(0.51)	(0.54)	(0.54)	(0.54)
Adj R^2	0.076***	0.076***	0.076***	0.079***	0.078***	0.078***	0.083***	0.082***	0.082***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Obs	178,636	178,636	178,636	178,636	178,636	178,636	115,995	115,995	115,995

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Table 4: Continued

Panel C: Next Week Spread Between Actual and Synthetic Stock Returns (AMS_Spread1) as Dependent Variable

	C1: Subsample with Implied Fee			C2: Controling for Implied Fee			C3: Subsample without Implied Fee > 1%		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
DecOMO/S	0.000			0.001			-0.003***		
	(0.00)			(0.00)			(0.00)		
DecOMC/S		0.000			0.000			-0.003***	
		(0.00)			(0.00)			(0.00)	
DecOMP/S			0.001			0.002**			-0.001**
			(0.00)			(0.00)			(0.00)
Impl_fee				-1.126***	-1.121***	-1.131***			
				(0.13)	(0.13)	(0.13)			
Amihud	0.000	0.000	0.000	0.001	0.001	0.001	0.002	0.002	0.002
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
CAAR(0)	-0.331***	-0.329***	-0.331***	-0.286***	-0.286***	-0.283***	-0.245***	-0.236***	-0.252***
	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
DecΔeqvol	0.000	-0.001	0.000	-0.001	-0.001	-0.001	0.000	0.000	0.000
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
DecΔopvol	0.001**	0.002***	0.001***	0.001	0.001**	0.001*	0.002***	0.002***	0.001***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Impl_vol	0.038	0.042	0.033	0.072***	0.077***	0.068***	0.032*	0.037**	0.019
	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)
B/M	0.003*	0.003*	0.003*	0.003*	0.002	0.003*	0.002	0.002	0.003
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Size	0.004*	0.004*	0.003*	0.002	0.002	0.001	-0.002	-0.002	-0.003*
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Momen	-0.003	-0.004	-0.003	-0.003	-0.004	-0.003	0.007	0.006	0.008
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Skewness	0.013	0.012	0.013	0.022	0.022	0.023	-0.011	-0.014	-0.010
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)
Intercept	-0.092*	-0.096*	-0.086	-0.058	-0.065	-0.052	0.067*	0.061	0.085**
	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.04)	(0.04)	(0.04)
Adj R^2	0.042***	0.041***	0.041***	0.073***	0.072***	0.071***	0.047***	0.047***	0.046***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Obs	178,636	178,636	178,636	178,636	178,636	178,636	115,995	115,995	115,995

Table 5: Signed Option Volumes and Stock Borrow Fees

Results of weekly Fama-MacBeth regressions of cumulative abnormal returns on the week after the measurement of O/S on deciles of the signed O/S measures OBC/S, OSC/S, CBC/S, CSC/S, OBP/S, OSP/S, CBP/S, and CSP/S. Implied-fee data is obtained from Muravyev et al. (2022) and then merged with the subsample used in Ge et al. (2016), which results in a sample that covers the period from July 2006 to August 2012. Subpanels A1, B1, and C1, run the baselines tests on this subsample, without additional controls. Subpanels A2, B2, and C2 include implied borrow fees as an additional control. Subpanels A3, B3, and C3 exclude hard-to-borrow stocks from the sample (i.e., stocks with implied fees greater than 1%). Like in Table 2, the samples used in these tests consist of firms with options traded on the ISE, and they exclude expiration weeks (third week of each month) and observations from the last quarter of 2008 to avoid the potential effect of the short-sale ban on option trading volume. Implied fees are averaged across the days of the week in which the O/S measures are computed. The dependent variables in Panels A, B and C are the same as used in Table 2. All the variables are defined in Appendix 1. We estimate a cross-sectional regression for each week, and then report the time-series means of the regression coefficients. Obs is the number of firm-weeks. ***, **, and * represent significance at the 1%, 5%, and 10% levels, respectively. Newey and West (1987) adjusted standard errors with four lags are reported in parentheses.

Panel A: Next Week Actual Stock Returns (CAAR1) as Dependent Variable

	A1: Subsample with Implied Fee			A2: Controlling for Implied Fee			A3: Subsample without Implied Fee > 1%		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
DecOBC/S	0.024***		0.030***	0.023***		0.028***	0.018*		0.024**
	(0.01)		(0.01)	(0.01)		(0.01)	(0.01)		(0.01)
DecOSC/S	-0.018***		-0.021***	-0.016**		-0.019***	-0.013*		-0.014*
	(0.01)		(0.01)	(0.01)		(0.01)	(0.01)		(0.01)
DecCBC/S	-0.010		-0.006	-0.010*		-0.007	-0.006		0.000
	(0.01)		(0.01)	(0.01)		(0.01)	(0.01)		(0.01)
DecCSC/S	-0.021***		-0.020***	-0.020***		-0.019**	-0.022***		-0.021**
	(0.01)		(0.01)	(0.01)		(0.01)	(0.01)		(0.01)
DecOBP/S		-0.025***	-0.028***		-0.024***	-0.027***		-0.024***	-0.028***
		(0.01)	(0.01)		(0.01)	(0.01)		(0.01)	(0.01)
DecOSP/S		0.020***	0.024***		0.020***	0.024***		0.012	0.014*
		(0.01)	(0.01)		(0.01)	(0.01)		(0.01)	(0.01)
DecCBP/S		-0.014**	-0.011*		-0.013**	-0.010		-0.018**	-0.016*
		(0.01)	(0.01)		(0.01)	(0.01)		(0.01)	(0.01)
DecCSP/S		-0.006	-0.001		-0.005	0.000		0.005	0.009
		(0.01)	(0.01)		(0.01)	(0.01)		(0.01)	(0.01)
Impl_fee				-2.260***	-2.227***	-2.158***			
				(0.78)	(0.79)	(0.79)			
Amihud	0.019*	0.019*	0.017*	0.020*	0.019*	0.018*	0.036**	0.036**	0.036**
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)	(0.02)	(0.02)
CAAR(0)	-0.942*	-0.978**	-0.962*	-0.953*	-0.973**	-0.963**	-0.448	-0.464	-0.420
	(0.50)	(0.49)	(0.49)	(0.50)	(0.48)	(0.48)	(0.63)	(0.62)	(0.63)
DecΔeqvol	-0.010	-0.009	-0.010	-0.010	-0.009	-0.010	-0.001	0.000	-0.001
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
DecΔopvol	0.005	0.005	0.007	0.004	0.004	0.007	0.005	0.005	0.008
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Impl_vol	-0.103	-0.177	-0.078	-0.019	-0.104	-0.009	-0.038	-0.063	0.012
	(0.47)	(0.47)	(0.47)	(0.47)	(0.46)	(0.47)	(0.54)	(0.54)	(0.54)
B/M	-0.011	-0.012	-0.013	-0.012	-0.013	-0.014	-0.020	-0.019	-0.020
	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.04)	(0.04)	(0.04)
Size	-0.007	-0.009	-0.002	-0.010	-0.013	-0.006	0.003	0.007	0.014
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Momen	-0.247	-0.249	-0.239	-0.243	-0.244	-0.235	-0.351	-0.344	-0.331
	(0.23)	(0.23)	(0.23)	(0.23)	(0.23)	(0.23)	(0.25)	(0.25)	(0.25)
Skewness	0.092	0.068	0.108	0.102	0.076	0.115	-0.011	-0.035	-0.018
	(0.45)	(0.45)	(0.45)	(0.44)	(0.44)	(0.44)	(0.50)	(0.49)	(0.50)
Intercept	0.281	0.329	0.182	0.337	0.403	0.258	-0.004	-0.082	-0.226
	(0.52)	(0.52)	(0.52)	(0.52)	(0.52)	(0.52)	(0.55)	(0.55)	(0.56)
Adj R^2	0.077***	0.076***	0.078***	0.080***	0.079***	0.081***	0.083***	0.083***	0.085***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Obs	178,636	178,636	178,636	178,636	178,636	178,636	115,995	115,995	115,995

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Table 5: Continued

Panel B: Next Week Synthetic Stock Returns (CASR1) as Dependent Variable

	B1: Subsample with Implied Fee			B2: Controling for Implied Fee			B3: Subsample without Implied Fee > 1%		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
DecOBC/S	0.022***		0.028***	0.021***		0.027***	0.017*		0.023**
	(0.01)		(0.01)	(0.01)		(0.01)	(0.01)		(0.01)
DecOSC/S	-0.017***		-0.020***	-0.015**		-0.018***	-0.012		-0.013
	(0.01)		(0.01)	(0.01)		(0.01)	(0.01)		(0.01)
DecCBC/S	-0.010*		-0.007	-0.011*		-0.008	-0.005		0.001
	(0.01)		(0.01)	(0.01)		(0.01)	(0.01)		(0.01)
DecCSC/S	-0.020***		-0.019**	-0.019***		-0.018**	-0.020**		-0.020**
	(0.01)		(0.01)	(0.01)		(0.01)	(0.01)		(0.01)
DecOBP/S		-0.025***	-0.028***		-0.024***	-0.027***		-0.023***	-0.026***
		(0.01)	(0.01)		(0.01)	(0.01)		(0.01)	(0.01)
DecOSP/S		0.020***	0.024***		0.020***	0.024***		0.011	0.013
		(0.01)	(0.01)		(0.01)	(0.01)		(0.01)	(0.01)
DecCBP/S		-0.014**	-0.011*		-0.013**	-0.010		-0.017**	-0.015*
		(0.01)	(0.01)		(0.01)	(0.01)		(0.01)	(0.01)
DecCSP/S		-0.006	-0.001		-0.005	-0.001		0.006	0.009
		(0.01)	(0.01)		(0.01)	(0.01)		(0.01)	(0.01)
Impl_fee				-1.139	-1.098	-1.032			
				(0.76)	(0.77)	(0.77)			
Amihud	0.019*	0.018*	0.017*	0.019*	0.019*	0.017*	0.035**	0.034**	0.034**
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)	(0.02)	(0.02)
CAAR(0)	-0.608	-0.648	-0.629	-0.662	-0.689	-0.676	-0.205	-0.213	-0.173
	(0.49)	(0.48)	(0.48)	(0.49)	(0.48)	(0.48)	(0.63)	(0.62)	(0.63)
DecΔeqvol	-0.009	-0.008	-0.009	-0.009	-0.008	-0.009	-0.001	0.000	-0.001
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
DecΔopvol	0.003	0.004	0.006	0.003	0.003	0.005	0.004	0.004	0.006
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Impl_vol	-0.144	-0.213	-0.116	-0.098	-0.176	-0.083	-0.071	-0.091	-0.024
	(0.47)	(0.47)	(0.47)	(0.47)	(0.46)	(0.47)	(0.54)	(0.54)	(0.54)
B/M	-0.013	-0.015	-0.015	-0.015	-0.016	-0.016	-0.022	-0.021	-0.022
	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.04)	(0.04)	(0.04)
Size	-0.011	-0.012	-0.005	-0.012	-0.014	-0.007	0.004	0.009	0.014
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Momen	-0.244	-0.246	-0.237	-0.240	-0.242	-0.232	-0.359	-0.352	-0.338
	(0.23)	(0.23)	(0.23)	(0.23)	(0.23)	(0.23)	(0.25)	(0.25)	(0.25)
Skewness	0.080	0.055	0.098	0.080	0.053	0.095	-0.002	-0.029	-0.009
	(0.45)	(0.45)	(0.45)	(0.45)	(0.45)	(0.45)	(0.51)	(0.49)	(0.51)
Intercept	0.376	0.412	0.267	0.402	0.449	0.306	-0.047	-0.141	-0.263
	(0.51)	(0.52)	(0.52)	(0.51)	(0.51)	(0.51)	(0.54)	(0.55)	(0.56)
Adj R^2	0.076***	0.076***	0.078***	0.079***	0.079***	0.081***	0.083***	0.083***	0.085***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Obs	178,636	178,636	178,636	178,636	178,636	178,636	115,995	115,995	115,995

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Table 5: Continued

Panel C: Next Week Spread Between Actual and Synthetic Stock Returns (AMS_Spread1) as Dependent Variable

	C1: Subsample with Implied Fee			C2: Controling for Implied Fee			C3: Subsample without Implied Fee > 1%		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
DecOBC/S	0.001***		0.001**	0.001**		0.001**	0.000		0.001
	(0.00)		(0.00)	(0.00)		(0.00)	(0.00)		(0.00)
DecOSC/S	-0.001*		-0.001**	-0.001		-0.001	-0.001**		-0.001**
	(0.00)		(0.00)	(0.00)		(0.00)	(0.00)		(0.00)
DecCBC/S	0.000		0.000	0.000		0.000	-0.001*		-0.001
	(0.00)		(0.00)	(0.00)		(0.00)	(0.00)		(0.00)
DecCSC/S	-0.001		-0.001	-0.001		-0.001	-0.001**		-0.001*
	(0.00)		(0.00)	(0.00)		(0.00)	(0.00)		(0.00)
DecOBP/S		0.000	0.000		0.001	0.000		-0.001**	-0.001**
		(0.00)	(0.00)		(0.00)	(0.00)		(0.00)	(0.00)
DecOSP/S		0.000	0.001		0.000	0.001		0.000	0.001
		(0.00)	(0.00)		(0.00)	(0.00)		(0.00)	(0.00)
DecCBP/S		0.000	0.000		0.000	0.000		-0.001*	-0.001
		(0.00)	(0.00)		(0.00)	(0.00)		(0.00)	(0.00)
DecCSP/S		0.000	0.000		0.000	0.001		-0.001	-0.001
		(0.00)	(0.00)		(0.00)	(0.00)		(0.00)	(0.00)
Impl_fee				-1.121***	-1.129***	-1.126***			
				(0.12)	(0.13)	(0.12)			
Amihud	0.000	0.000	0.000	0.001	0.001	0.001	0.002	0.002	0.002
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
CAAR(0)	-0.334***	-0.330***	-0.333***	-0.291***	-0.284***	-0.287***	-0.244***	-0.251***	-0.247***
	(0.04)	(0.04)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
DecΔeqvol	0.000	0.000	0.000	-0.001*	-0.001**	-0.001*	0.000	0.000	0.000
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
DecΔopvol	0.002***	0.002***	0.002***	0.002***	0.001***	0.001***	0.001**	0.001***	0.002***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Impl_vol	0.041	0.036	0.038	0.079***	0.072***	0.074***	0.033**	0.028*	0.037**
	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)
B/M	0.003*	0.003*	0.003*	0.002	0.002	0.002	0.002	0.002	0.002
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Size	0.004*	0.003*	0.003*	0.002	0.001	0.001	-0.001	-0.002	-0.001
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Momen	-0.003	-0.003	-0.003	-0.003	-0.003	-0.002	0.008	0.008	0.007
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Skewness	0.013	0.012	0.010	0.022	0.022	0.020	-0.010	-0.006	-0.009
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)
Intercept	-0.095*	-0.083*	-0.085*	-0.066	-0.046	-0.048	0.043	0.059	0.037
	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.04)	(0.04)	(0.04)
Adj R^2	0.042***	0.042***	0.044***	0.072***	0.072***	0.074***	0.048***	0.047***	0.050***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Obs	178,636	178,636	178,636	178,636	178,636	178,636	115,995	115,995	115,995

Table 6: Predictability of Unsigned Option Volumes Around Earnings News

This table reports in Panel A the returns of portfolios of stocks formed based on deciles of the option-to-stock (O/S) volume ratio around quarterly earnings announcements. Panel B repeats the analysis of Panel A but uses the put-to-call (P/C) volume ratio as a sorting variable. In Panels A1 and B1, portfolios are formed on the day before earnings announcements (t=-1), while in Panels A2 and B2 they are formed on the announcement days (t=0). Each panel reports the returns for the top (High) and bottom (Low) decile portfolios formed using the corresponding sorting variable, as well as the difference between the extreme portfolios (High-Low). Value-weighted returns are reported for the portfolios on the sorting date, as well as two subsequent windows. Rows (1) to (3) report value-weighted mid-quote stock returns in excess of the risk-free rate, and rows (4) to (6) report value-weighted synthetic stock returns in excess of the risk-free rate. Synthetic stock returns are computed as percent changes in synthetic stock prices, which in turn are computed as the mid-point of the upper and lower no-arbitrage bounds defined in Section 2.2. The sample period covered in this analysis is from 1996 to 2021, as it only uses option data from OptionMetrics. ***, **, and * represent significance at the 1%, 5%, and 10% levels, respectively. Newey and West (1987) standard errors are reported in parentheses.

Panel A: Predictability of O/S Around Earnings Announcements

		Panel A1: Sort on day before earnings (t=-1)				Panel A2: Sort on earnings day (t=0)			
Return	Row	Period	Low O/S	High O/S	High-Low	Period	Low O/S	High O/S	High-Low
Actual	(1)	-1	0.043	0.048	0.004	0	-0.001	0.196**	0.197**
			(0.054)	(0.062)	(0.077)		(0.064)	(0.082)	(0.098)
Actual	(2)	[0,2]	0.436***	-0.479***	-0.915***	[1,3]	0.333***	-0.339***	-0.672***
			(0.104)	(0.161)	(0.211)		(0.124)	(0.109)	(0.158)
Actual	(3)	[1,3]	0.191	-0.357***	-0.548***	[2,5]	0.179*	-0.164	-0.344**
			(0.136)	(0.132)	(0.166)		(0.106)	(0.106)	(0.132)
Synthetic	(4)	-1	0.050	0.044	-0.006	0	0.000	0.211**	0.210**
			(0.049)	(0.063)	(0.076)		(0.066)	(0.083)	(0.097)
Synthetic	(5)	[0,2]	0.452***	-0.495***	-0.948***	[1,3]	0.351***	-0.365***	-0.716***
			(0.112)	(0.160)	(0.216)		(0.117)	(0.115)	(0.164)
Synthetic	(6)	[1,3]	0.209	-0.388***	-0.597***	[2,5]	0.225**	-0.176	-0.401***
			(0.134)	(0.141)	(0.170)		(0.106)	(0.114)	(0.142)
Actual - Synthetic	(1) - (4)	-1	-0.007	0.003	0.010	0	-0.002	-0.015	-0.013
			(0.013)	(0.007)	(0.018)		(0.016)	(0.015)	(0.024)
Actual - Synthetic	(2) - (5)	[0,2]	-0.017	0.016	0.033	[1,3]	-0.018	0.026	0.044
			(0.025)	(0.057)	(0.072)		(0.019)	(0.027)	(0.034)
Actual - Synthetic	(3) - (6)	[1,3]	-0.041	0.011	0.052	[2,5]	-0.046**	0.012	0.058**
			(0.027)	(0.042)	(0.050)		(0.022)	(0.028)	(0.027)

Panel B: Predictability of P/C Around Earnings Announcements

		Panel B1: Sort on day before earnings (t=-1)				Panel B2: Sort on earnings day (t=0)			
Return	Row	Period	Low P/C	High P/C	High-Low	Period	Low P/C	High P/C	High-Low
Actual	(1)	-1	0.487***	-0.237***	-0.724***	0	0.875***	-0.764***	-1.567***
			(0.097)	(0.058)	(0.116)		(0.091)	(0.077)	(0.127)
Actual	(2)	[0,2]	0.305***	-0.136	-0.442***	[1,3]	0.222**	0.066	-0.169
			(0.113)	(0.124)	(0.144)		(0.090)	(0.127)	(0.149)
Actual	(3)	[1,3]	0.072	-0.095	-0.167	[2,5]	0.000	0.332***	0.274**
			(0.098)	(0.123)	(0.147)		(0.110)	(0.110)	(0.107)
Synthetic	(4)	-1	0.484***	-0.237***	-0.721***	0	0.832***	-0.756***	-1.531***
			(0.102)	(0.057)	(0.120)		(0.088)	(0.076)	(0.118)
Synthetic	(5)	[0,2]	0.320***	-0.116	-0.435***	[1,3]	0.272***	0.018	-0.267
			(0.117)	(0.130)	(0.141)		(0.078)	(0.157)	(0.168)
Synthetic	(6)	[1,3]	0.094	-0.084	-0.177	[2,5]	0.043	0.353***	0.259**
			(0.103)	(0.126)	(0.151)		(0.107)	(0.117)	(0.117)
Actual - Synthetic	(1) - (4)	-1	0.003	0.000	-0.003	0	0.044***	-0.009	-0.036
			(0.015)	(0.009)	(0.021)		(0.015)	(0.018)	(0.026)
Actual - Synthetic	(2) - (5)	[0,2]	-0.014	-0.021	-0.006	[1,3]	-0.050**	0.048	0.098
			(0.024)	(0.023)	(0.031)		(0.022)	(0.067)	(0.062)
Actual - Synthetic	(3) - (6)	[1,3]	-0.030	-0.009	0.021	[2,5]	-0.043**	-0.021	0.015
			(0.022)	(0.013)	(0.020)		(0.022)	(0.027)	(0.023)

Table 7: Range of Implied No-Arbitrage Bounds Around Earnings Announcements

This table reports the difference between the upper and lower no-arbitrage bounds, as a percent of the mid-point, for the portfolios of stocks created in Table 6. The no-arbitrage price range (NAPR) is defined as NAPR= $(S^{*,U} - S^{*,L})/MP$, where $MP = (S^{*,U} + S^{*,L})/2$ is the mid-point (which is also the definition of the option-implied price in Equation (4)), and the no-arbitrage bounds $S^{*,U}$ and $S^{*,L}$ are as defined in Section 2.2. Panel A sorts stocks into deciles using the O/S ratio, and Panel B repeats the analysis but uses the P/C ratio as a sorting variable. The table shows the NAPR not only for the sorting date but also the days surrounding it. The table also shows the difference between the portfolios of stocks in the top and bottom deciles for each of the sorting variables ("High-Low"). The sample period is from 1996 to 2021. ***, **, and * represent significance at the 1%, 5%, and 10% levels, respectively. Newey and West (1987) standard errors (NWSE) are reported in parenthesis.

Panel A: Range of Implied Bounds for Deciles of O/S Around Earnings Announcements

	Low O/S	2	3	4	5	6	7	8	9	High O/S	High - Low	NWSE
NAPR at t=0 when sorting by O/S at t=-1	1.745	1.532	1.361	1.199	1.048	0.989	0.945	0.887	0.824	0.777	-0.968***	(0.124)
NAPR at t=0 when sorting by O/S at t=0	1.889	1.565	1.379	1.228	1.072	0.991	0.930	0.859	0.854	0.793	-1.095***	(0.140)
NAPR at t=-1 when sorting by O/S at t=-1	1.767	1.530	1.351	1.174	1.039	0.975	0.936	0.861	0.816	0.757	-1.011***	(0.132)
NAPR at t=-1 when sorting by O/S at t=0	1.829	1.559	1.387	1.231	1.062	0.989	0.940	0.854	0.837	0.755	-1.075***	(0.139)

Panel B: Range of Implied Bounds for Deciles of P/C Around Earnings Announcements

	Low P/C	2	3	4	5	6	7	8	9	High P/C	High - Low	NWSE
NAPR at t=0 when sorting by P/C at t=-1	2.014	1.370	1.058	0.950	0.885	0.858	0.877	0.910	1.033	1.260	-0.696***	(0.081)
NAPR at t=0 when sorting by P/C at t=0	2.016	1.358	1.018	0.928	0.894	0.854	0.892	0.954	1.102	1.336	-0.626***	(0.068)
NAPR at t=-1 when sorting by P/C at t=-1	2.046	1.364	1.054	0.934	0.854	0.834	0.860	0.906	1.041	1.269	-0.712***	(0.079)
NAPR at t=-1 when sorting by P/C at t=0	2.008	1.378	1.018	0.919	0.873	0.838	0.871	0.948	1.090	1.329	-0.629***	(0.064)

Table 8: Predictability of Unsigned Option Volumes Around 8-K Filings

This table reports in Panel A the returns of portfolios of stocks formed based on deciles of the option-to-stock volume (O/S) ratio, around non-earning 8-K filings, which include only the first filing of each month and only if there were no earnings news from one week before to one week after the filing date. Panel B repeats the analysis of Panel A but uses the put-to-call (P/C) ratio as a sorting variable. In Panels A1 and B1, portfolios are formed on the day before the 8-K filing (t=-1), while in Panels A2 and B2 they are formed on the filing day (t=0). Each panel reports the returns for the top (High) and bottom (Low) decile portfolios formed using the corresponding sorting variable, as well as the difference between the extreme portfolios (High-Low). Value-weighted returns are reported for the portfolios on the sorting date, as well as two subsequent windows. Rows (1) to (3) report value-weighted mid-quote stock returns in excess of the risk-free rate, and rows (4) to (6) report value-weighted synthetic stock returns in excess of the risk-free rate. Synthetic stock returns are computed as percent changes in synthetic stock prices, which in turn are computed as the mid-point of the upper and lower no-arbitrage bounds defined in Section 2.2. The sample period covered in this analysis is from 1996 to 2021 as it only uses option data from OptionMetrics. ***, **, and * represent significance at the 1%, 5%, and 10% levels, respectively. Newey and West (1987) standard errors are reported in parentheses.

Panel A: Predictability of O/S Around 8-K Filings

		Panel A1: Sort on day before 8-K (t=-1)				Panel A2: Sort on 8-K day (t=0)			
Return	Row	Period	Low O/S	High O/S	High-Low	Period	Low O/S	High O/S	High-Low
Actual	(1)	-1	0.015	0.083	0.068	0	-0.094**	0.078	0.171
			(0.038)	(0.068)	(0.072)		(0.044)	(0.117)	(0.112)
Actual	(2)	[0,2]	0.118	-0.405***	-0.523***	[1,3]	-0.303	-0.193*	0.110
			(0.096)	(0.149)	(0.160)		(0.400)	(0.108)	(0.425)
Actual	(3)	[1,3]	0.033	-0.407***	-0.440***	[2,5]	0.010	-0.480*	-0.489*
			(0.095)	(0.123)	(0.145)		(0.090)	(0.280)	(0.276)
Synthetic	(4)	-1	0.016	0.064	0.048	0	-0.097**	0.090	0.187*
			(0.037)	(0.068)	(0.074)		(0.043)	(0.117)	(0.109)
Synthetic	(5)	[0,2]	0.116	-0.401***	-0.518***	[1,3]	-0.347	-0.202*	0.145
			(0.094)	(0.151)	(0.158)		(0.400)	(0.116)	(0.429)
Synthetic	(6)	[1,3]	0.050	-0.426***	-0.475***	[2,5]	-0.013	-0.501*	-0.488*
			(0.092)	(0.127)	(0.144)		(0.104)	(0.292)	(0.281)
Actual - Synthetic	(1) - (4)	-1	-0.001	0.019	0.020	0	0.004	-0.012	-0.016
			(0.013)	(0.031)	(0.030)		(0.009)	(0.012)	(0.010)
Actual - Synthetic	(2) - (5)	[0,2]	0.001	-0.004	-0.005	[1,3]	0.044	0.009	-0.036
			(0.014)	(0.020)	(0.020)		(0.029)	(0.016)	(0.035)
Actual - Synthetic	(3) - (6)	[1,3]	0.000	-0.030	-0.030	[2,5]	0.023	0.021	-0.001
			(0.036)	(0.063)	(0.069)		(0.026)	(0.021)	(0.027)

Panel B: Predictability of P/C Around 8-K Filings

		Panel B1: Sort on day before 8-K (t=-1)				Panel B2: Sort on 8-K day (t=0)			
Return	Row	Period	Low P/C	High P/C	High-Low	Period	Low P/C	High P/C	High-Low
Actual	(1)	-1	0.580***	-0.442***	-1.022***	0	0.258***	-0.535***	-0.779***
			(0.095)	(0.070)	(0.119)		(0.034)	(0.094)	(0.095)
Actual	(2)	[0,2]	-0.384	-0.036	0.348	[1,3]	0.217**	-0.071	-0.216**
			(0.281)	(0.114)	(0.291)		(0.087)	(0.111)	(0.108)
Actual	(3)	[1,3]	-0.263	0.009	0.271	[2,5]	0.118	0.022	-0.037
			(0.192)	(0.106)	(0.204)		(0.092)	(0.126)	(0.123)
Synthetic	(4)	-1	0.578***	-0.452***	-1.030***	0	0.238***	-0.531***	-0.772***
			(0.091)	(0.073)	(0.125)		(0.031)	(0.095)	(0.096)
Synthetic	(5)	[0,2]	-0.368	-0.035	0.334	[1,3]	0.220**	-0.132	-0.246**
			(0.274)	(0.116)	(0.289)		(0.085)	(0.130)	(0.118)
Synthetic	(6)	[1,3]	-0.228	-0.002	0.226	[2,5]	0.120	0.013	-0.037
			(0.190)	(0.110)	(0.206)		(0.093)	(0.140)	(0.130)
Actual - Synthetic	(1) - (4)	-1	0.002	0.010	0.008	0	0.020***	-0.004	-0.007
			(0.020)	(0.009)	(0.022)		(0.005)	(0.008)	(0.015)
Actual - Synthetic	(2) - (5)	[0,2]	-0.016	-0.001	0.014	[1,3]	-0.003	0.062	0.031
			(0.021)	(0.015)	(0.025)		(0.008)	(0.045)	(0.039)
Actual - Synthetic	(3) - (6)	[1,3]	-0.034	0.006	0.040	[2,5]	-0.002	0.009	0.000
			(0.026)	(0.023)	(0.036)		(0.018)	(0.037)	(0.033)

Table 9: Predictability of Signed Option Volumes Around Earnings and 8-K Filings

This table presents one-day equal-weighted returns following measurements of put-call open buy (OB) and open sell (OS) ratios, like in Cremers et al. (2022). Open buy(sell) ratios are calculated as put open buy (sell) volume as a percentage of total open buy (sell) volume. Returns are presented in percentages after dividing stocks into quintiles daily based on open buy (sell) ratios. We report mid-quote stock returns (Actual), synthetic stock returns (Synthetic), and their spread (Actual-Synthetic). Synthetic stock returns are computed as percent changes in synthetic stock prices, which in turn are computed as the mid-point of the upper and lower no-arbitrage bounds defined in Section 2.2. Results are presented for measurement dates on days preceding news days (Subpanels A1 and B1) and for measurement dates on news days (Subpanels A2 and B2). Panel A focuses on earnings announcements and Panel B focuses on non-earning 8-K filings (i.e., the first 8-K filing of each month not surrounded by an earnings announcement from one week before to one week after the filing date). News events are sorted quarterly into quintiles based on the sorting variable of interest, i.e., either OB or OS. The sample period is from May 2005 to December 2021, as this analysis uses signed option volume from ISE. ***, **, and * represent significance at the 1%, 5%, and 10% levels, respectively. Newey and West (1987) standard errors are reported in parentheses.

Panel A: Earnings Announcements

	Open Sell (OS)							Open Buy (OB)					
	Subpanel A1: Days before news							Subpanel B1: Days before news					
	Low OS	2	3	4	High OS	High-Low		Low OB	2	3	4	High OB	High-Low
Actual Return	0.092	0.089	0.215**	0.176**	0.230***	0.138*	Actual Return	0.147	0.131*	0.207***	0.103	0.256***	0.109
	(0.071)	(0.075)	(0.104)	(0.080)	(0.076)	(0.081)		(0.090)	(0.072)	(0.069)	(0.073)	(0.074)	(0.090)
Synthetic Return	0.091	0.082	0.205*	0.166**	0.227***	0.137*	Synthetic Return	0.149	0.124*	0.200***	0.094	0.248***	0.099
	(0.071)	(0.074)	(0.105)	(0.081)	(0.077)	(0.082)		(0.090)	(0.073)	(0.069)	(0.072)	(0.075)	(0.090)
Actual - Synthetic	0.001	0.007*	0.010*	0.010**	0.003	0.002	Actual - Synthetic	-0.002	0.007*	0.007	0.008*	0.008***	0.010*
	(0.005)	(0.004)	(0.005)	(0.004)	(0.004)	(0.006)		(0.005)	(0.004)	(0.006)	(0.005)	(0.003)	(0.006)
	Subpanel A2: News days							Subpanel B2: News days					
	Low OS	2	3	4	High OS	High-Low		Low OB	2	3	4	High OB	High-Low
Actual Return	1.237***	0.788***	0.397***	-0.128	-0.581***	-1.818***	Actual Return	0.994***	0.701***	0.463***	0.038	-0.449***	-1.444***
	(0.117)	(0.118)	(0.136)	(0.144)	(0.103)	(0.128)		(0.161)	(0.146)	(0.129)	(0.127)	(0.116)	(0.204)
Synthetic Return	1.227***	0.782***	0.394***	-0.131	-0.583***	-1.809***	Synthetic Return	0.982***	0.698***	0.456***	0.032	-0.453***	-1.435***
	(0.115)	(0.118)	(0.136)	(0.146)	(0.103)	(0.128)		(0.160)	(0.146)	(0.127)	(0.127)	(0.115)	(0.203)
Actual - Synthetic	0.011*	0.006	0.003	0.003	0.002	-0.008	Actual - Synthetic	0.012*	0.002	0.007	0.006	0.004	-0.008
	(0.006)	(0.004)	(0.005)	(0.008)	(0.005)	(0.006)		(0.007)	(0.006)	(0.005)	(0.005)	(0.005)	(0.005)

Panel B: 8-K Filings

	Open Sell (OS)							Open Buy (OB)					
	Subpanel A1: Days before news							Subpanel B1: Days before news					
	Low OS	2	3	4	High OS	High-Low		Low OB	2	3	4	High OB	High-Low
Actual Return	0.118***	0.088**	0.074	0.026	0.131*	0.013	Actual Return	0.165***	0.149**	0.048	0.007	0.045	-0.120***
	(0.040)	(0.040)	(0.071)	(0.065)	(0.069)	(0.078)		(0.048)	(0.063)	(0.063)	(0.056)	(0.039)	(0.042)
Synthetic Return	0.119***	0.090**	0.073	0.02	0.125*	0.005	Synthetic Return	0.166***	0.143**	0.047	0.002	0.05	-0.116***
	(0.041)	(0.039)	(0.071)	(0.064)	(0.070)	(0.081)		(0.048)	(0.062)	(0.061)	(0.056)	(0.040)	(0.042)
Actual - Synthetic	-0.001	-0.002	0.001	0.006	0.006	0.008	Actual - Synthetic	-0.001	0.006	0.001	0.005**	-0.006	-0.004
	(0.005)	(0.005)	(0.003)	(0.003)	(0.005)	(0.009)		(0.005)	(0.004)	(0.004)	(0.002)	(0.004)	(0.007)
	Subpanel A2: News days							Subpanel B2: News days					
	Low OS	2	3	4	High OS	High-Low		Low OB	2	3	4	High OB	High-Low
Actual Return	0.676***	0.608***	0.168	-0.111	-0.243	-0.919***	Actual Return	0.772***	0.599***	0.276*	-0.204	-0.315**	-1.088***
	(0.151)	(0.113)	(0.173)	(0.144)	(0.157)	(0.166)		(0.128)	(0.153)	(0.152)	(0.169)	(0.139)	(0.144)
Synthetic Return	0.673***	0.607***	0.157	-0.118	-0.248	-0.921***	Synthetic Return	0.770***	0.599***	0.272*	-0.21	-0.326**	-1.097***
	(0.150)	(0.115)	(0.173)	(0.143)	(0.151)	(0.165)		(0.128)	(0.152)	(0.151)	(0.166)	(0.135)	(0.142)
Actual - Synthetic	0.003	0.001	0.011***	0.006	0.005	0.002	Actual - Synthetic	0.002	0	0.004	0.007	0.011	0.009
	(0.005)	(0.004)	(0.003)	(0.005)	(0.008)	(0.006)		(0.006)	(0.007)	(0.003)	(0.005)	(0.007)	(0.006)

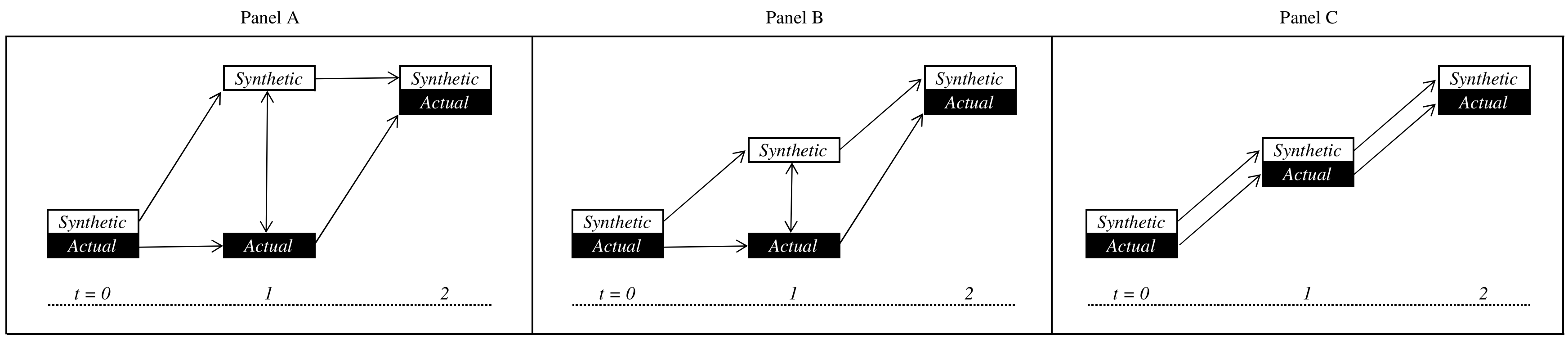

Figure 1: Actual and Synthetic Stock Prices and a Positive Private Signal

This figure illustrates the interaction between *actual* and *synthetic* stock prices when there is a positive private signal to be traded on. Panel A shows the case in which the informed investor prefers to trade in options, and noise trading is absent from the options market. To take advantage of a positive signal received at time $t = 1$, the informed investor purchases call options and sells put options. This pushes up the synthetic (option-implied) stock price immediately to the fundamental value of the stock. As the new information gets revealed to the stock traders, the actual stock price adjusts to the fundamental value at time $t = 2$. Panel B illustrates the case with the presence of noise traders in the options market. In this case, prices adjust only partially towards their efficient levels. Panels A and B assume that the stock and options markets are not fully interconnected (i.e. for American Options). Panel C illustrates the case in which the two markets are fully interconnected, informed investors can trade in both markets, and noise traders exist in both markets (i.e. for European Options). In this case, synthetic and actual stock prices are always equal to each other, and there is no information in options that is not simultaneously reflected in the stock. By construction, in Panel C, the predictability of actual stock returns is exactly the same as that of synthetic stock returns.

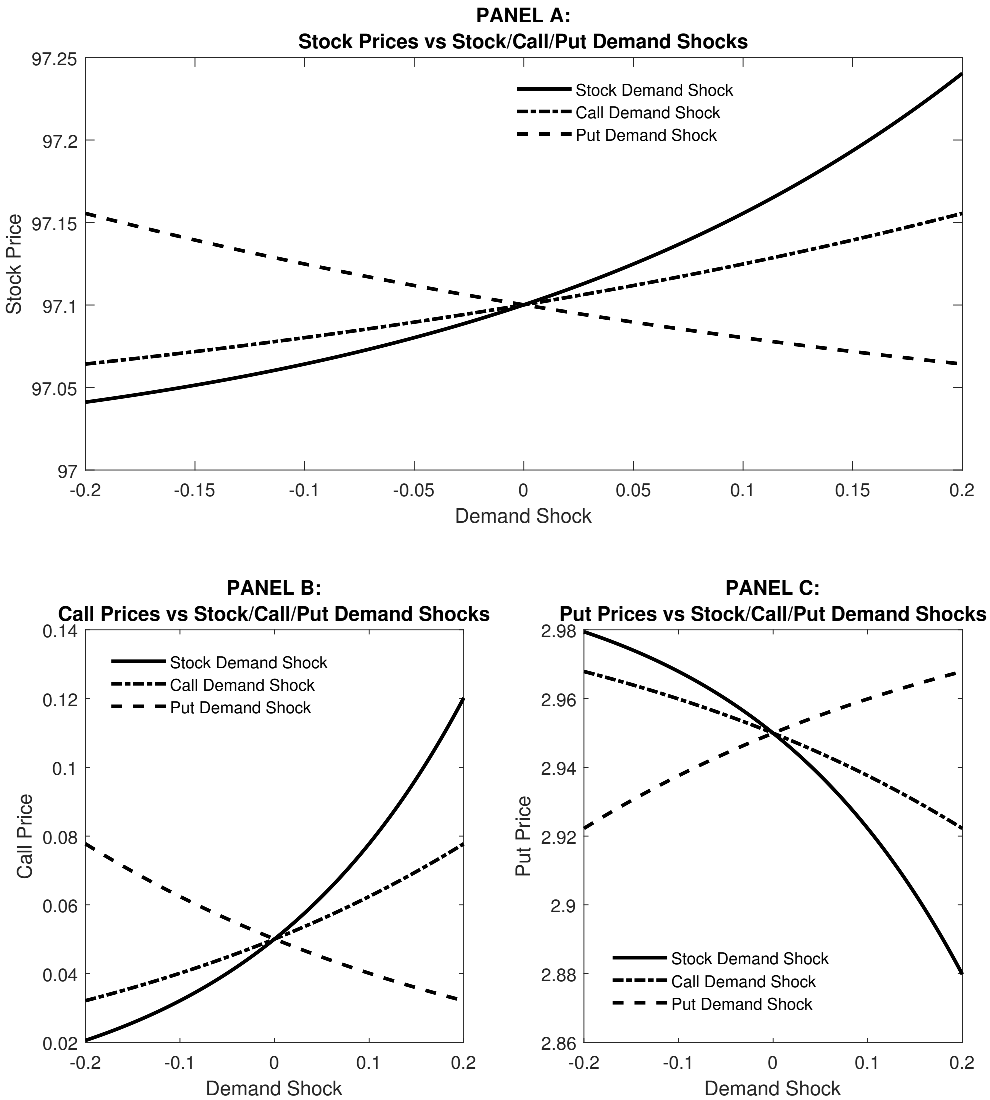

Figure 2: Stock Prices, Options Prices and Noise Trading Shocks

This figure plots the prices of the stock, the call option, and the put option, as a function of uninformed demand shocks given a positive signal, i.e., $\theta = 1$. Panel A plots the price of the stock as a solid line for stock demand shocks z, while keeping the call and put demand shocks at $\nu_c = 0$ and $\nu_p = 0$. The dash-dotted line represents the stock price as a function of call demand shocks ν_c, while holding the stock and put demand shocks at $z = 0$ and $\nu_p = 0$. The dotted line represents the stock price as a function of put demands shocks ν_p, holding the stock and call demand shocks at $z = 0$ and $\nu_c = 0$. Panels B and C repeat the exercise but plotting the prices of the call and put options, respectively. The parameter values used to generate these plots are as follows: $F_H = 103$, $F_L = 97$, $K = 100$, $\omega = 0.7$, and $\gamma = 1.5$.

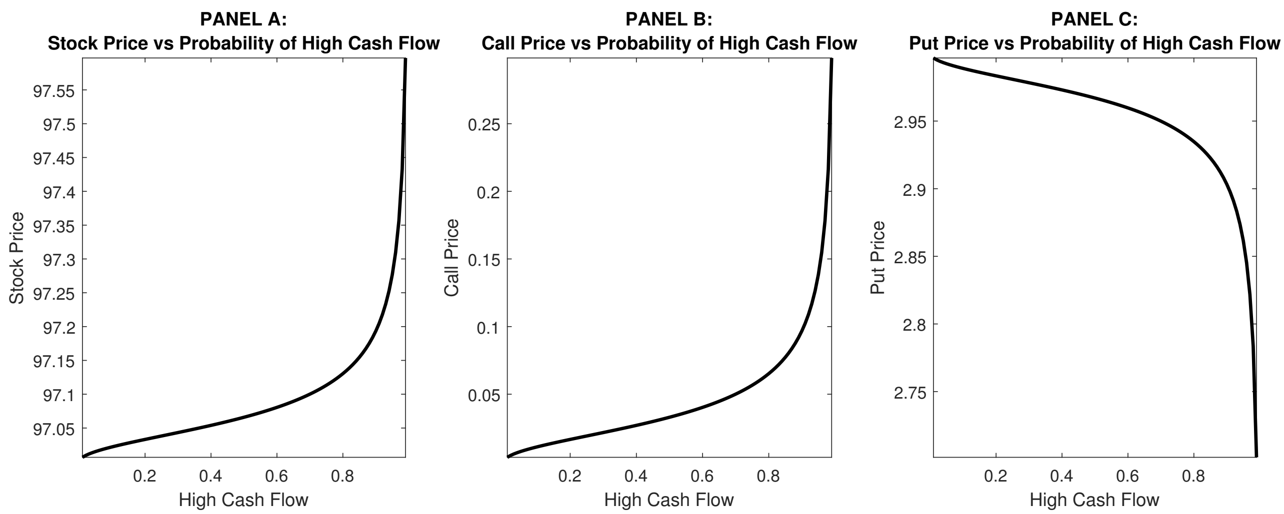

Figure 3: Comparative Statics on Probability of High Cash Flow

This figure shows the sensitivity of the stock price (Panel A), the call price (Panel B), and the put price (Panel C), to changes in the probability of the high cash flow (ω), given a positive signal ($\theta = 1$). Figure 2 shows that the curves intersect at the point in which all the noise trading shocks are equal to zero (i.e., $z = 0$, $\nu_c = 0$, and $\nu_p = 0$). This figure plots these intersection points for different values of ω. The parameter ω can be interpreted as the quality of the positive signal. Its baseline value is $\omega = 0.7$, and this figure takes it to vary in the interval $]0, 1[$. The remaining parameter values are kept unchanged: $F_H = 103$, $F_L = 97$, $K = 100$, and $\gamma = 1.5$.

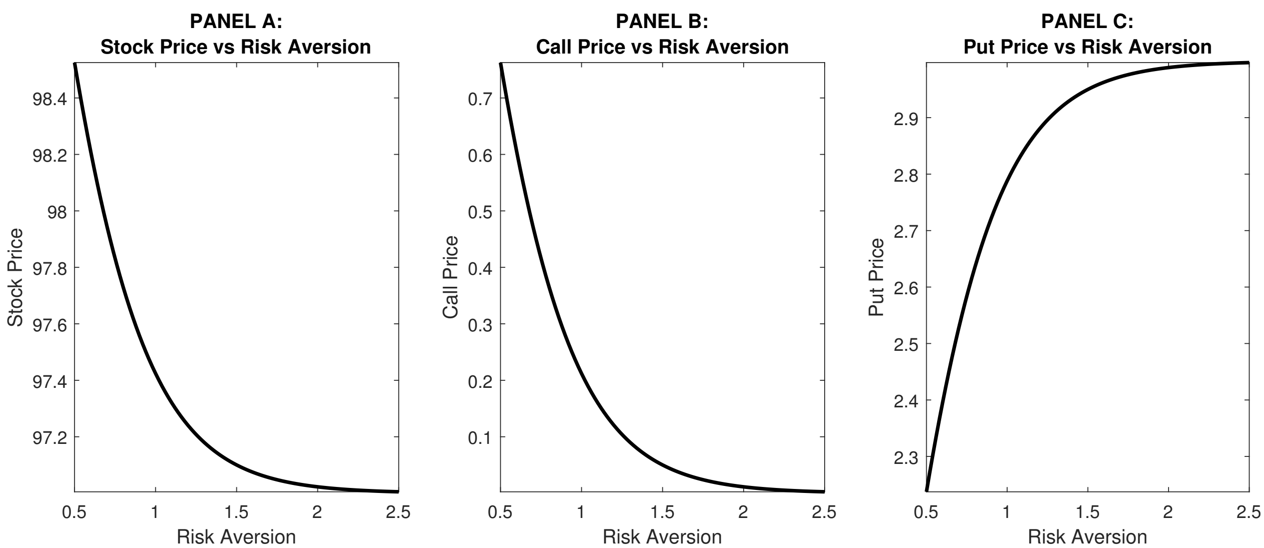

Figure 4: Comparative Statics on Risk Aversion

This figure shows the sensitivity of the stock price (Panel A), the call price (Panel B), and the put price (Panel C), to changes in the risk aversion parameter (γ), given a positive signal ($\theta = 1$). Figure 2 shows that the curves intersect at the point in which all the noise trading shocks are equal to zero (i.e., $z = 0$, $\nu_c = 0$, and $\nu_p = 0$). This figure plots these intersection points for different values of γ. Its baseline value is $\gamma = 1.5$. The remaining parameter values are kept unchanged: $F_H = 103$, $F_L = 97$, $K = 100$, and $\omega = 0.7$.

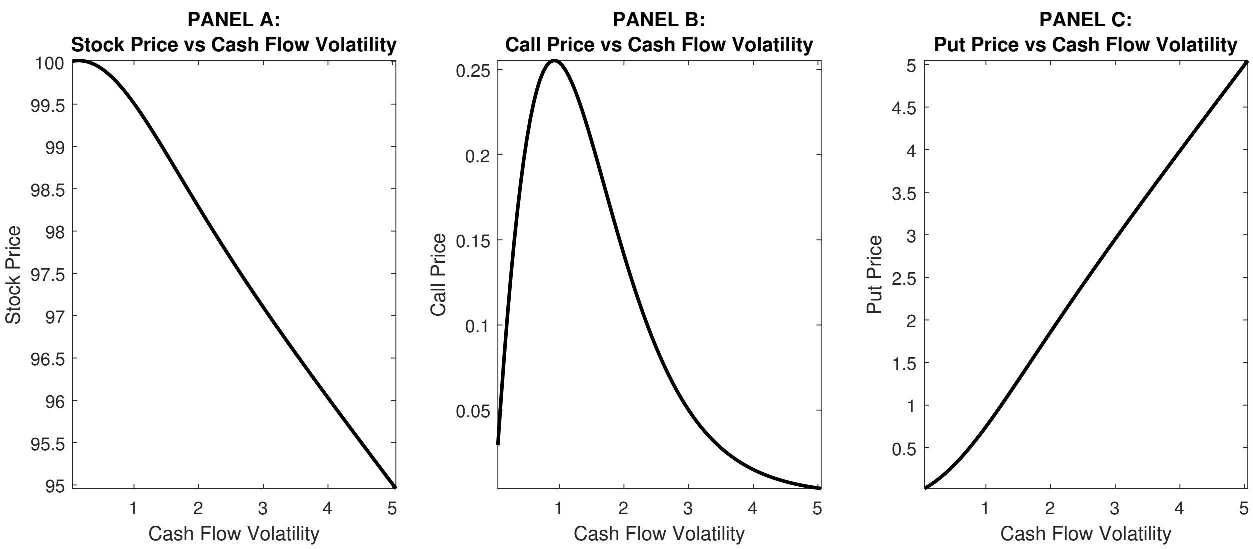

Figure 5: Comparative Statics on Cash Flow Volatility

This figure shows the sensitivity of the stock price (Panel A), the call price (Panel B), and the put price (Panel C), to changes in the volatility of the firm cash flows at time $t = 2$. The cash flow volatility is captured by the distance between the high and the low cash flows, i.e., $F_H - F_L$. The variation in cash flows is kept symmetric in relation to the strike price $K = 100$, which means that $F_H = K + a$ and $F_L = K - a$. In the baseline model, $F_H - F_L = 103 - 97 = 6$, which corresponds to the case $a = 3$. This figure plots the results for values of a between 0.05 and 5.05. Figure 2 shows that the curves intersect at the point in which all the noise trading shocks are equal to zero (i.e., $z = 0$, $\nu_c = 0$, and $\nu_p = 0$). This figure plots these intersection points for different values of a. The remaining parameter values are kept unchanged: $K = 100$, $\omega = 0.7$, and $\gamma = 1.5$.

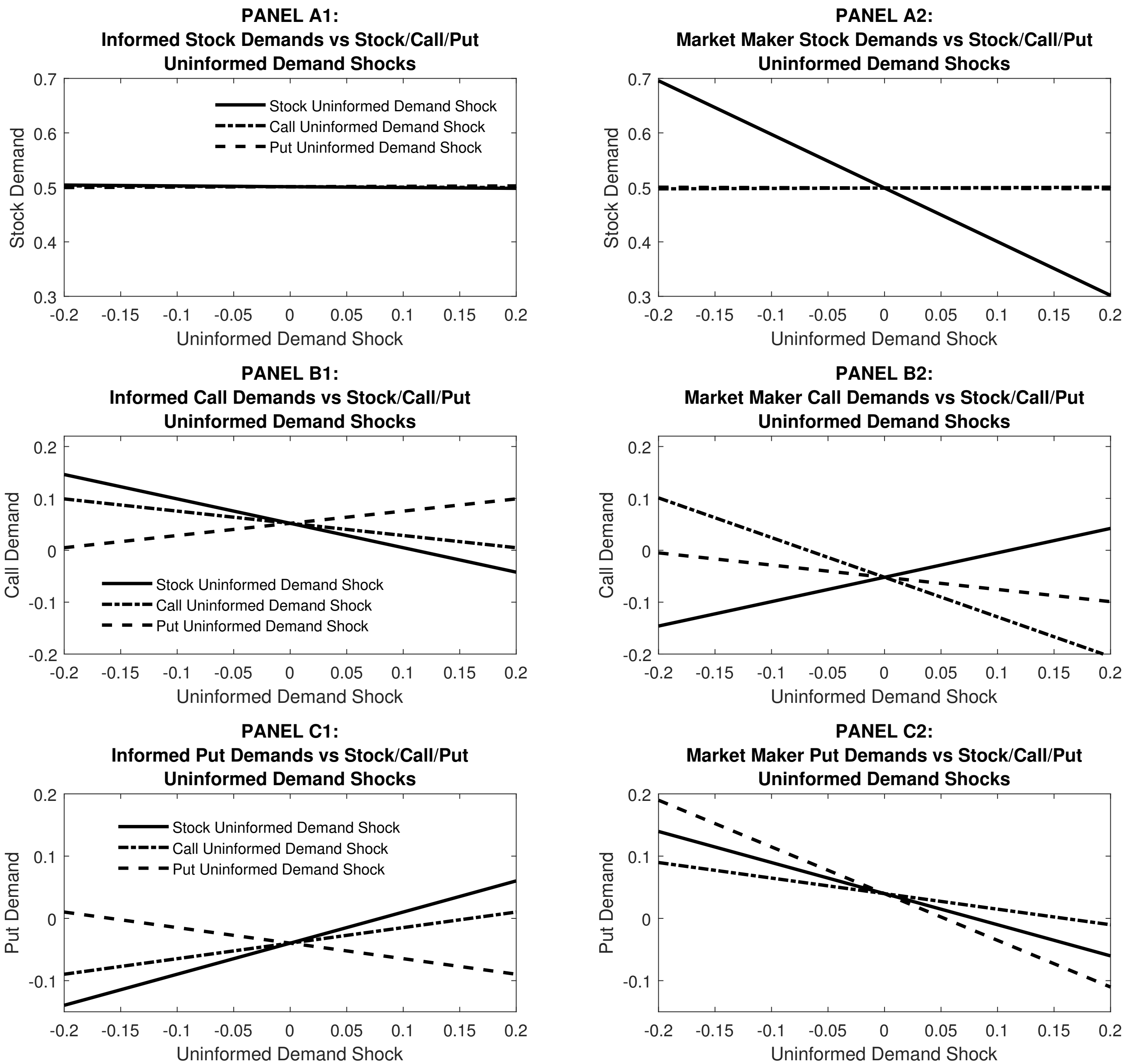

Figure 6: Comparative Statics on Demand Functions

This figure plots in panels A1, B1 and C1 the informed stock, call and put demands as functions of uninformed demand shocks, given a positive signal, i.e., $\theta = 1$, respectively. For all panels, the solid line represents the uninformed stock demand shocks z, while keeping the uninformed demand shocks fixed at $\nu_c = 0$ and $\nu_p = 0$, respectively. The dash-dotted lines represent the informed call demand as a function of the uninformed call demand shocks ν_c, while holding the uninformed stock and put demand shocks at $z = 0$ and $\nu_p = 0$. The dotted line represents the informed put demand as a function of the uninformed put demand shocks ν_p, holding the uninformed stock and call demand shocks at $z = 0$ and $\nu_c = 0$. Panel A2, B2 and C2 depict the same analysis, but for the market maker. The parameter values used to generate the plot are as follows: $F_H = 103$, $F_L = 97$, $K = 100$, $\omega = 0.7$, and $\gamma = 1.5$.

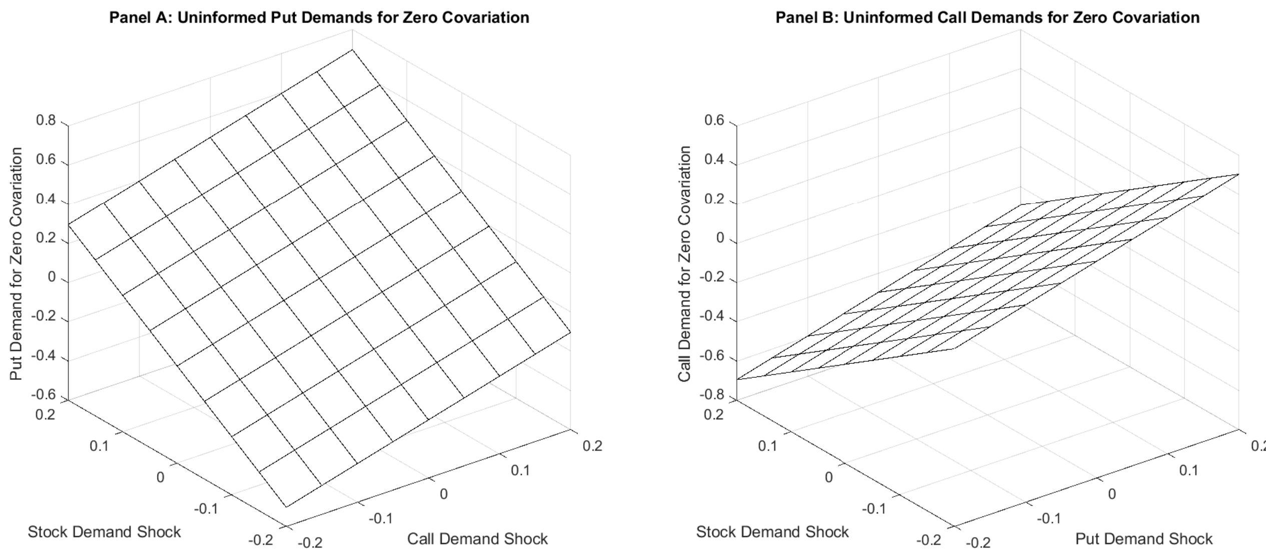

Figure 7: Indifference Surfaces for Call and Put Demand Shocks

This figure shows the uninformed put (Panel A) and call (Panel B) demands that lead to $S_1 = E(S_1)$, keeping the other two demand shocks fixed. These surfaces are the same for calls and puts, because the zero-covariance condition is the same for both. The only difference between the panels is about which demands we fix, and which we solve for, to obtain zero covariance. For Panel A (B), we fix both the stock and call (put) demand shocks and solve for the put (call) demand that matches the zero covariance condition. The sign of covariance above or below the surfaces depends on the option we use. It is negative for coordinates (z,ν_c,ν_p) above the surface in Panel A if we use a call option, but positive for a put option. In Panel B, the covariance is positive above the surface if we use a call option, but negative for a put option. The signs of the covariances are the opposite below the surfaces.

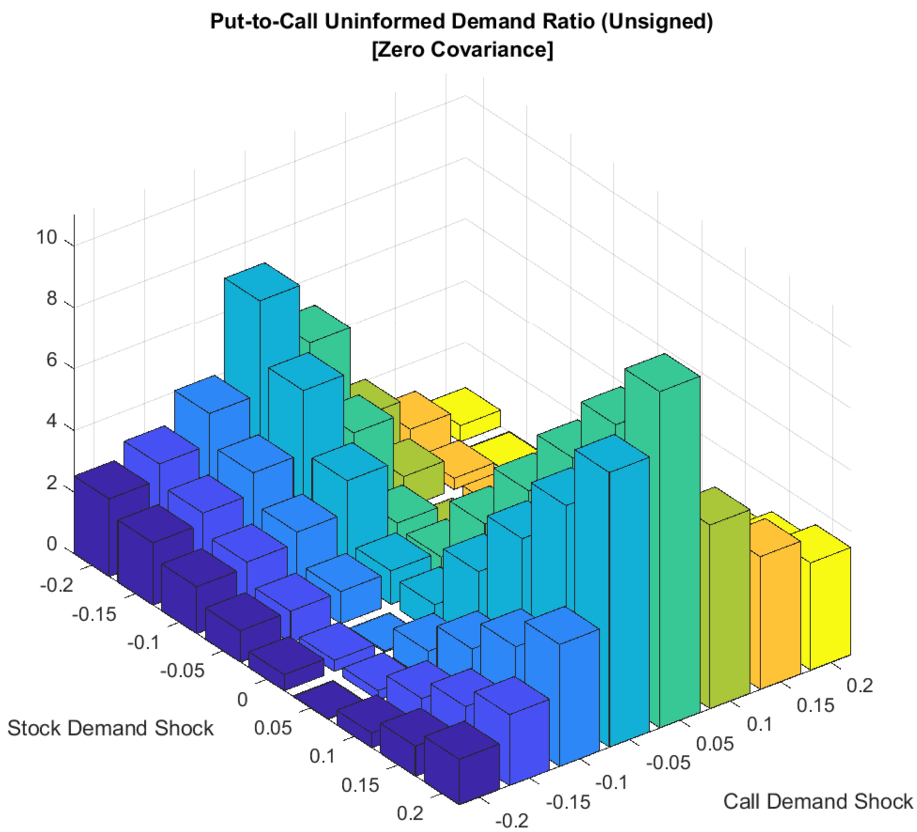

Figure 8: Put-Call Volume Ratio

This figure plots the absolute uninformed put/call (P/C) ratio defined as:|put volume|/|call volume|. The put volume used to estimate the P/C ratio is obtained by fixing both the uninformed stock demand shock and the uninformed call demand shock, and solving for the uninformed put demand subject to the constraint $(S_1 - E(S_1)) = 0$. The same holds for the call volume, but this time fixing the uninformed stock demand shock and the uninformed put demand shock, and solving for the uninformed call demand subject to the constraint $(S_1 - E(S_1)) = 0$. The parameter values used to generate the plot are as follows: $F_H = 103$, $F_L = 97$, $K = 100$, $\omega = 0.7$, and $\gamma = 1.5$.

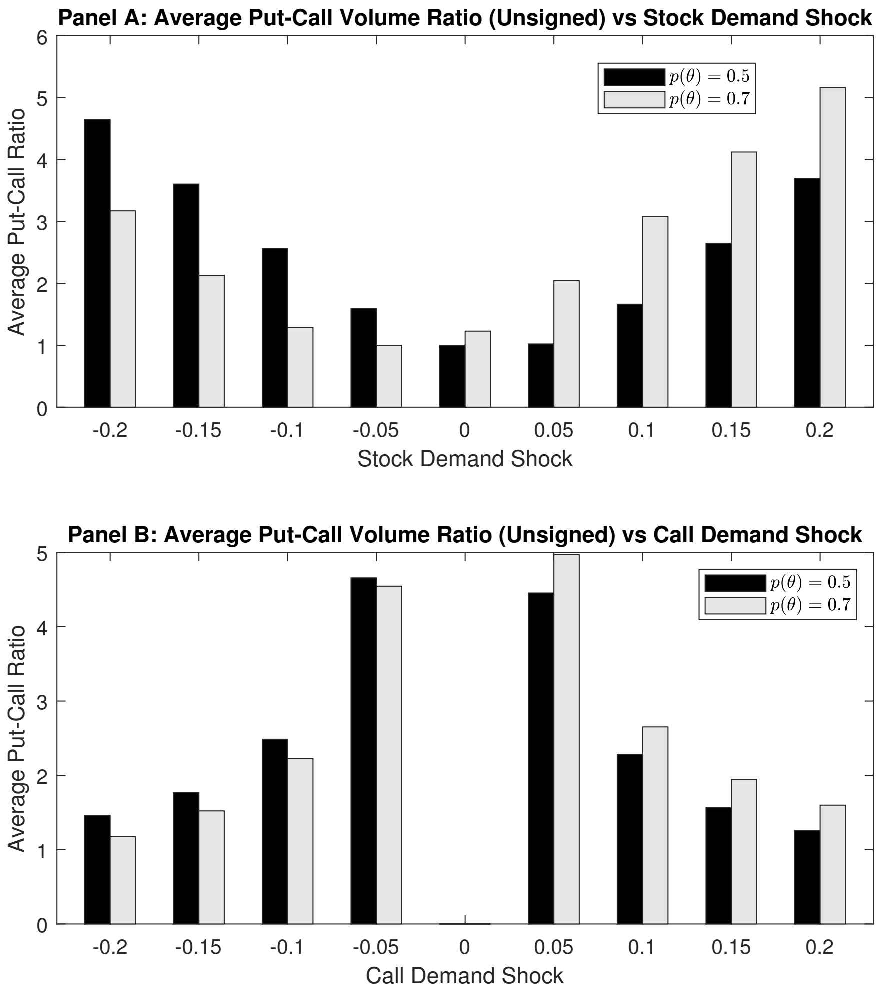

Figure 9: Put-Call Volume Ratio for Informative and Uninformative Signals

The upper and lower panels of this figure plot the average absolute put/call (P/C) ratio defined as:|average put volume|/|average call volume| for the case in which the informed investor receives or not a private signal before time 1. For both panels, the black bars depict the P/C ratio for the case in which there are no informed investors in the economy $p(\theta) = 0.5$ while the gray bars depict the case in which the informed investor receives positive information $p(\theta) = 0.7$ right before time 1. Panel A depicts the average P/C ratio having the stock demand shock on the x axis. Panel B depicts the average P/C ratio having the call demand shock on the x axis. The parameter values used to generate the plot are as follows: $F_H = 103$, $F_L = 97$, $K = 100$, $\omega = 0.7$, and $\gamma = 1.5$.

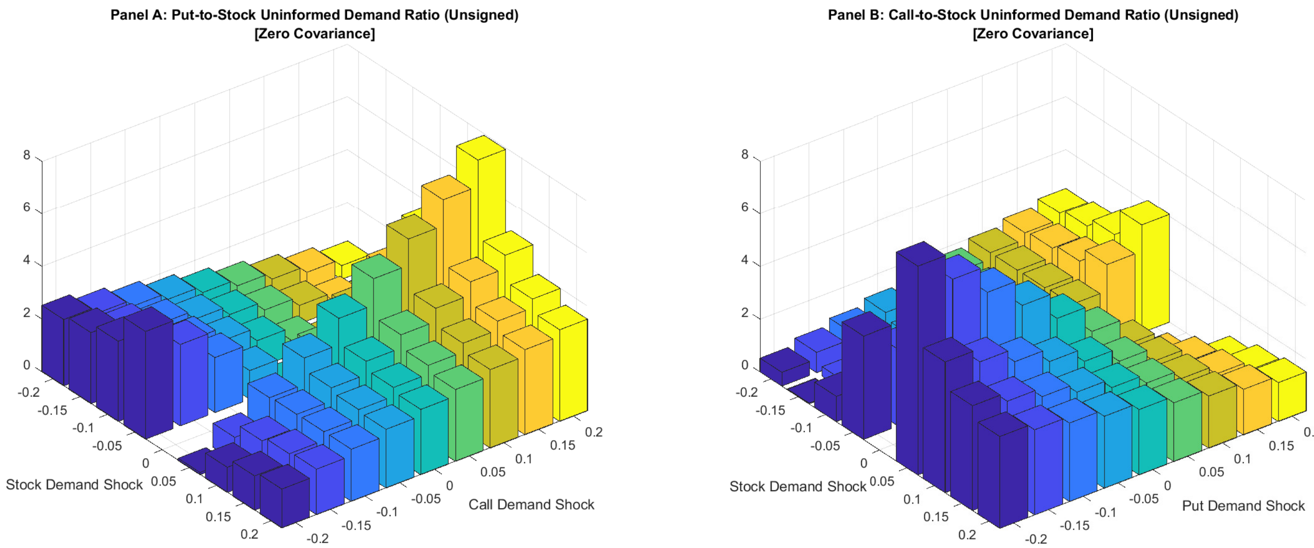

Figure 10: Option-to-Stock (O/S) Volume Ratio

This figure plots the uninformed option-to-stock (O/S) volume ratio defined as: |Put volume|/|Stock volume| in Panel A, and |Call volume|/|Stock volume| in Panel B. Similar to the demand functions, the option volumes used in the figure are obtained by optimization. The put (call) volume is obtained by fixing both the uninformed stock demand shock and the uninformed call (put) demand shock, and solving for the uninformed put (call) demand subject to the constraint $(S_1 - E(S_1)) = 0$. The parameter values used to generate the plot are as follows: $F_H = 103$, $F_L = 97$, $K = 100$, $\omega = 0.7$, and $\gamma = 1.5$.